Meaning-making and Well-being of

Chinese Working Dementia Caregivers:

Implementation of Experience Sampling Method

by

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Abstract of thesis entitled

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Background: The well-being of Chinese working dementia caregivers demands urgent and critical attention due to their extraordinary suffering from burdens in their daily care, work, and personal life. As the backbone of informal care, adult child dementia caregivers bear significant burdens in all aspects of their daily lives. Besides various coping approaches, meaning-making as an existential approach can be extremely helpful in devastating experiences occurring to these caregivers. However, the existing literature investigating the meaning-making of stressful experiences in dementia caregivers has limited scope in the unidimensional aspect (i.e., caregiving experiences only), while the work and personal life experiences have been completely overlooked. A meaning-making choice model is proposed to reflect the actual daily living experiences of an adult-child working dementia caregiver and how meaning-making of multiple stressful experiences compensates for their compromised well-being.

Objectives: This study aims to investigate the meaning-making choice of Chinese working dementia caregivers and how the choices affect their hedonic and social well-being. Furthermore, the study will address how the dyadic relationship between a caregiver and a care recipient, a critical cultural element in Chinese society, interacts with the meaning-making choice in affecting well-being.

Methods: To address the research objectives, the 15-day-ESM structured questionnaire survey was conducted with a cohort of 100 working dementia caregivers recruited in Guangzhou, Guangdong, China, from October 2020 to August 2021. The ESM survey included phases of baseline (30 minutes), ESM (2 to 3 times per day), and follow-up (10 minutes). The average ESM well-being scores were calculated to accurately simulate working caregivers' daily experiences over a period of time. The mediation models and moderated mediation models were adopted to investigate the impact of caregivers' meaning-making choices and dyadic relationships on their well-being.

Results: First, the findings identify the choice of making higher meanings in care, work and personal experiences is associated with the best hedonic and social well-being. Holding higher meaningful levels on more aspects (i.e., care and work aspects) leads to better well-being than that on fewer aspects (work aspect only, or none aspect). Second, the validated meaning-making choice model gives guidance on meaning-making practices and informs theoretical development on meaning-making in working dementia caregivers. Third, ESM data examines hedonic well-being more stably and accurately than that of the cross-sectional design. Last, the dyadic relationship is critical in the proposed meaning-making choice model.

Discussion: The proposed meaning-making choice model incorporating the dyadic relationship implies that caregivers' well-being can be sustained or improved with high meaning-making levels and/or better dyadic relationships. More specifically, better dyadic relationships can help maintain a higher well-being level for those who with a higher meaning-making level, and even more greatly benefit the well-being of those with a lower meaning-making level. The meaning-making choice model indicates that achieving the balance among meaningful care, work and life experiences yields better well-being outcomes. The mobile-based ESM design is advantageous in detecting hedonic well-being during daily living experiences with more robust measurement, better sensitivity, feasibility, usability, and ecological validity and reliability. (493 words)

Keywords: Meaning-making, Well-being, Chinese Working Dementia Caregivers, Experience Sampling Method

Declaration

I hereby declare that this dissertation entitled "meaning-making and well-being of Chinese working dementia caregivers: implementation of experience sampling method" represents my own work, except where due acknowledgment is made, and that it has not been previously included in a thesis, dissertation or report submitted to this University or to any other institution for a degree, diploma or other qualification.

Signature: Shuangzhou Chen

Date: May 1st, 2022

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List of Abbreviations

Abbr.	Abbreviation(s); Abbreviated
PwD	people living with dementia
WDC(s)	working dementia caregiver(s)
ESM	experience sampling method
WHO	World Health Organization
OECD	The Organisation for Economic Co-operation and Development
GDP	gross domestic product
DALYs	disability (adjusted) life years
YLLs	the sum of years for life lost
IADL	instrumental activities of daily living
ADL	activities of daily living
HWB	hedonic well-being
SWB	social well-being
SPM	the Stress Processing Model
SDT	the Self-Determination Theory
MFT	the Meaning Frame Theory
TM	the Theory of Meaning
TD	the Decision Theory
TMD	the Theory of Meaning and Decision
CT	the Choice Theory
MMC	meaning-making choice(s)
DR	dyadic relationship(s)
PANAS	Positive and Negative Affect Schedule
PA	positive affectivity
NA	negative affectivity
PHQ	Patient Health Questionnaire-9
ZBI	Zarit Burden Interview
OSI	Occupational Stress Index
CG	caregiver(s)
CR	care recipient(s)
URCS	Unidimensional Relationship Closeness Scale
FMTC	Finding Meaning through Caregiving Scale
PAC	Positive Aspects of Caregiving
WAMI	Work and Meaning Inventory
MLQ	Meaning in Life Questionnaire
NPI	Neuropsychiatric Inventory
ANOVA	analysis of variances
SPSS	Statistical Package for the Social Sciences

Chapter 1 Introduction

Among chronic non-communicable diseases with increasing prevalence, dementia is estimated to have the largest economic and social impact, surpassing the combined impact of cancer, heart disease, and stroke (World Health Organization, 2019). These impacts are even more significant in countries of low-income and middle-income status, such as China (World Health Organization, 2018). The prevalence rate of dementia in China was estimated at 7.4% among older adults aged 60 years or above in 2017, and projected to have 36 million older adults suffering from dementia in 2050, making China one of the countries with the highest dementia prevalence in the world (Y. Wang et al., 2019). The economic burden of dementia in China is estimated to be US$ 111 billion in 2030, accounting for a 10% share of the global costs, thus bearing the largest burden of dementia disease around the world (Xu et al., 2017).

Whilst the affordable and sustainable care for people living with dementia (PwD) is posing great challenges to both medical and social care systems, dementia care casts a paramount burden on informal caregivers, who could potentially become the “invisible second patients” (Brodaty & Donkin, 2009). Family caregivers of PwD in general are facing great stress and burden from caregiving tasks, and suffering from both adverse physical and mental issues. Depression, negative affectivities, and poorer physical health are broadly reported among dementia family caregivers compared with non-caregivers (Cheng, 2017; World Health Organization, 2019), more severe than caring for other chronic diseases (Cheng, 2017; Fujihara et al., 2019; Li et al., 2017). The percentages of family caregivers in the total population range from 5% (i.e., China, Japan, Germany) to 20% and above (i.e., Canada, Israel, United States, United Kingdom) (International Alliance of Carer Organizations, 2021). A national report in the United States indicated that more than half (51%) of the informal caregivers of PwD are adult children, followed by spouses/partners (12%), grandchildren (8%), siblings (7%), parents (6%), and other relatives (6%) (AARP, 2015). So far, there is no representative data manifesting the percentages of each caregiving role among Chinese dementia caregivers. During the effective period of China’s one-child policy effectively taken from 1979 to 2015, the caregiving burden has undoubtedly grown into a serious concern when fewer working adult-child caregivers are available to respond to rapidly increasing demands from PwD in both urban and rural areas (K. Y. Chan et al., 2013). Adult-child caregivers are experiencing stressful experiences in their daily caregiving. At the same time, they also suffer stress from their work and the deviations from their original personal interests and development. Stressful experiences from aspects of daily caring, working and personal life can be more challenging than caregiving stress alone (Black et al., 2010; F. Chen et al., 2018). Caregivers suffer from not only caregiving burdens, but also physical and psychological burdens from work stressors and the absence of personal interests and development (Fujihara et al., 2019). Caregivers suffer greatly from the overwhelming double burden or triple stressful experiences from care, work, and personal life and then can easily develop depression and neuropsychiatric symptoms (Cheng, 2017).

People generally have an inherent need to make meanings out of stressful experiences. The absence of meaning can be catastrophic to caregivers living with challenging events. Common adverse consequences without meaning-making comprise of existential stress (V. E. Frankl, 1963; Park, 2020), negative subjective states (e.g., boredom, apathy, emptiness, or self-doubt) (Krok et al., 2021; May & Yalom, 1989), negative emotions (e.g., aggression, depression and addiction) (V. E. Frankl, 1985; Milman et al., 2020), and compromised well-being (Csikszentmihalyi & Larson, 2014). Meaningfulness can be extremely helpful in devastating experiences such as the challenges posed in the situations where the demands of care, work, and personal pursuits that keep people under pressure. Meaning-making has been empirically proven with its usefulness in dealing with highly stressful life experiences (e.g., traumatic experiences, hopelessness, grief experiences) (Baumeister, 1991; V. Frankl, 1963; Park, 2010), and motivating one's care performance (e.g., care for PwD, care for children with autism, and care for spouses who have chronic diseases) (Allison & Arrigg, 2012; N. M. Lou & Noels, 2019; Lubetsky et al., 2014), increasing value awareness, dealing with work and life stress (Quinn & Toms, 2018), and increasing hedonic and social well-being (Csikszentmihalyi & Larson, 2014).

For working dementia caregivers (WDCs), the aspects of care, work and personal life have not been readily addressed in past empirical studies. Existing literature has focused on one of two aspects, in correspondence to care-work conflicts, care-life conflicts, and work-life conflicts (Alpass et al., 2017; Clarke, 2019; NurFatihah et al., 2013). Balancing activities of care, work and life are found to benefit caregivers' well-being. However, the balance of all three aspects of care, work, and life simultaneously has been ineffectively addressed. Besides making meanings of all aspects, WDCs who live with diverse conditions and backgrounds have choices in prioritizing making sense of some of these activities. While it is certain that meaning-making leads to improved health outcomes, simultaneous meaning-making of all daily living stressors has been much less conclusive both theoretically and empirically (Gillies & Neimeyer, 2006; Park & Folkman, 1997; Segerstrom et al., 2003). Therefore, WDCs well-being can be a heterogeneous outcome corresponding to meaning-making choices of daily living stressors (Quinn & Toms, 2018). In addition, well-being is generally unstable and volatile during extreme circumstances, such as high-demanding dementia care tasks, high working pressure, and an urge to develop personal interests within a limited timeframe in an occupied schedule (Csikszentmihalyi & Larson, 2014). Traditional measurements of well-being with either cross-sectional or longitudinal surveys do not accurately record daily fluctuations and changes in well-being. In comparison, within-subject analyses can precisely depict variations of well-being within each adult-child WDC.

To underscore these unresolved barriers in measuring the daily experiences of WDCs, two major goals need to be achieved: 1) a theoretical framework should be developed to respond to the research inquiries, and 2) an appropriate measurement should be adopted accordingly. To begin with, a theoretical framework can be proposed to interpret the decisional process of existentialistic experiences. A meaning-making choice model, incorporating self-determination theory (Noguchi, 2019), meaning framework theory (Leontiev, 2013), stress process model (Pearlin et al., 1990), and theory of meaning (Davidson, 1980), can be proposed to understand how WDCs deal with the stressful burden and seek meanings from the situational experiences in caregiving, working, or even personal affairs, and how their well-being is consequently affected. The conceptualization of meaning-making choice refers to the rational choice for a working caregiver to make for meaning-making among his or her care, work, or personal life experiences. The emerging notion of work-life balance inspired the careful selection of meaning-making strategies via various settings to improve caregivers' well-being (Jolanki, 2015; McPherson et al., 2010). The meaning-making choice model in this sense becomes more important to the caregivers living with various stressful sources when they want to maintain a decent and balanced life with quality and healthy well-being. Second, the experience sampling method (ESM) is applied to address the measurement of critical health outcomes. ESM differs from inter-person, cross-sectional, panel, and longitudinal designs.

Traditional cross-sectional or panel studies are designed to examine how individuals exhibit differences through trait-related characteristics, assuming that each individual has a stable trait, preferences, and behavioral pattern. ESM, on the other hand, is designed to efficiently summarize how the dynamic states of feelings, behaviors, and attitudes fluctuate by assuming that within-person variances of individual feelings (i.e., well-being) broadly exist and are influenced by naturally occurring events or daily routine incidents (Dimotakis et al., 2013). To address the research inquiries that are concerned, ESM is one of the fittest methods that concisely describe the well-being trajectory.

To understand how meaning-making choices can benefit WDCs, their well-being needs to be evaluated as well. The concepts of well-being have long been established and documented. Hedonic well-being refers to subjective well-being, happiness, and positive affectivities and emotions (Chamberlain & Zika, 1988; Debats et al., 1993; Diener, 1984). Social well-being refers to the evaluation of one's conditions and societal functioning in the aspects of "social integration," "social acceptance," "social contribution," "social actualization," and "social coherence" (Keyes, 1998; Naci & Ioannidis, 2015). Despite scarce literature demonstrating how meaning-making choices across multiple contexts influence caregivers' well-being states, abundant studies that address the well-being of dementia caregivers give inspirational ideas for understanding how meaning-making choices impact caregivers' well-being. Literature in the past has successfully identified the associations between meaning-making of care with hedonic well-being (M. Steger, 2012). More recent studies urged the importance of meaningful work and its relationship with social well-being (M. F. Steger et al., 2012; M. F. Steger, 2018). Meaning in life has been a big topic discussed in the fields of psychology, philosophy, and health, in which the significance of meaning-making to people's well-being has as well been emphasized. Hedonic and social well-being has been commonly found as extremely crucial among all well-being for evaluating the effectiveness of meaning-making (M. Steger, 2012). Overall, meaning-making of care, work and personal life all have impacts on a working caregiver's hedonic and social well-being.

However, the literature has not addressed the uniqueness of cultural and ethnic contexts as meaning-making is outlined as the general strategy for all human beings to overcome extreme challenges. Despite the fact that the entire world is aging rapidly, China, as a developing country, has also been one of the fastest aging countries in the world holding 18.47% of the world's aging population (Nichols et al., 2019). China has one of the highest dementia prevalence rates (6.61%) in the world and the largest PwD population (9.5 million) since 2015 (Prince et al., 2017). Meaning-making of dementia caregiving experience in the Chinese context can be affected by not only common characteristics and psychological components, but also unique values and beliefs embedded in the Chinese culture, such as filial piety, dyadic relationships, and living arrangement in the cultural context of family union (Yeh et al., 2013). This study aims to investigate the meaning-making choices of Chinese WDCs and how the choices affect their hedonic well-being and social well-being in the proposed theoretical framework of meaning-making choice.

In Chapter 2, a collection of existing literature will be reviewed, followed by the research gap identified in Chapter 3. In Chapter 4, an integration of the theoretical framework will be proposed based on the research gap. In Chapter 5, research questions and hypotheses were raised. In Chapter 6, the methodological issues will be planned and discussed. In Chapter 7, major findings will be generated and demonstrated. In Chapter 8, issues relevant to research questions will be discussed based on the findings.

Chapter 2 Literature Review

The literature review chapter summarizes dementia and working dementia caregivers (WDCs) in the context of China; meaning-making for dementia caregiving, working, and personal life; and well-being. Chinese WDCs who confront caregiving, working, and living challenges are in great need for making sense of the demanding experiences in such a challenging and fast-paced environment. The meaning-making of care, work and life further poses influences on their well-being to make them feel sustainable and motivational to carry on caregiving, working, and living tasks. The focus of this chapter covers general dementia care, informal dementia care provided by WDCs, meaning-making of dementia care, work and life, and the associations between meaning-making and well-being.

2.1 Dementia and Caregiving

2.1.1 Aging and Dementia

Many chronic diseases, including dementia, are reported to be more prevalent in the aging population compared to the general population (Alzheimer's Association, 2018). Global aging has been observed ubiquitously since recent years as the life expectancy is increased from an average of 67.2 year-old in 2010 to 72.2 in 2015 (WHO, 2016), while the proportion of the aging (65 years old and older) population being 962 million in 2017 has been projected more than doubled as much as 2.1 billion in 2050 according to WHO (WHO, 2017). Meanwhile, the concept of a super-aged society has come to our attention. The so-called super-based society refers to a society that has an aging population of more than 20% of the total population (Koohsari et al., 2018). More than ten super-aged societies (e.g., Netherland, France, Sweden, Portugal, Slovenia, and Croatia) are estimated by the end of 2020, and 34 are expected by 2030 (e.g., Korea, the US, the UK, New Zealand, and Hong Kong), according to the report from the United Nations (2019). Impaired economic and social development in these super-aged countries are projected to drag down the world's economic growth (Nichols et al., 2019; Shimoyamal et al., 2018). The aging trend observed in the aging era and super-aged societies give rise to the demanding efforts in advocating environmental support and policy interventions from the government and counteracting the challenges given by chronic issues, long-term health care and support, caregiving needs in addition to health disparities, environmental equity, and activity-friendly urban challenges (Koohsari et al., 2018). Besides worldwide societies such as France, Sweden, the UK, and the U.S., many Eastern and South-Eastern Asian societies will soon join the union of aging societies with a growing awareness of aging. For example, in terms of the increasing percentage of older adults' share in the total population from 2019 to 2050, Korea will have the largest increase of 23.0%, followed by Singapore 20.9%, Taiwan province 19.9%, Macao SAR 17.7%, and Hong Kong SAR 17.2% (United Nations et al., 2019). The number of Chinese older adults will reach 487 million, or 35% of its total population, in 2050 (China National Committee on Aging, 2006). The aging issues have brought global concerns to the economy, environment, society, and public health (Nichols et al., 2019). Dementia has been one of the leading causes of death among the aging population, followed by arthritis, diabetes, heart diseases, cancer, and stroke (Nichols et al., 2019).

Dementia, an umbrella term mainly consisting of Alzheimer's Disease, has been defined as a syndrome with deterioration in memory, cognitive thinking, and behavioral functioning and the abilities required in daily activities (Alzheimer's Association, 2018). Dementia can be found in different specific subtypes: Alzheimer's disease taking 70% of all dementia cases, which include vascular dementia, Lewy bodies dementia, and a collection of symptoms constituting frontotemporal dementia. Dementia and its subtypes prevalently exist in the aging population, and are noticeably observed across the world. According to the estimation, 81% of the PwD are aged 75 or older (Gaugler et al., 2019). Overall, one in every ten people age 65 or older suffers from dementia. The percentages of PwD increase by age. Among the age group of 65 to 74, there are 3% living with dementia, 17% in the 75-to-84-age group, and 32% among people aged 85 and older (Rajan et al., 2019; Tom et al., 2015). There are 50 million people suffering from dementia worldwide, nearly 60% of whom are living in low-income and middle-income countries (Livingston et al., 2017). An increase of 10 million dementia cases is estimated per year. The affected population worldwide is projected as 82 million in 2030 and 153 million in 2050 (Alzheimer's Association, 2019). The prevalence of dementia rises in aging countries as well. Typical aging countries, such as Japan, Italy, and Germany, are averagely estimated 2% of dementia prevalence, while the Slovak Republic, Turkey, and Mexico have approximately 1%.

OECD countries are the most fast-aging ones in the world, such as Korea, Chile, Brazil, China, Columbia, and Costa Rica, with an average prevalence of 1.5% (OECD, 2017). Based on the previous estimation of the high prevalence of dementia among the aging population, many aging countries and super-aged societies will tackle this issue seriously in the near future. Pharmacological interventions, such as the prescription of antipsychotics, have been proven to greatly reduce the prevalence of dementia. In a longitudinal view, the aging countries prescribed antipsychotic medications to 3% of people aged 65 to 69 in 2015 and 12% to people aged over 90.

In comparison to the average dementia prevalence of 5.50% to 7.00% in the countries with the fastest growth in prevalence of dementia, China has grown to 6.19% in 2020 (Hong Kong 7.2% and Macau 4.5%), increasing since 1990, accounting for 25% of dementia cases in the world (Nichols et al., 2019; World Health Organization, 2019; Y.-T. Wu et al., 2018). The age group of 55 to 90 year-olds has doubled its dementia prevalence rate in a 5-year interval. Compared to Chinese men, women have a ratio of 1.65 in suffering from dementia, and 2.37 Alzheimer's disease (K. Y. Chan et al., 2013; J. Jia et al., 2014; Y. Zhang et al., 2012). Formal dementia care relies on sources of nursing homes, day-care centers, government-owned and private agencies for long-term care, and community-based centers. However, these supportive sources appear to be non-useful or unfeasible at this moment due to the insufficiency of dementia care services, manpower, useful interventions, or inaccessibility. In addition, dementia casts great challenges on the health budget by consuming 1.09% of the global GDP and at the cost of approximately 168 billion, including direct medical costs (32.5%; e.g., outpatient costs), direct non-medical costs (15.6%; e.g., health care equipment), and indirect costs (51.9%; e.g., monetary loss due to unemployment) (J. Jia et al., 2018). The overall health cost was estimated to rise to 249 billion (18.7% of the global health costs) in 2020, and 1.89 trillion (20.7%) in 2050 in China. China now has not yet come up with a well-developed care system for dealing with dementia, due to not only the nature of dementia care, such as poor public awareness regarding dementia, high costs of care, policy challenges, and legislative difficulties, but also caregivers' insufficient education, inadequate knowledge, and misbelief on dementia and dementia care (Z. Chen et al., 2017). Since the start of the 13th Five-Year Plan, the Chinese Government and its Minister of Health have implemented policy plans to have early screenings for the PwD population and improve the accessibility to healthcare services for dementia (L. Jia et al., 2019).

Without professional services, such as early screening, preventive approaches, or supportive care, the consequences of dementia, according to the Lancet report (2019) include deaths, excess mortality, and demands for health services. The estimated lifetime risks of having dementia symptoms at the age of 45 and onward indicated one-fifth (20%) for women and one-tenth (10%) for men (Gaugler et al., 2019). The lifetime risk is defined as the probability of someone developing the particular adverse condition(s) after a certain age. The lifetime risk of dementia at the age of 45 is estimated to occur at one in five for women and one in ten for men. The general risks for both sexes were alarming at the age of 65 (Alzheimer's Association, 2019). The percentage of changes in death rates of dementia for the age group of 65 and above between 2000 to 2017 was the highest as of 145%, in comparison to breast cancer with 0.5%, prostate cancer 1.9%, heart disease 8.9%, stroke 12.7%, and HIV with 6.6% (Gaugler et al., 2019). The burden of disease is defined as the public health impacts of disease before one's death. Dementia among many death-causing diseases, in terms of disease burden, was ranked 12th most burdensome disease or injury in 1990 and rose to 6th in 2016 by the calculation of disability-adjusted life years (DALYs). Dementia was also the 4th highest disease or injury according to the calculation of the sum of the number of years of life lost due to premature mortality (YLLs) (Gaugler et al., 2019; Mokdad et al., 2018). According to another survey, among all the leading causes of death for older adults, dementia ranks dementia now ranks the 1st in the UK, the 2nd in Australia, and the 6th in the US (Alzheimer's Association, 2019).

2.1.2 Dementia Informal Caregiving

Informal care greatly compensates for the incredible costs spent on direct medical and social care, (WHO, 2016; 2019). Informal care is surprisingly undersupplied due to the increasing number of PwD disproportionately outnumbering the available processional and formal services that are accessible (Winblad et al., 2016). With insufficient professional resources and nursing capacity, informal care has been highly demanded by families of PwD and long-term care facilities (Gräsel, 2002; Papastavrou et al., 2007). Informal caregivers contributed substantially to global dementia care, with 82 million caregiving hours annually, being equivalent to the working hours of 40 million full-time workers in 2015 and will increase to an equivalence of 65 million workers by 2030 (Ander Wimo, Serge Gauthier, Martin Prince, 2018). Informal caregiving provides 400 billion US dollars' worth of societal and economic impact, taking up 40% of the global dementia costs (Ander Wimo, Serge Gauthier, Martin Prince, 2018). The intensity of dementia care brings a financial burden to those informal caregivers. Dementia care, among all lifetime informal care, demands 70% of the total budget and its costs overthrow the average household income (Alzheimer's Association, 2018). According to the study provided by the Alzheimer's Association (2018), only 48% of caregivers reported financial readiness in daily caregiving, and 66% reported over-burdensome of monthly spending on care. Dementia care is time-consuming. The average weekly caregiving hours for an adult-child caregiver was about 22 hours (Alzheimer's Association, 2019), being equivalent to a part-time job with the worth $12.64 per hour (Gaugler et al., 2019). Given the great amount of time devoted to caregiving work for PwD, common dementia caregiving tasks involved: 1) assisting with instrumental activities of daily living (IADLs), such as "household chores, shopping, preparing meals, providing transportation, arranging for doctor's appointments, managing finances and legal affairs, and answering the telephone"; 2) assisting the person in taking medications correctly; 3) assisting the person in complying with treatment recommendations; 4) helping with personal activities of daily living (ADLs), such as "bathing, dressing, grooming and feeding and helping the person walk, transfer from bed to chair, use the toilet and manage incontinence"; 5) behavioral managing including "aggressive behavior, wandering, depressive mood, agitation, anxiety, repetitive activity and nighttime disturbances". 6) seeking and taking support services such as "support groups and adult day service programs"; 7) arrangement of nursing homes or assistive support care; 8) hiring and supervising others who provide care; 9) other tasks, such as: "providing overall management of getting through the day, addressing family issues related to caring for a relative with Alzheimer's disease, including communication with other family members about care plans, decision-making and arrangements for respite for the main caregiver, managing other health conditions (i.e., "comorbidities"), such as arthritis, diabetes or cancer, providing emotional support and a sense of security" (Sansoni et al., 2013). In the population of informal caregivers, offspring (70%) and spouses (12%) are the backbone taking care of PwD. Amongst offspring caregivers, adult children are the primary family caregivers with a share of 71% in comparison to other roles of grandchildren (11%), siblings (10%), and other relatives (8%) (AARP, 2015). Given the fact that most PwD in their early and medium stages are taken care of primarily by family members at home where professional services are not available or accessible (Caldwell et al., 2014), adult children are the major force tackling the degenerative symptoms of dementia. More specifically, they are issues of basic living skills, such as eating, showering, clothing, moving out of bed or chair, toileting, housekeeping, cooking, shopping, commuting, and maintaining medication. In addition, PwD's emotional and behavioral symptoms require long-term familial support and companionship (Alwin et al., 2010; Boylstein & Hayes, 2012).

The informal caregivers in their daily caregiving scenarios frequently confront high physical, emotional, and financial demands for their frequent and prolonged dementia care (World Health Organization, 2018). Regardless of meeting care demands or not, caregiving burden and stress can be perceived and lead to declined wellness in physical health (Sinha et al., 2017), emotions (Sansoni et al., 2013), economic costs to both households and society (Hollander et al., 2009), and psychological well-being and wellness (Jensen et al., 2015; Pinquart & Sörensen, 2006; Schulz & Beach, 1999), subsequent depression (Aryal, 2017), morbidity (Tuomola et al., 2016), and compromises in caregiver's resilience to keep informal caregiving sustained (Tuomola et al., 2016) due to the burden and stress caused in the dementia caregiving. Caregiving stress can be even more intense when the needs for long-term care have been consistently unmet, undesirable, inaccessible, ineffectively implemented, or mistrusted by family members (Tapia Muñoz et al., 2019). Chinese dementia caregivers have been greatly suffering from physical and psychological burdens. China, with the largest population and fast-growing aging population, is facing a similar situation where informal caregiving bears the most burden in the care system for PwD. China will soon become a member of the super-aged society in 30 years. Hong Kong SAR, being part of China, has its percentage of the aging population (65+) rising from 11.7% in 2017 to 27% in 2031 (Chi et al., 2005) and the prevalence of dementia was 7.2%, according to the survey in 2018 (Y.-T. Wu et al., 2018). In Mainland China, the percentage of the aging population (60+) was 12.73% in 2018 and the prevalence of dementia was 5.3% according to a survey in 2018 (Y.-T. Wu et al., 2018), and will take up 3% of the world dementia population by 2030 (Xu et al., 2017). Chinese in general hold the belief of "aging in place," with which people prefer informal or family caregiving rather than formal and professional care from the community and the government (W. C. H. Chan et al., 2017). Chinese informal caregivers thus tend to take more responsibility for taking care of PwD than those from other cultures. Dementia caregiving provided by Chinese adult-child caregivers with such intensity and financial challenges leads to endless consumption of time, physical power, and psychological burden and devastating wellness states, such as role strains, stress and anxiety, and depressive symptoms (X. Zhang et al., 2018). Chinese caregivers with additional responsibilities, such as employment and personal development, confront even worse situations.

2.1.3 *Working Dementia Caregivers and Work-Life Balance*

Adult-child WDCs being the backbone of dementia caregivers are bound to full-time or part-time work in addition to their care tasks. Work-related stress has tremendous impacts on adult-child WDCs' care performance. Working caregivers are found with more caregiving stress and role conflicts in care than regular full-time caregivers (C. Ryan et al., 2021). Caregiving, on the other hand, influences one's work participation, work ethics, and work motivation (Fujihara et al., 2019). According to the Alzheimer's Association (2019), caregivers of people with living dementia, compared with other diseases, are more often found to coming to work late, leaving employment early, or dozing off during working hours. There are some other cases, where caregivers cut working hours, transitioning from full-time work to part-time work, occasionally receive a warning about working performance, turn down a promotion, or even quit the job (Alzheimer's Association, 2019).

Conflicts between work and personal life significantly occur to adult-child WDCs as well. The imbalance between work roles and family roles breaks the dynamic demands of personal life, professional work, and family roles (Ford et al., 2007), leads to the absence of financial and emotional support, and takes personal life as an extra burden to be ignored (X. Liu et al., 2017). On the other hand, caregiving burden and stress have a significant impact on dementia caregivers' personal lives (Xu et al., 2017). More evidence suggests that life-work or care-life conflicts inevitably compromise one's personal time and aspiration, and are associated with poorer well-being because simultaneous work and care are harmful to the mental (Alpass et al., 2017; NurFatihah et al., 2013), physical (Kimura et al., 2015; NurFatihah et al., 2013) and social functioning (NurFatihah et al., 2013) of WDC. Work-life balance has been an important topic worldwide since the 1970s, when the emerging working cultures, gender roles in work, and competitive job market negatively infringed other aspects of life among the working class and their families (Kalliath & Brough, 2008). Given the fact that Chinese WDCs have been overlooked in the evolvement of literature regarding work-life balance (Clarke, 2019), adult-child WDCs face challenges of "double burden" or "triple burden." The term "double burden" refers to the burden perceived in both employment and domestic responsibilities (such as caregiving or household chores) for which caregivers need to balance (Moen, 1992; S. Smith & Converse, 2012). The "triple burden" involves burdens generated from personal affairs and pursuits besides care responsibilities and work duties. Therefore, the WDCs are facing a great undue burden with stress, role strains, and guilt for not helping the PwD with their full strengths (Li et al., 2017; Sun, 2014), yet cannot avoid similar challenges that exist in other countries, such as stress and depressive symptoms (Clarke, 2019; X. Zhang et al., 2018).

Chinese WDCs with diverse characteristics and beliefs have different perspectives on balancing conflicts between care, work and personal life. To begin with, Confucius' term "filial piety" has been deeply rooted in Chinese culture. Filial piety means to "show love, respect, and support to one's parents" (Khalaila & Litwin, 2012), and is not merely an external expression as in ritual respect to parents living with dementia, but also an innate attitude (Sung, 2001). Filial piety consists of several aspects, including an awareness of rewarding parents who bore the rearing burden (Sung, 2001), reciprocity of the care back to parents (Sung, 2001), and an obligation towards one's ancestors (D. Y. Ho, 1994; Hsu, 1998) regardless of their own busy work and life. Filial obligation leads to excessive concerns about a parent's dependency and underreported negative emotions and feelings of incompetence among caregivers (Li et al., 2017). Furthermore, the responsibility of family dementia care has been greatly emphasized in Chinese culture, which expects family members, spouses and children, to carry out the dementia care tasks (W. C. H. Chan et al., 2017). Moreover, the stigma attached to dementia prevents families from seeking professional care, adding more psychological strains and burdens to informal caregivers. Stigmatization towards individuals diagnosed with dementia has been commonly observed in the Chinese culture and discourages one's potential willingness to receive further prevention, diagnosis, and interventions (Hofstraat & van Brakel, 2016; Kung, 2003; Nyblade et al., 2019). Sever stigma undermines one's willingness to seek assistance, aggregates social withdrawals, and negatively influences one's well-being in daily living (Nyblade et al., 2019). Other social influences, such as "the social perception of the diseases, the availability in familial support, coping resources and strategies, and potential involvement in caregiving", on stigma plays an important role in affecting caregivers' commitment to caregiving (Kung, 2003). Even though a small portion of families seek help from neurological and psychiatric treatment (Sun, 2014), the prevalent belief considers dementia to be nothing other than the consequence of the normal aging process (Sun, 2014). With little knowledge besides ICD-10 and Chinese dementia guidelines (American Psychiatric Association, 2013; J. Jia, 2016; Rajan et al., 2019), dementia to the Chinese general public is considered a misdiagnosis rather than mental diseases. Last, support for dementia caregiving by family members and communities is accepted, whereas the professional assistance was believed not useful and stigmatized (Pang & Lee, 2017).

2.2 Hedonic Well-being and Social Well-being

Without well-equipped coping strategies countering overwhelming stressors from caregiving tasks, employment pressure, and absence of personal development, a poor-balanced life of working dementia caregivers leads to compromised well-being whereas a well-balanced life among sustainable care, appropriate employment and happy living is often associated with a high level of well-being (Austen & Ong, 2010; OECD, 2006; Trust, 2015). Meaning-making as an existential means helps caregivers mitigate their perceptions of stress and positively influence their state of well-being (Quinn, Clare, & Woods, 2012; Schonwetter et al., 2003). Well-being generally refers to the state of being comfortable, healthy, happy, or prosperous (Diener & Suh, 2003). The improved state of well-being influences WDCs' caregiving performance and indirectly benefits PwD (M. F. Steger et al., 2006). Well-being has a broad scope of connotations, including emotional well-being, social well-being, mental well-being, spiritual well-being, physical well-being, and financial well-being (Bradburn, 1969; Brdar, 2014). For working dementia caregivers, their daily perceived burden can negatively influence their emotional well-being. More literature also indicates that the employment burden adversely affects social well-being (Bradburn, 1969; Brdar, 2014). Of course, the daily burden would implicitly cause physical well-being or financial well-being. In the sense of stress-processing context, emotional well-being and social well-being are the more direct resultant of coped stressors, and thus are the two most crucial indicators of working dementia caregivers' well-being.

Steger (2018) in his writings pointed out that hedonic (subjective) well-being indicators are associated with meaning. The well-being that was originally and has still been most commonly referred to as hedonic well-being (HWB). Hedonic well-being refers to interchangeably with subjective well-being (Diener, 1984), happiness (Debats et al., 1993), positive affective and emotions (Chamberlain & Zika, 1988), and even vitality and life satisfaction (Chamberlain & Zika, 1988; Keyes, 2002; M. F. Steger et al., 2009). The subjective (hedonic) well-being, by its definition, consists of three components: frequent positive affectivity, infrequent negative affectivity, and cognitive evaluations such as life satisfaction, or in other words, include cognitive and affective evaluations of life as a whole (Diener, 1984). Scholars such as Flugel (1925) and Bradburn (1969) studied how people feel in their daily lives as in terms of hedonic well-being. Another similar term subjective well-being (SWB) was coined by Diener (1984), and was later used interchangeably with hedonic well-being (M. F. Steger, 2018). Hedonic or subjective well-being covers a broad scale of pleasant feelings and emotions, sense of satisfaction, and reduced negative emotions. More importantly, hedonic well-being facilitates one's judgment of meaningfulness (Hicks & King, 2007; King et al., 2007; M. F. Steger, Shin, et al., 2013).

Social well-being is another dimension of well-being status. Social well-being refers to "the appraisal of one's circumstance and functioning in society in the aspects of social integration (relationship to society and community), social acceptance (construal of society through the character and qualities of other people as a generalized category), social contribution (one's social value), social actualization (potential and trajectory of society), and social coherence (perception of the quality, organization, and operation of the social world)" (Keyes, 1998; Naci & Ioannidis, 2015). Keyes's (1998) model of social well-being, therefore, extends Ryff's model (1998) focusing on intrapersonal focus and eudaimonic view of well-being to further include the interpersonal perspective. From the view of social well-being, WDCs bearing overwhelming caregiving tasks are deviated from building a sense of social integration, social acceptance, social contribution, social actualization, and social coherence for being isolated from the community and the society (B. Ho et al., 2003).

Hedonic well-being and social well-being independently co-exist in human beings' well-being system (Gallagher et al., 2009). These two types of well-being are predominantly critical to the holistic life experience of Chinese WDCs (Lattanzi-Licht et al., 1991). However, the associations between well-being and specific meaning-making choice made by Chinese WDCs have rarely been studied. In light of the dearth in strategies of improving well-being and self-awareness in the context of mental health flourishment, the distinction between types of well-being needs to be identified while how this well-being in the real-time is affected by meaning-making approaches should be timely recorded (M. F. Steger, Shin, et al., 2013).

2.3 Meaning-making of Working Dementia Caregivers

WDCs experience stress from all walks of daily living experiences, such as care, work, and personal development. Therefore, making meanings of these different stresses can benefit the well-being of a WDC in many ways.

2.3.1 Making Meaning in General

The development of meaning-making has been transitioning from a reactive perspective of meaning maintenance, such as the style of coping stresses, to a proactive perspective of meaning creation. Stress-coping perspective has been well documented since the 1960s by assuming human beings have the instinct to search for meaning in the sense of spiritual motivation for self-transcendence (Baumeister, 1991; Davidson, 1974; V. Frankl, 1981; Frankl, Victor, 1959; Rubinstein, 1989). Despite casting negativity on issues related to dementia caregiving and treating them as stressors, a more uplifting perspective of seeking positivity rises in caregiving experiences. Steger (2012) indicated that the meaning-making process in the past literature was considered relatively passive, assuming the appraisals of meaning violation or conflicts were subconsciously taken without prior detection. This empowerment sheds light on the positive aspect of extreme suffering as a deep reflection of human nature: how do we remedy the conflicts or violations in our understanding of our behavioral, cognitive, and emotional experiences in the caregiving context. Positive aspects of caregiving have drawn scholars' attention after a retrospect of "lack of attention to the positive dimension of caregiving seriously skews perceptions of the caregiving experience and limits our ability to enhance theory of caregiving adaption" (Kramer, 1997, p. 218), with empirical support and validations (Quinn & Toms, 2018; D. S. F. Yu et al., 2018). Positive aspects of caregiving, aiming to address the neglected part, lead to finding positivity and meanings in the caregiving experience (Quinn & Toms, 2018; D. S. F. Yu et al., 2018). The same question has been raised repetitively throughout humanistic inquiry "what is the meaning of life" in the fields of philosophy, psychology, history, anthropology, and sociology (Baumeister, 1991).

Finding meaning can proactively and effectively prevent or intervene the stress. The definition of "meaning" in general refers to many abstract and big concepts without a unified understanding. Common concepts pertinent to "meaning" in both theoretical and empirical studies include meaning in life (Czekierda et al., 2017; King et al., 2016), sense of coherence (M. F. Steger et al., 2006; Wong, 2013), global meaning and situational meaning (Park, 2010; Park & Folkman, 1997), subjective meaningfulness (Park & George, 2013), search for meaning (Dilworth-Anderson & Gibson, 2002), stress-related growth (Roepke et al., 2014; Triplett et al., 2012). Searching for meaning in life has long been inquired in philosophical work since ancient wisdom (Charles, 2000; M. Rosen, 1984). Frankl (1963) believed that meaning in life is a crucial ingredient in human well-being and flourishing while finding meanings and purpose in one's life sustains a sense of survival in devastating experiences (Ryff & Singer, 2003; Seligman, 2011; M. F. Steger, Kashdan, & Oishi, 2008). Based on his own experiences in the concentration camp, Frankl (1963) assumes life with absurd, painful, and dehumanized situations can still nurture meanings. According to Frankl (1985), the pathways through spiritual care and self-transcendence will enhance meaning in life and well-being, especially when no pathways to well-being are not available. Another social psychologist Baumeister (1991) enquired to "find out how people attempt to make sense of their lives, why people desire meanings in their lives, how the meanings function, what forms they take, and what happens when life loses meaning." Baumeister (1991) further stated that finding meaning in the midst of a stressful experience, such as dementia caregiving, enables the transition from adversity to "benefit finding." People then accept the unfortunate sufferings and extract positive changes from adverse experiences, such as "depersonalization, moral deformity, bitterness, and disillusionment if he survives and is liberated" (Frankl, Victor, 1959). More recent research focuses on the assimilation or accommodation processes via which individuals have an indigenous tendency to make meanings of the stressful experiences (Park, 2010). In the meaning-making process proposed by Park (2010), assimilation occurs when the appraised meanings in situational conditions change to conform with the global condition, whereas accommodation involves changes in global meanings being termed with situational beliefs or goals (Park, 2011). Assimilation is more commonly perceived in the practical experience of resolving meaning discrepancies. On the other hand, accommodation is more powerful in reorienting global meanings into better adjustment (Brandtstädter, 2009). The meaning-making process in the model indicates the information and assessments required can be found to create meanings in life (M. Steger, 2012).

2.3.2 Making Meaning of Dementia Caregiving

Dementia caregiving is the primary challenge to deal with for WDCs. Part-time or full-time care for an older adult is called "the silent productivity killer" (Pitsenberger, 2006), let alone the dementia caregiving that needs more attention and time. The foremost common strategy is stress coping and management. The stress coping models that are used to identify and reduce perceived stress have been well documented in past literature for years. Among them all, the stress process model (Pearlin et al., 1990), stress and coping model (Knight et al., 2017; Lazarus & Folkman, 1984), diathesis-stress model (Gatz et al., 1996; Schulz & Beach, 1999) have identified similar types of stressors counting caregivers' experiences in the context of stress coping: 1) individuality in characteristics and subjective perceptions, 2) objective caregiving stressors, 3) coping styles and social support, and 4) conflicts in social roles (Schaie & Willis, 2010). The stress coping process relies on a coping mechanism that is automatically triggered in a passive manner by the stresses and challenges in one's daily experience (M. F. Steger, 2012).

The application of meaning-making in the context of informal dementia caregiving has been addressed since the late 1980s (Allen, 1987; Dura et al., 1991; Hutchinson et al., 1997; Karel et al., 2007; Rubinstein, 1989; Sparks et al., 1998). Meaning-making research has been gradually found useful to the well-being of informal dementia caregivers (J. Lim et al., 2012; McGovern, 2011; McLennon, 2008; Quinn, Clare, McGuinness, et al., 2012; Quinn, Clare, & Woods, 2012; Tretteteig et al., 2017). Many Chinese caregivers regard the virtue of showing love, respect, support, and responsibilities shown to PwD as a part of the meaning-making process (Kohn, 2004). In the Chinese context, the meaning-making of dementia caregiving experiences can be influenced not only by personal experiences and values, but also the Chinese values, such as the tradition (filial piety, Xiao - 孝, Li - 禮, harmony), the culture (Confucianism, Buddhism, and Daoism), relations (Guanxi, implicit rules, and face – 面子) (Yeh et al., 2013). Making meaning of dementia caregiving in the given context allows caregivers to have an existential interpretation of caregiving experiences, reducing stress perceived, and leads to promotion of well-being for them to continue with caregiving over time to indirectly contribute to PwD's conditions (Quinn et al., 2010). Besides caregiving tasks, work and personal life are comparably important in the life of a WDC. Chinese WDCs may choose to make meanings of work and life in addition to care because they have much to care and want to make their life more meaningful through not only the caregiving experience but also other living experiences as well. Younger WDCs are more likely to make sense of their work and personal life other than solely care experience because all these aspects matter to them. The absence in any of the meanings in care, work, and life will be detrimental to their well-being (Kalliath & Brough, 2008; Rosso et al., 2010).

2.3.3 Meaning-making of Work

Working tasks are taking up as much time as care jobs for a WDC who wants to sustain the work-care balance. Working while caregiving is extremely challenging (Bernard & Phillips, 2007). Full-time working and caregiving are both demanding, time-consuming, and energy taxing. Since the last century, WDCs have been reported to experience more difficulties than other caregivers (Clarke, 2019; Department of Health, 1999). Government policies, for instance in Europe, have intervened to reconcile employment and care in order to prevent labor market participation and the pension systems from being shattered (Department of Health, 2009). The Care Act in the UK supports caregivers' basic rights for a healthy work-life balance, flexible working schedule, and protected pension to secure informal caregiving support (Bettio & Verashchagina, 2010). These types of policies have been naturally established in several European countries supporting WDCs (Bettio & Verashchagina, 2010). Other UK programs, such as the Carers Allowance Act, financially support caregivers who contribute much more than what they are paid at employment, and somehow encourage WDCs to have more autonomy in handling both care tasks and jobs at employment. Provided with all the policy support, WDCs are still facing a growing concern of struggling and wandering between the roles of a caregiver, an employee, and an individual with personal pursuit (Clarke, 2019).

Meaning-making of stressful working experiences can be beneficial to WDCs. The payback of employment can be appreciated as the means to obtain sufficient income for affordable professional services and care assistance (Moon & Dilworth-Anderson, 2015). Approximately 10% of the working population were found to be caregivers (Stuckey & Gwyther, 2003). WDCs confront the dilemma of balancing conflicting demands of working and caregiving responsibilities. WDCs who spend a great amount of time caring often report being late for work, leaving early, taking time off, taking a leave of absence, or changing to part-time work (United Nations et al., 2019). Their well-being is consequently compromised due to conflicts between working and caring duties (Hoff et al., 2014).

Making one's work meaningful serves a high purpose of not only mitigating the stress and anxiety from the working tasks, but also promote their work satisfaction and their discretionary and unpaid working hours to greatly benefit the output of the companies (Dik et al., 2009; Wrzesniewski et al., 1997). In the long run from the employer's perspective, making meanings of work allows more faith in employees in management, better work ethics, more efficient team functioning (Wrzesniewski et al., 1997), greater vocational self-clarity and choice comfort (Wrzesniewski et al., 1997), and perceptions of existential meanings in a holistic life (Dik et al., 2008). From the employee's view, meaningful work experience benefits WDCs in many ways such as improvement of work satisfaction, work commitment, performance, motivation, work values and engagement (Mercure & Vultur, 2010). From the dementia caregiver's perspective, making sense of work helps reduce working stresses and enables them to focus more on their work and caregiving tasks (Jolanki, 2015; Sakka et al., 2019). For WDCs in China, work is an important part of daily social life that also influences WDCs' financial and psychological aspects. Chinese WDCs with meaningful purposes generally have stronger motivational values in cultivating working relationships, a sense of career development, practical needs for improving both employment and financial conditions, autonomy and goal-setting in their employment by actively making more meanings out of their working experiences (Yet-Mee et al., 2008). These Chinese WDCs appear to be individualistic, materialistic, hedonic, and entrepreneurially oriented in their working settings (W. C. H. Chan, 2014; S. Rosen, 1990). Lacking meaning in employment is destructive to people's self-respect, and leads to low satisfaction, motivation, and work ethics (Deranty & MacMillan, 2012).

2.3.4 Meaning-making of (Personal) Life

Making meanings of work and care does not summarize the full story of a meaningful life. Making meaning of personal living or life, on the other hand, matters to everyone. Personal life is not useless and cannot be ignored by everyone including WDCs. Making meanings of personal life is necessary for the sense of balancing care, work, and life. Despite their major caregiving tasks and jobs that bring challenges and burdens to WDCs, their limited personal activities and development can as well produce burdens. Therefore, personal life is as well indispensable for WDCs. A meaningful life, other than work and care, broadly encompasses personal interests, family, social or leisure activities. For instance, work-life balance for the working class has been addressed for decades. Work-life balance is defined as "the balance of time allocated for work and other aspects of life for a working individual" (Kalliath & Brough, 2008). The UK work-life legislation was developed to "promote and establish a healthy work-life balance and to remain alert to societal changes, such as the aging population" (Trade Union Congress, 2010). The National Dementia Strategy also advocated raising awareness, facilitating assessment, improving services, and strengthening the quality of support (Department of Health, 2009) in order to preserve one's personal life at best (Schaufeli & Bakker, 2004). The idea of work-life balance originated from the growing demands for a contemporary multi-dimensional lifestyle to equally contribute to personal development, professional career, and family life in UK and US during the 1970s to 1980s (Pitsenberger, 2006), and still has been a concerning social and academic topic in the 21st century where lies a fast and busy living pace (Cinamon & Rich, 2002). Literature has pointed out the importance of work-life balance in people's daily life (Mclaughlin & Muldoon, 2016; Mishra et al., 2014), and more recent literature has identified the demands for work-life balance among dementia caregivers (Sakka et al., 2019).

In order to make meaning of life to overcome the burden and consequent negative outcomes generated from stressors, an individual will try to make appropriate meaning-making choices to benefit their caregiving, working, and living experiences by reasonably allocating available resources and energy. Meaning-making of individual stressful experiences has been documented in past studies. Meaningful care provided for dementia patients in the family setting results in better adaptation and promoted emotional experiences (Tretteteig et al., 2017). Meaningful work has demonstrated a better resilience under great work pressure in achieving desirable work performance. In addition, making sense of work enables one to establish a sense of work-life balance (Kalliath & Brough, 2008; M. F. Steger et al., 2012a). Lastly, useful meanings attached to the personal life events services serve as a buffer factor confronting stressors of daily life challenges (M. Steger, 2012). People make different meanings of daily activities depending on their understanding, cognition, personality, etc. In addition, working dementia caregivers may have different preferences in making sense of a combination of daily living stressors. Therefore, these working dementia caregivers need to make choices in, for instance, prioritizing which type of events should be made sense of, or making sense of one or two types of activities simultaneously while ignoring the other aspects, or enduring stressors rather than making any meaning out of them, or making sense of all daily living activities. However, the concept of "meaning-making choice" has not yet been discussed, given dementia caregivers are facing all aspects of stressors from care, employment, personal development, social relationships, martial or parent-child relationships, financial challenges, and meaninglessness (Gilhooly et al., 2016; Sun, 2014b). In addition, the meaning-making process in the existing literature so far has still been generally conceptualized yet not tailored to the setting of dementia caregiving, the basic common sense that supports a holistic life experience and urges meaningful life besides finding meanings in work and care among WDCs (Jolanki, 2015; Sakka et al., 2019; M. F. Steger, 2018)

2.3.5 Meaning-making Choice of Care, Work and Life

The awareness of making choices and coordinating meaning-making of care, work, and personal life is necessary, as each of these aspects is vitally important in a holistic human experience. Making meanings of all experiences during one's life seems almost impossible, while ignoring some critical life experiences can be detrimental to one's well-being. For instance, even though one makes meanings of care can be self-soothing in the care experience (Tretteteig et al., 2017), the burden from work or other personal business can still be perceived (Clarke, 2019). Solely making meanings of care can be less useful than the choice of proactively making meanings of work and personal life as well. The choices in making meaningful care, work and personal life simultaneously can be strategic as a necessary decision-making process that can be potentially beneficial to WDC's well-being.

The meaning-making choice can be characterized as the decisional choice of making independent meanings of caregiving experience, working experience, and living experience (Fowler & Roberts, 2019). The choices of meaning-making vary across different daily scenarios of activities and attached meanings. By having diverse independent choices of meaning-making, some WDCs find it burdensome or desperate in their work and not being able to achieve any of their personal purpose in life; whereas someone seeks meanings of the challenges from their care and work with adjusted expectations and life purpose (Boyle et al., 2010). WDCs sometimes find meanings based on their own experiences or certain activities, such as finding a sense of achievement and rewarding from caring PwD; getting promoted at work; obtaining personal achievement or reaching one's life purpose via daily activities (Clarke, 2019; Park, 2020). Meanings can as well be obtained when dementia care is considered part of life purpose; or care and work lead to the same benefits; a workaholic WDC takes work more meaningful than care and personal life (Burns et al., 2005). In this sense, meaning-making needs to incorporate the combination of care, work and life among dementia caregivers (Fouché & Martindale, 2011). While work is compulsory for WDCs, making sense of working experiences determines their well-being during work and indirectly influences how they pursue their care and life of quality (Sakka et al., 2019). Therefore it is reasonable to extend to the notion of care-work-life balance, in which all these independent activities are intertwined in daily living scenarios for meaning-making choices. Moreover, meaning-making choices are useful in finding a care-work-life balance that further reduces perceived stress (Bernstein et al., 2019; Fujihara et al., 2019) and enables multi-dimensional living experiences with an emphasis throughout the life trajectory (Fouché & Martindale, 2011). Even though meanings and well-being have been both theoretically and empirically associated in literature since the 1960s (Diener, 1984; V. E. Frankl, 1963; Ryff, 1989), the applications of meaning-making choices and well-being have been poorly studied in dementia care research (Byrne & MacKinlay, 2012; Hanssen & Kuven, 2016; Jones et al., 2019; Nay et al., 2015; Quinn, Clare, & Woods, 2012). The combinations of making meanings from different sources of care, work, and life, lead to different levels of well-being. Thus, the impact of meaning-making choices on the well-being of dementia caregivers should be investigated.

2.3.6 Dyadic Relationship

The dyadic relationship between a WDC and a PwD is closely related to one's perceptions and attitude toward care, work and personal life. Caregivers with a high relationship quality report better perceptions of caregiving stress and burden, caregiving experiences, and satisfactory well-being (Rippon et al., 2020). Low family-work conflicts with intimate and rewarding dyadic relationships are beneficial to one's work-life balance and employee performance that further reduce occupational stress and improve well-being (Karakaş & Tezcan, 2019; Soomro et al., 2018). A desirable family relationship has the advantage in promoting one's attitude toward work-life balance, daily personal living and well-being (Sirgy & Lee, 2018). Therefore, the relationship between the WDC and PwD can moderate the association between stress generated from care-work-life conflicts and meaning-making for improved well-being. Surprisingly, the dyadic relationship has not yet been emphasized in one's meaning-making process concerning conflicts between care, work and personal life. Past meaning meaning-making literature has not taken the dyadic relationship seriously because the research participant has been focusing on a single agent while relationships with others have not yet been the concern (Park, 2010).The dyadic relationship becomes even more important nowadays in studies of positive psychology addressing both meaning-making and well-being of caregivers (L. L. Davis et al., 2011; Nagpal et al., 2015; Norton et al., 2009). In the context of dementia caregiving, caregivers' attitude and understanding of the caregiving experience, their knowledge about dementia symptoms and their PwD, and viewpoints on personality changes of PwD are known to be highly correlated to or as an integral part of their relationships with the PwD (Hinton et al., 1999). Current studies emphasize the importance of dyadic relationships interacting with the caregiver's meaning-making experiences, and well-being (S. Smith & Converse, 2012).

Chapter 3 Research Gap

A number of research gaps are identified based on the critical review of the literature.

The phenomenon of meaning-making choices practically demonstrates a multi-dimensional life of WDCs to be made sense of. Meaning-making choice of care, work, and life has not yet been discussed in the entire population of WDCs, including Chinese WDCs (Bernstein et al., 2019; W. C. H. Chan et al., 2017; Fujihara et al., 2019). The meaning-making of various aspects in different contexts affecting one's well-being has been controversial and without a conclusive consensus. Heterogeneous meaning-making choices that link to well-being status in WDCs' daily life can be further investigated (Park, 2010; Quinn et al., 2010; Quinn & Toms, 2018). For example, some people experienced a higher level of hedonic and social well-being from self-transcendent and altruistically meaningful sources than self-centered or materialistic sources. While some others reported a moderate or lower level of well-being with confounding variables even when their meaning-making level is sufficiently high (Schnell, 2009). Empirical evidence has indicated the dyadic relationship can be a critical variable moderating the associations between meaning-making and wellness (Braun et al., 2009; D. W. Davis et al., 2015; Tretteteig et al., 2017). The review on positive aspects of the caregiving experience implies that dementia caregivers can have a significant improvement in their perceived self-growth and psychological well-being (Cheng et al., 2016; D. S. F. Yu et al., 2018).

Moreover, the investigation of choices of meaning-making of care, work, and life has been incompletely carried out in the literature. First, even though a person can be expected to make meanings of any stressful experience in an existential way, it is still questionable that people can make meanings of every stressful experience simultaneously. In addition, even if one can make meanings of care, work, and personal experiences at the same time, those attempted meanings do not necessarily lead to the ultimately made meanings (Gillies & Neimeyer, 2006; Park & Folkman, 1997; Segerstrom et al., 2003). Third, caregivers with different choices in making meanings of care, work, and/or personal life are found to present unequal levels of well-being outcomes (Quinn & Toms, 2018). Last, well-being is generally unstable and volatile during extreme circumstances, such as dementia care tasks, high working pressure, and an urge to develop personal interests within a limited timeframe (Csikszentmihalyi & Larson, 2014). Traditional between-subject analyses examine differences in well-being between groups with types of characteristics, however the analyses do not accurately record daily fluctuations and changes in well-being. In comparison, within-subject analyses can precisely depict variations of well-being within each adult-child WDC during their caregiving experiences.

Furthermore, the dyadic relationship influences the relationship between the meaning-making choice and well-being. The PERMA model proposed by Seligman (2011; 2012) mentioned not only well-being and meanings in the meaning-making framework, but also the role of the dyadic relationships as a critical interaction term within the framework. The relationships, along with types of activities, are the most common factors mediating the relationship between meanings and well-being according to Steger (2013). The moderation effect of dyadic relationships influencing the impact of meaning-making on the well-being of WDCs can be tested with empirical evidence.

Last but not least, to accurately depict WDCs' well-being in consideration of their various daily experiences, the Experience Sampling Method (ESM) as a relatively innovative methodological approach could be implemented. ESM has a few advantages in studying this topic: 1) ESM fits better with the nature of well-being. Well-being, including hedonic and social well-being, is a momentary indicator being easily influenced by meanings made and other factors such as temporary caregiving tasks, work or living arrangements. Meaning-making choices may change over time leading to changes in well-being. 2) ESM can avoid the non-random delivery of questions in most surveys that are scheduled to be addressed by recording momentary feelings and thoughts during the randomly assigned time (Judge et al., 2010). 3) Data collected frequently over the period of time about the momentary lives of participants can better reflect a reliable state of their experiences, assuming people are living in dynamic and fluctuating experiences, in comparison to the longitudinal or cross-sectional studies, which assume people behave and think stably with the data from the retrospective reports that may cause recall bias (Alzheimer's Association, 2018); 4) ESM can more efficiently assess within-individual variance as the trajectory of well-being experience of each participant can be different (Judge et al., 2010).

Chapter 4 Theoretical Framework

This chapter covers the development of theories on meaning-making and well-being, theoretical framework, and factors in meaning-making choices in addressing the aforementioned research gaps. A proposed theoretical framework (see **Figure 1**) is thus called for to bridge one's meaning-making choice and well-being in consideration of dyadic relationships.

4.1 Meaning-making and Well-being

Meanings are indispensable in sustaining one's well-being because meaning-making is the human's innate motivation in an extremely challenging environment according to Frankl (1963).

4.1.1 Evidence about Meaning-making

The meaning-making is subjective and complicated in the identification of meaningful sources from care, work and life. The process of meaning-making demands meaning affirmation and violations (Heine et al., 2006). Meaning affirmation in working, caregiving, and living scenarios motivates people to make or regain their sense of meaning while meaning violation creates an opportunity of reviewing an unfamiliar meaning for further affirmation to strengthen one's perspective of meaning (C. M. Steele, 1988). Next, the meaning maintenance model suggests a need for meaning through projecting a mental representation of the expected perception of things or the world (Heine et al., 2006). One fact that can be inferred from the meaning maintenance model is that people are most likely to seek meaning when they consistently encounter meaning disruptions, or struggle to improve the adverse meaning arousals that appear to accompany such disruptions (M. F. Steger, 2012). Therefore, the actions seeking meanings and the perception of mental distress can be positively correlated (Park, 2010; M. F. Steger, 2012; M. F. Steger, Kashdan, Sullivan, et al., 2008). Meanwhile, existing research implies that the perception of well-being can be lacking among people with high meaningfulness in life (Cohen & Cairns, 2012; M. F. Steger, Kashdan, & Oishi, 2008), or people from Eastern cultures (M. F. Steger, Kawabata, Shimai, et al., 2008).

4.1.2 Evidence about Meaning-making and Well-being

Positive perception of well-being was accompanied by meanings attached to one's experiences according to theories. For instance, motivational or need theories assume people have better well-being if they tend to seek meanings, or perceive worse well-being if they do not make meanings of stressful experiences (Heckhausen et al., 2010). According to the theory of human need, people have basic and universal preconditions for meaning-making in forms of life. Yet, the incapacity in making meanings leads to poor well-being (Doyal & Gough, 1991; Gough & Way, 2004). Most three common scenarios are in the following: 1) people's innate need to make sense of things to maintain the level of well-being, 2) natural instinct to make meanings to meet basic needs if with poor well-being, and 3) a need to make meanings while generating well-being (Heckhausen et al., 2010). People who successfully meet their needs are in their natural states of maintaining a certain level of well-being, whereas people who feel upset and disappointed for not meeting their needs or are in the process of striving to meet the needs. Meaning-making from this perspective meets the internal needs to transcend the self and the life, and thus further influences one's well-being with met needs and made meanings (Heckhausen et al., 2010). The choices in making meanings of care, work, and/or life can be prevalently made among WDCs. Empirical studies have reported the meaning-making of care and work, or care and life (W. C. H. Chan, 2014; Güngör & Uçman, 2020; Kalliath & Brough, 2008; Krok et al., 2021). However, current meaning-making literature has not covered more than two aspects.

Hedonic and social well-being are often evaluated when people make meanings of various stressful experiences (Diener et al., 1999; Gallagher et al., 2009; M. F. Steger, Shin, et al., 2013). For instance, meaning in life is found to be correlated with hedonic or subjective well-being due to individuals' engagement in eudaimonic reasoning and action in relation to feeling about daily mental health and social functioning (Bauer et al., 2019; Heintzelman, 2018). Caregivers' psychological well-being is affected by meaning made on positive aspects of caregiving experiences (Quinn et al., 2019). Meaningful work has been found to improve one's psychological, subjective, and social well-being (Chancellor et al., 2015; Janicke-Bowles et al., 2019; Kaplan et al., 2014; Keyes, 1998; Rieger et al., 2014). However, three major issues remain unresolved. First, the literature leads to a vague conclusion regarding the impact of life meanings and care meanings on one's well-being. There is little information on what types of subjective well-being being influenced by meaningful experiences. Second, making meanings of the multi-dimensional experiences involving care, work, and/or life has been poorly investigated. While both meaning-making and well-being types were separately identified, their associations have been weakly examined in empirical evidence among the general population as well as WDCs. Moreover, WDCs in China generally appreciate autonomous development, the quality of work and life, and meaningful experiences in their work and life as an achievement (Cheung et al., 2018). Last, familial and cultural elements, such as the dyadic relationship, can potentially influence the impact of meaning-making on well-being (Joshanloo et al., 2018; Keyes, 2013). In Chinese culture, WDCs are expected to fulfill their caregiving obligations following the traditional culture of filial piety, while undermining individual needs for work and personal life. On the other hand, work ethics is greatly valued so that compromised work performance, even due to the overwhelming care burden, is not socially expected and leads to declining social well-being. An appropriate dyadic relationship can strengthen one's meanings made in their work as well (C. Lim & Putnam, 2010). Overall, a strong sense of filial piety and family-oriented care results in little trust in formal assistance from the communities and professionals, as well as worsened social withdrawal and isolation. It is arguably that care-work conflicts or care-life conflicts may constantly exist due to the cultural norm and expectations. For example, a filial caregiver is expected to sacrifice their working hours and personal matters for caring for the older person in the family, whereas employment requires a working caregiver to adequately fulfill the jobs (Kim-Prieto & Miller, 2018; Lun & Bond, 2013). A caregiver who gives up all his or her personal life for wholehearted care. However, lacking personal development may generate stress. Therefore, a good balance between care, work and personal life is closely associated with desirable well-being. The maximal elimination of care-work conflicts and care-life conflicts greatly benefit well-being.

4.2 Proposed Framework

A theoretical framework is proposed to illustrate the choices in meaning-making of various experiences and types of well-being. The proposed framework is supported by the theoretical background to illustrate each major component, empirical evidence connecting those components, and a potential moderator that impacts these connections.

4.2.1 *Meaning-making*

Contemporary knowledge of meanings and meaning-making shifted from collective ideation of philosophical inquiry of human meanings to existential experiences within individuals in the absence of absolute principles of meaningful experiences. Frankl (1963) suggested that meaning-seeking is human's instinctive action in the extremely challenging environment. The assumption for inherently making meanings is that humans being spiritual creatures with purpose and goals have an innate drive for living meaningful lives based on their individual values (V. E. Frankl, 1963; M. F. Steger, 2018). Battista and Almond (1973) believed a well-described mental representation is needed to depict our rationale of value systems, purpose setting, and meaning-making. With a similar view, Klinger (1977) took a step forward believing that living creatures, such as human beings, all have a natural tendency to set goals to acquire resources for a living. The materialized resources cannot meet the spiritual pursuit of human beings' natural inclination and later develop into more abstract concepts such as purpose and goals. Baumeister (1991) suggested meaning occurs only when a specific set of aspects are realized: having a goal or purpose, self-worth, feeling good about self, values that can justify decisions and behaviors, and efficacy and self-competence.

The Stress Processing Model (SPM) continued to contribute to the meaning-making theories. The SPM in the context of caregiving addresses the supportive resources and support to buffer the negative interactions with the stressors during caregiving experiences (Whitlatch et al., 2001). According to Pearlin (1990), SPM involves several key components: 1) background and context, namely "age, gender, race, education, socioeconomic status, health history, family and friend network, and living arrangement"; 2) primary stressors such as objective stressors (namely "cognitive status, functional status, and behavior problems") and subjective stressors (namely "role captivity, perceived distress, perceived dependency"); 3) secondary strains, including role strains (namely "family role strain, job or work role, social or recreational role, and dyad strain") and intrapsychic strains (namely "self-esteem, mastery and self-efficacy"); 4) mediators, including internal mediators (namely "personality, hardiness, life orientations, spirituality, and core values") and external mediators (namely "social support, knowledge of the illness, and financial resources"); 5) outcomes, including quality of life, depression, anxiety, and physiological reactions. Park (2010), based on Pearlin (1990) and Folkman (1997)'s stress process models, stated in the review that the meaning system has an appraisal mechanism to reduce the discrepancy between the global and situational meanings. According to Park (1997), the meaning-making process is predicated on the idea that individuals experience stress, such as dementia caregiving and burden, when they report discrepancies or inconsistencies between the perceived global meaning (i.e., what they believe and desire) and the appraised meaning regarding a specific condition. The distress yielded from the meaning discrepancies propels one to undertake the source of stress and dissolve the consequent negative emotions (Park, 2010). In the context of making meaning of care, work, and life, there is a need to expand the meaning-making tool to interpret the experiences rather than tackle stress because not all experiences are necessarily stressful, such as a mundane and boring life, or numbness in any of the experiences. Meaning-making continues to improve one's well-being when the stress no longer needs to be reduced (Park, 2010).

Based on Leontiev (2013)'s work, Noguchi (2019) developed the Meaning Frame Theory (MFT) and further argued that the multi-dimensions on frames of inherent life purpose, motivational meaning, and relational meaning lack one more dimension of the frame, which is the sources of meaning in the course of human's experience. The MFT emphasizes that multiple stressors simultaneously exist and have a comprehensive impact on one's well-being (Noguchi, 2019). Individuals living in a complicated set of contexts have a unique multi-dimensional structure and trajectory of life as well as their meaning systems. The meaning-making system covering care, work, and life can evolve over time during the course of one's life, developed with appropriate and diverse meaning-making strategies. MFT with the reference point focuses on the multi-framed meaning and accounts for the potential meanings to be made in one's meaning system, while assuming the meanings are made from diverse aspects in a dynamically developing pattern (Noguchi, 2019).

Self-Determination Theory (SDT), referring to each person's ability to make meaning and improve one's well-being, assumes humans have the motivation and purpose of pursuing self-growth with self-determination for a better-perceived outcome of wellness (Deci & Ryan, 2012; R. M. Ryan et al., 2008). According to SDT, the meaning-making of experiences to strengthen wellness can be achieved through autonomy and competence. Autonomy, rather than constraints, motivates people to make their own choices and decisions in the surrounded environment, and sometimes even generates a sense of growth against external influences. Competence gives people positive feedback on their behaviors, thoughts, or things and consequently increases their intrinsic motivation and sense of fulfillment. Relatedness between people, such as attachment, and affect master-related outcomes (Ng et al., 2012). However, SDT fails to address the inherent mechanism of meaning-making choices. The meaning-making choice among care, work and life being a part of the psychology of choice refers to the individual ability to make decisions when presented with two or more options according to the Theory of Meaning (TM) (Tversky & Kahneman, 1981). TM is useful not merely in the semantic knowledge of meaning where it was originated but also associated with propositional attitudes to judge the validity of thoughts and beliefs from the personal perspective (P. Yu, 1979). TM helps us understand how ones' beliefs, desires, intentions and meanings are derived from certain incidents. For example, in the caregiving context, one needs to work for earnings and pursue one's personal life goals, the meaning or beliefs attached to certain incidents of those experiences can be evaluated respectively – whether an incident is meaningful or not. However, the theory of meaning alone cannot reflect intrapsychic choices between various meanings. Being theoretically implied, the sources of meanings are the critical reference points to make meanings by endowing values, significance, purpose, and comprehensive utility to meanings of care, work, and life (Baumeister, 1991; Martela et al., 2018). Burden, stress, and relationship issues within care, work, and life are attached with meanings from a self-transcendent and altruistic perspective (Schnell, 2009; M. F. Steger, Shin, et al., 2013). It has been pointed out that meaning can be made as an indicator appropriately in the contribution to a broad range of well-being, such as hedonic and social well-being (Gallagher et al., 2009; M. F. Steger, Shin, et al., 2013). Therefore, meaning-making choices occur when multiple sources of meanings are made. For example, a working caregiver needs to take care of the PwD and secure a full-time job with a sense of meaningfulness. In this case, the caregiver may choose what and how to make meanings out of such an experience.

4.2.2 Meaning-making Choice

Echoing the proposed concept of "meaning-making choices," decisional theories together offer a structural and conceptualized framework illustrating the possible mechanism undergone when individuals make meaning-making choices. Davidson (1980) further incorporated both TM and the Decision Theory (TD) to identify cognitive decisions to seek an optimal choice from an individualistic perspective (MacCrimmon, 1968; Slovic et al., 1977). TD is conceptually related to the field of game theory in its normative branch of the concept and refers to “the process of analyzing the outcome of decisions or determining the optimal decisions given constraints and assumptions” (MacCrimmon, 1968; Slovic et al., 1977; K. Steele & Stefánsson, 2015). TD assumes that normative decisions are processed with rational accuracy and optimality. The practical application of this prescriptive approach (“how people ought to make decisions”) is called “decision analysis and is aimed at finding tools, methodologies, and software (decision support systems) to help people make better decisions” (K. Steele & Stefánsson, 2015). One or more choices usually confront many uncertainties yet will be chosen at the highest total expected value from the utilitarian perspective (Malpas, 1992). Given that meaning-making is not completely a rational decision-making process, there are still some reasoning elements in this mechanism. The fundamental purpose of meaning-making is to alleviate the uncomfortable experiences generated from the stressors. People may subjectively develop different meaning-making strategies, yet their meaning-making approaches all serve to reduce the negative affectivities and health outcomes suffering from challenging experiences. The integration of TM and TD identifies the cognitive and psychological process of making meanings rather than behavioral reactions associated with the choices. However, this “unified theory” has a weak explanatory power in associating the choices of meaning-making with one’s well-being status (Davidson, 1980).

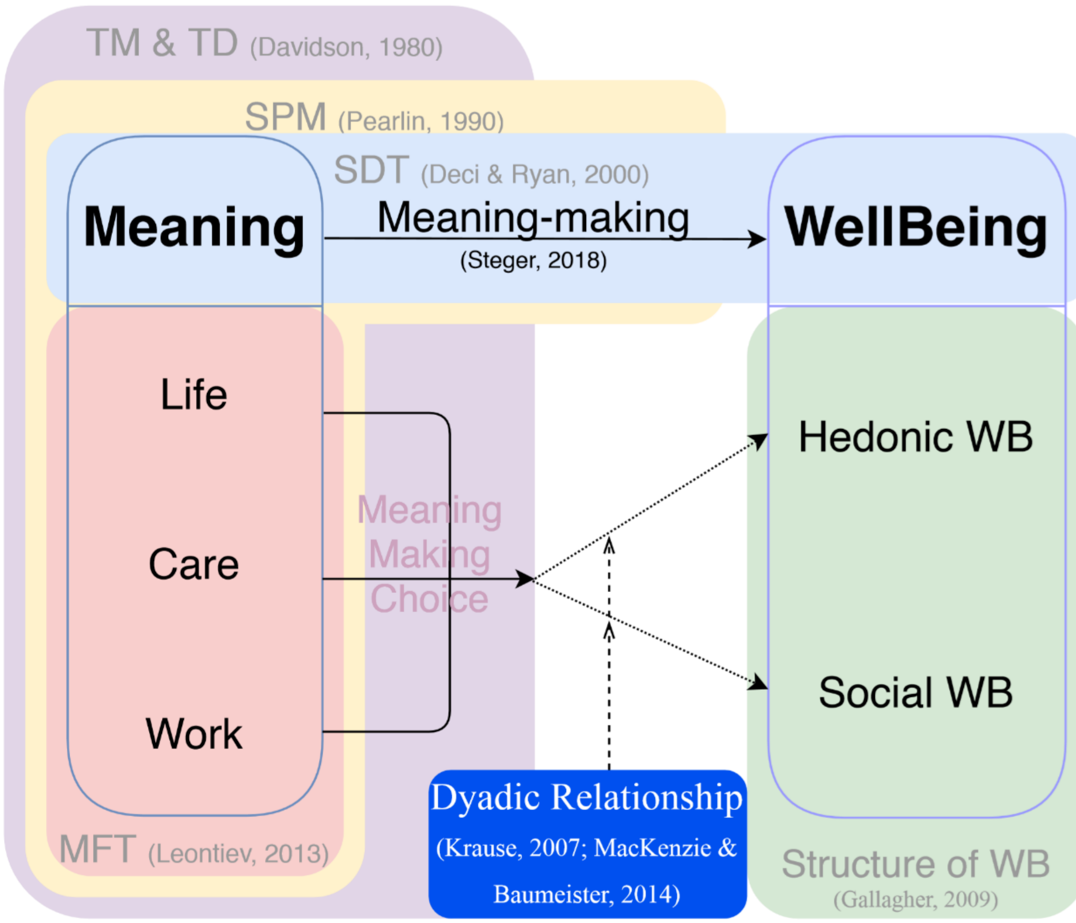

Figure 1. Proposed Theoretical Framework of MMC for WDCs

To bridge the association gap between meaning-making choices and health outcomes, the Glasser (1999) suggested that the Choice Theory (CT), on top of TM and TD, advocates the "total behavior" of making a choice includes four components: acting, thinking, feeling, and physiology. The first two components, acting and thinking, are initiated and regulated by individual efforts; while the latter two components, feelings and physiology, are the resultant of how people choose to act and think (Glasser, 1999). Thus, meaning-making can be conceptualized as a subjective process of making a choice over various aspects of meaning-making levels. In practice, to make sense of various adverse experiences, working dementia caregivers have their own choices and priorities depending on the stressors that are confronted and preferred aspects for meaning-making. Therefore, patterns of choices of making sense of care, work, and personal life can be observed among working dementia caregivers. Certain desirable patterns of meaning-making choices produce a high level of positive affectivities and health outcomes. In comparison, undesirable patterns of meaning-making choices can lead to a less satisfactory wellness level.

The typology of meaning-making choices aims to classify all possible choices based on observed data. Applying observed mean to differentiate high/low levels of each meaning-making aspect, there are a total of eight possible categories of meaning-making choices: 1) high meaning-making levels on care, work, and life, 2) high on care and work, 3) high on care and life, 4) high on work and life, 5) high on care only, 6) high on work only, 7) high on life only, 8) low meaning on all aspects. Working dementia caregivers make different meaning-making choices based on their own characteristics, cognition, past understanding and experiences, and with to reach balances between aspects of daily living activities, such as care-work balance, care-life balance, and work-life balance. The rationale of categorization is based on the criteria of the mean score of each meaning-making measurement in different aspects, whereby a high level of meaning-making indicates meaningfulness while a lower level indicates non-meaningfulness. For example, we have caregivers who make meanings of all aspects in care, work and life (choice 1), or caregivers making meanings of care and work (choice 2), or care and life (choice 3). On the other side, we have caregivers who don't make any meaning (choice 8), or only make meanings of work (choice 6), life (choice 7), or care (choice 5). While there is a unique category, in which caregivers make meaning of work and life (i.e., not making meaning of care; choice 4), while not making meanings of care. Therefore, we have three main types of caregivers given their choices on meaning-making preferences. According to the idea of seeking a balance between care, work, and life, caregivers who can make meanings of various aspects are considered to have a better level of well-being. In this sense, the first type of caregivers with possibly better well-being are the ones who make meaning-making choice categories of 1, 2 and 3. The second type is the ones with worse well-being making choice categories 5, 6, 7, and 8. The last unique type includes the ones making the choice category 4. Such typology associated with well-being levels can be interpreted with the hedonic terms "flourishing" and "languishing" (Keyes, 1998). In this way, caregivers of the first type are considered flourishing caregivers (positive type), the second type are languishing caregivers (negative type), and the neutral type is called neutral caregivers (mixed type).

Care Meaning	Work Meaning	Life Meaning	
		High	Low
High	High	1 Well-balanced	2 Work to care
	Low	3 Live well to care	5 Filial
Low	High	4 Self-centered	6 Workaholic
	Low	7 Self-carer	8 Hopeless
Classes	Flourishing caregivers: 1, 2, 3		
	Neutral caregivers: 4		
	Languishing caregivers: 5, 6, 7, 8		

Figure 2. Classes of meaning-making choices

4.2.3 Meaning-making Choices and Well-being

To supplement the theoretical application of both TM and TD in well-being research, the Glasser (1999) suggested the Choice Theory (CT) advocating the "total behavior" of making a choice includes four components: acting, thinking, feeling, and physiology. Among these components, acting and thinking can be directly handled and managed by individuals; while feelings and physiology can be influenced as the result of how people choose to act and think (Glasser, 1999). Meanwhile, having appropriate types of acting or thinking in choices leads to a sound state of well-being and emotional feelings according to the theory. Therefore, a better state of well-being and the dyadic relationship between the agent and the associated person will be found within the congruent choice of "total behavior". On the other hand, undesirable relationships lead to compromised emotions, whereas a satisfactory relationship serves as a precondition for lasting happiness in life. In order to reach an expected state of well-being, according to the utilitarian view, there should be certain types of choice combinations that can be prevalently discovered.

Meaning in personal life, the foremost part among the meaning in general, according to Noguchi (2019), is associated not only with hedonic well-being (emotional affective from support) (Hicks & King, 2009; Krause, 2007), but also with social well-being. Firstly, meaningful life is relevant to hedonic well-being, such as positive affectivity and emotional experiences, subjective well-being, psychological affective (Chamberlain & Zika, 1988), joy, vitality (M. F. Steger et al., 2006), happiness (Debats et al., 1993), positive emotions (King et al., 2006; Pinquart, 2002), and life satisfaction (Ryff, 1989) in both dimensions of hedonic well-being (M. F. Steger et al., 2006). Secondly, meaning in life is relevant to the perception of belonging in the family and the community (Lambert et al., 2013), personal control of life, locus of control (Ryff, 1989), and positive perception of self and world (Sharpe & Viney, 1973), and sense of involvement (Aron et al., 2004).

Meaning in care comes the second after meaning in life. Making meanings out of caregiving among informal dementia caregivers has been studied since decades ago as most dementia patients at the early and middle stages are taken care at home, usually by their family members (Knapp et al., 2007; Quinn et al., 2010). Therefore, caregiving of demented family members, such as parents living with dementia, has been regarded as an extremely stressful process. While the stress-coping model proposed by Pearlin (1990) has been criticized for its emphasis on the negative side of the experiences (Kramer, 1997), the positive aspect of caregiving in recent literature has been identified, such as uplifts and gratification in terms of hedonic well-being. Making meanings of the disease and caregiving experiences have been empirically supported in association with positive and negative cognitive and emotional perceptions (Noonan et al., 1996; Rubinstein, 1989; Zika & Chamberlain, 1992). Existential interpretation of the caregiver's identities, roles, and responsibilities in relation to wellness outcome (Farran et al., 1991) indicates its impact on well-being as a form of coping with mitigating misfortunes and stresses (Pearlin et al., 1990). Meaning in care also needs to deal with the social aspects of the caregiving, such as other people's opinions on dementia care (Ar & Karanci, 2019), familial traditions on the caregiving responsibilities (Boylstein & Hayes, 2012), community support allowing caregivers provide effective care (Globerman, 2006), and so on. Social well-being will suffer if these social aspects are unavailable to make no meaning at all (Davies & Nolan, 2006).

Meaning in work has as well been one of the important aspects in the WDCs' regular life, given the fact that dementia caregivers are facing challenging demands and many of them may give up on their full-time employment or transition to part-time work. According to the most recent survey, there are still 43% of dementia caregivers struggle to maintain their working patterns due to caregiving. Otherwise, 18% changed to part-time, 16% took a leave of absence, and 8% turned down a promotion (Alzheimer's Association, 2019). Overall, the majority (83.3%) of dementia caregivers appeared to be working while providing caregiving to people living with dementia. Meaningful work, according to Steger (2012), has at least two primary impacts on well-being: 1) Hedonic well-being: greater good motivations that pose a positive impact on the employees and others (Rosso et al., 2010). 2) Social well-being (positive relationships at work and with others, emotional support, and communication) (Noguchi, 2019).

Both work and care cast burdens on WDCs who may need further assistance in alleviating their stressful experiences (Fujihara et al., 2019; Jolanki, 2015; Sheehan & Nuttall, 1988). To give attention to the needs of this particular population, both work and care should be addressed in both service support and psychological motivation and persuasion. The notion of work-life balance emphasized the necessity to pay attention to desirable working experiences and balanced personal life. The approaches to improve both working and living experiences with better experiences involve not only time management and skills, but also intrapsychic motivation and the attitude in interpreting the experiences (Fouché & Martindale, 2011; Kalliath & Brough, 2008). On the other side, caregivers who value personal development and pursuits are confronted with the challenges in taking good care of both aspects simultaneously (Cassolato et al., 2010; Jolanki, 2015). Overall, we can find the attempt to make sense of multiple aspects of care, work, and life is something people can choose rather than negative experiences one passively adapts to. Making meanings of those experiences depends on people's choices and how well the person expects to feel.

4.2.4 Dyadic Relationship as a Moderator

The existing theoretical background does not address meanings being made on familial and social levels, such as a dyadic relationship between WDCs and PwD, familial tradition, or societal culture. Dementia caregivers in order to make meanings cannot only care about themselves, but also the care recipients, family members, work colleagues, and other people in their personal life. Thus individuals are facing much more complicated life scenarios than solely making meaning of care tasks.

The role of the dyadic relationship in the meaning and well-being could be addressed as a major component between meaning and well-being (Kirby et al., 2018; Park, 2005, 2014; M. F. Steger, 2018; Williams et al., 2014). The dyadic relationship between the WDC and the PwD shows the closeness and strength of the relationship. However, the importance of relationships in meaning and well-being has been thus far absent in the majority of literature on meaningful care-work-life and well-being. The relevant literature on WDCs is still lacking.

In the empirical literature, the relationship has been found as an important role in the meaning-making of dementia caregiving even though the pre-existing and ongoing relationship between the WDC and PwD has been neglected in many theoretical models (Montgomery & Williams, 2001). The stress process model considered the relationship as a background factor with little impact on the experience of caregiving (Lewis & Meredith, 1988). However, it has been found the quality of the pre-caregiving and current relationship determines how caregivers interpret and make meanings of the caregiving experience (Quinn et al., 2009). The dyadic relationship in particular is associated with well-being when people make meanings of challenging experiences in their lives (Glasser, 1999). In addition, on all levels of perceived caregiving experience, the dyadic relationship is associated with one's intrinsic motivation and psychological health and well-being (Tretteteig et al., 2017).

4.3 Research Questions

Based on the proposed theoretical framework, four major research questions are raised concerning well-being of working dementia caregivers and its influential factors – meaning-making choices and dyadic relationship.

<u>Research Question 1: is meaning-making choice associated with well-being?</u>

- H1.1: participants with the meaning-making choice 1 to 3 are flourishing caregivers with higher positive affectivities and lower negative affectivities.
- H1.2: participants with the meaning-making choice 4 are neutral caregivers with higher positive affectivities, and moderate negative affectivities.
- H1.3: participants with the meaning-making choice 5 to 8 are languishing caregivers with lower positive affectivities and higher negative affectivities.

Research Question 2: are the meaning-making choice associated with different hedonic and social well-being?

- H2.1: participants with the meaning-making choice 1 to 4 have higher positive affectivities than the meaning-making choice 5 to 8.
- H2.2: participants with the meaning-making choice 1 to 3 have lower negative affectivities than the meaning-making choice 4 to 8.
- H2.3: participants with the meaning-making choice of 1, 2, 4 and 6 have higher social well-being than the meaning-making choice 3, 5, 7 and 8.

Research Question 3: are the meaning-making choice associated with improved hedonic well-being and social well-being?

- H3.1: the meaning-making choice 1 to 4 predict improved positive affectivities than the meaning-making choice 5 to 8.
- H3.2: the meaning-making choice 1 to 3 predict declined negative affectivities than the meaning-making choice 4 to 8.
- H3.3: the meaning-making choice 1, 2, 4 and 6 predict improved social well-being than the meaning-making choice 3, 5, 7 and 8.

Research Question 4: does dyadic relationship moderate the association between the meaning-making choice and improved hedonic and social well-being?

- H4.1: among participants who have improved dyadic relationship, the ones with the meaning-making choice 1 to 4 are more likely to have improved positive affectivities than the meaning-making choice 5 to 8.
- H4.2: among participants who have improved dyadic relationship, the ones with the meaning-making choice 1 to 4 are more likely to have improved negative affectivities than the meaning-making choice 5 to 8.
- H4.3: among participants who have improved dyadic relationship, the ones with the meaning-making choice 1, 2, 4 and 6 are more likely to have improved social well-being than the rest.

Chapter 5 Methodology

5.1 Design

This study adopted the quantitative and longitudinal design by using panel data with multiple time points to deliberately describe the changes in well-being in response to different meaning-making choices. This 15-day design was web-based and included three major phases: baseline, ESM, and follow-up phases (**Figure 3**).

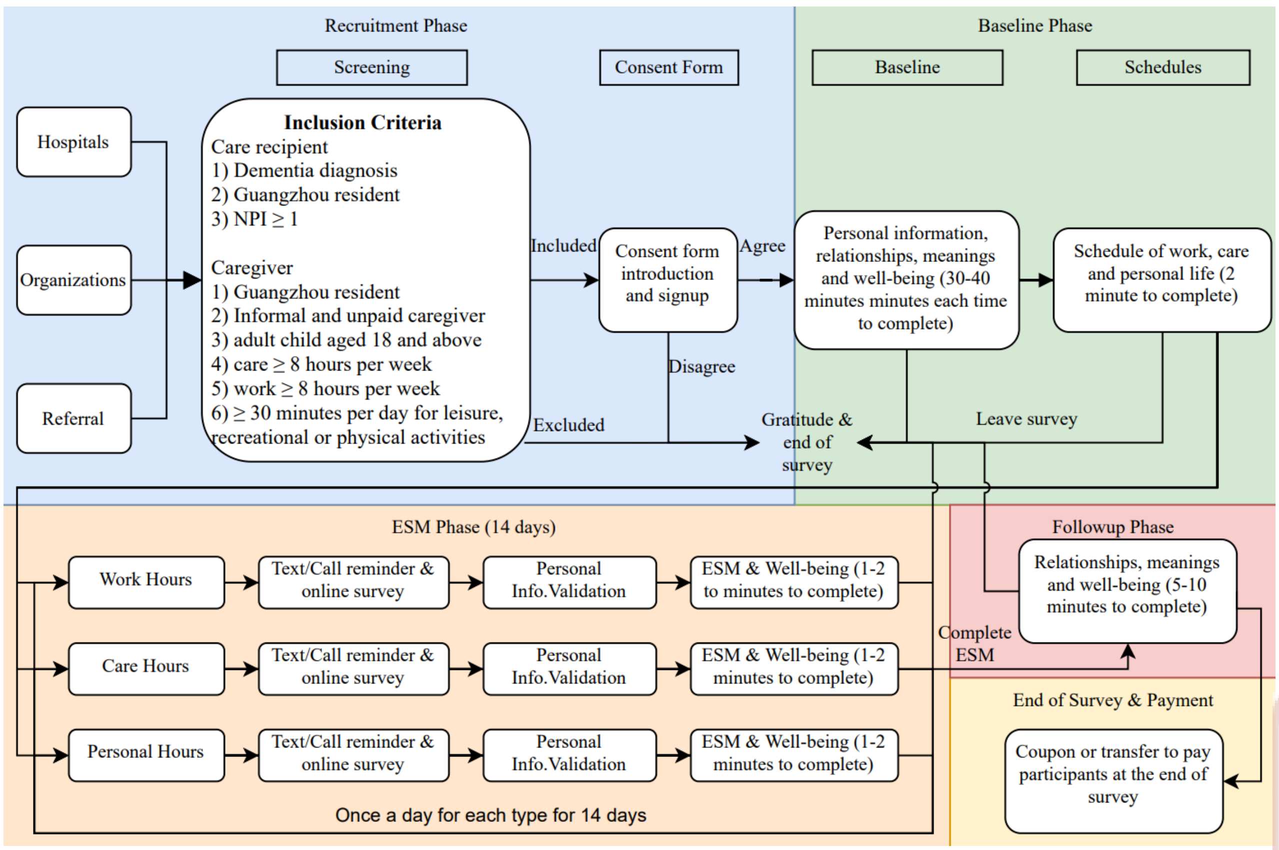

Figure 3. Project flow

Prior to the formal survey, an instruction page (including the project aim, project main content and survey flowchart), an informed consent page, and a recruitment screening page with inclusion criteria were provided via an electronic link to participants using an SMS message or an email. Participants who were willing to participate in the survey were required to review all the information, pass the inclusion criteria and sign the electronic inform consent for records. The electronic fliers including a brief introduction of the project, inclusion criteria, contact information, and a QR code directing to the instruction page were distributed via offline methods (hospitals, community centers, and personal contact) and online channels (instant chat messages, emails, phone calls, and video conferences). Researchers were available to answer questions whenever a participant had any relevant issues, questions, suggestions, and concerns.

Baseline as the first phase of the design collected fundamental information on the first day of the 15-day period. The baseline survey was conventionally taken place on a Sunday immediately followed by the 14-day period of ESM and follow-up surveys. The baseline survey collected basic personal information, meaning-making level, well-being, and the dyadic relationship. Personal information included socioeconomic information of both the PwD and the WDC. For the PwD, we collected their gender, year of birth, current residence (live at home; stay at the hospital; live in the nursing home, elderly care center, welfare center, etc.), the year since first dementia diagnosis, checklist items of ADL and IADL, clinical dementia stage, severity rates and disturbance rates on the checklist of neuropsychiatric inventory, past and current dyadic relationship between the PwD and the WDC. For the WDC, this study recorded their gender, year of birth, marital status, education level, religious belief, length of care, cohabiting with the older adult, caregiving roles, economic status in care, length of adaptation into care, caregiving burden, job stress, meaning-making scales on care, work, and personal life, wellness indices on positive affectivity, negative affectivity, social well-being, and depressive symptoms. The baseline survey takes 30 to 40 minutes approximately to finish.

Right after the baseline survey (usually taken on a Sunday), the Experience Sampling Method (ESM) assessment was scheduled on the coming Monday. ESM, also known as Ecological Momentary Assessment (EMA) in some studies, refers to as an intensive longitudinal research method designed to ask participants to report their momentary thoughts, feelings, behaviors, and environment on multiple occasions intermittently overtime on a given topic. ESM is used to obtain self-reports for a representative time period in the lives of dementia caregivers by sampling randomly during a fixed period of time (Csikszentmihalyi & Larson, 2014; Hektner et al., 2007). ESM usually consists of three major designs: signal contingent (variable time-based), interval contingent (fixed time-based), and event contingent. Signal contingent or variable time-based design asks questions randomly through the day/time period notified by the signals delivered through the mobile device (Riordan et al., 2015). Interval contingent or fixed time-base design asks questions at the specified fixed timepoint every day (Riordan et al., 2015), or time periods in sections of a day, (e.g., the morning, the afternoon, and the evening) (Conner, 2015) with or without a reminder. The questions are useful for measuring routine incidence that is not "susceptible to recall bias" (Conner, 2015). The last type of design is event contingent involving participatory self-report at the occurrence of certain incidences. The self-report in each assessment involved the momentary experience that occurred at the time of prompted messages. Recorded information of each assessment included the time and date, location, activity of engagement, perception of stress, and momentary state of well-being. This study adopted the even contingent method as to address event-specific information of interests. In the baseline study, baseline psychosocial demographics, stress from care and work, dyadic relationships in the past and on the baseline, meaning-making tendencies, and wellness were collected. The ESM design for each participant was conducted for two weeks (see Figure 2) including time, location and activities at the moment of the prompt reminders, and wellness scales. Firstly, the meaning scales included finding meanings through caregiving, working, and personal life experiences. Wellness scales included positive affectivity, negative affectivity, and social well-being. Because the assessments were conducted in mainland China, we sent out reminder messages via local instant message apps (e.g., WeChat and QQ) with a reusable link at the individually scheduled time period each day. Participants were sent reminder prompt linking to ESM assessments three times daily during workdays and 2 to 3 times (most participants don't work on weekends) daily on weekends. Each assessment took about 1 to 2 minutes to complete. The designed ESM phase in this study lasted for 14 consecutive days, being equivalent to two weeks. According to conventional measures and empirical evidence, a period between three days to one month was commonly designed in the ESM studies. Most studies investigating emotional well-being and affectivities adopted a 14-day ESM design (Bos et al., 2018; Kwasnicka et al., 2021).

The follow-up interview adopted a revised short version of the baseline survey and was given to each participant on the last (the 15th) day right after the ESM period. The follow-up survey consisted of dyadic relationships during the ESM period, meaning-making of care, work, and life, and wellness in positive affectivity, negative affectivity, and social well-being. The follow-up survey took about 5 to 10 minutes to accomplish. At the same time, we collected participants' self-reported perceptions of ESM's feasibility and usability experiences. In addition, we selected some of the participants for followup interviews to cross-validate our findings. (Kwasnicka et al., 2021)

5.2 Participants

Participants were recruited from the community centers, hospitals, individual and social network connections in the urban region of Guangzhou, Guangdong, China. According to sample size calculation, this study requires at least 11 participants (equivalent to 385 or more data points – 385/38 = 10.1) to have a confidence level of 95% that the estimated value is within ±5% range of the real values.

The inclusion criteria for potential participants are as follows: 1) providing informal, unpaid, or family care in the mainland; 2) with Guangzhou Hukou; 3) are full-time or part-time employed; 4) care-recipient diagnosed with dementia in all stages; 5) report to have personal life besides care and work; 6) the care recipient had some level of ADL (≥ 1), IADL (≥ 1) or NPI (≥ 1) issues.

The exclusion criteria, on the other hand, encompass 1) providing formal care or paid care to PwD; 2) participants not having Guangzhou Hukou; 3) not employed; 4) the care recipient does not have a diagnosis of dementia disease; 5) report to have no personal life besides care and work; 6) the care recipient had no ADL (= 0), IADL (= 0), and NPI (= 0) issue at all.

5.3 Data collection and confidentiality

Data were collected in the city of Guangzhou, Guangdong, China. Guangzhou lies in southern China near the head of the Pearl River, and is the capital city of Guangdong province. The total jurisdiction area of Guangzhou is 7,434 square km with a population of 15.3 million. Guangzhou has been famous for its manufacturing and tourism since the 1950s, starting from light manufacturers (i.e., electronics, textiles, newsprint, processed foods, and firecrackers) transitioning to heavy manufacturers (i.e., machinery, chemicals, iron and steel, and cement, shipbuilding, automobile, etc.). Guangzhou has also been renowned for its international trade businesses exporting sugar, fruits, silk, tea, herbs, mechanical, electrical, and electronic products to the world market. Guangzhou is a vibrant city highlighted with its lively historical and cultural elements. Local citizens appreciate diverse values and lifestyles. The community centers in Guangzhou have carried out the "dementia-friendly community" project since 2018 under the guidance of the Ministry of Civil Affairs. Symptoms of dementia and dementia care have been well-educated in the local communities and well-received by local residents.

Each participant was given an information sheet in either paper-based or electronic form. Once a participant decided to take part in the survey, a consent form was provided to read and sign. The participation is on a voluntary basis, with which participants can decide to leave the study anytime at their discretion. At the same time, potential participants were given a screening checklist using the inclusion criteria, which included (1) the care recipient was diagnosed with dementia or cognitive impairment; (2) the care recipient presented neuropsychiatric symptom(s) in the past month; (3) both the caregiver and care recipient were Guangzhou citizens with Hukou; (4) the caregiver being an adult child of the care recipient provided informal and unpaid care to the care recipient; (5) the caregiver had spent at least 8 hours a week in care; (6) the caregiver had spent at least 8 hours a week at work or employed jobs; (7) the caregiver had spent at least 30 minutes per week in personal affairs, such as leisure, recreational or physical activities. Eligible participants should have met all seven criteria mentioned above.

All data were collected between October 2010 to September 2021. The data of baseline, ESM, and follow-up phases were all collected on an online survey platform called Qualtrics. The Qualtrics system is highly customizable on various operating systems, such as Mac OS and Windows systems on computers and laptops, or Android systems and IOS systems on digital phones and tablets (Figure 3).

Ethical approval was obtained from the Human Research Ethics Committee at the University of Hong Kong (HREC No. EA200080). Participants' confidentiality was ensured. All sensitive and private information was handled carefully without sharing for any other purposes. Research ID will be created as the identifier of each case for further follow-up studies. The data were stored in password-protected storage devices, an external hard drive and a backup server in the laboratory. Data were backed up monthly in the main data server.

5.4 Measurements

5.4.1 Criteria Variables: daily living activities

PwD commonly suffered from difficulties in daily living activities. Principal measurements included activities of daily living (ADL) and instrumental activities of daily living (IADL). ADL measured abilities required for basic self-care tasks in one's daily living scenarios. ADL included basic activity items, such as "walking, feeding, dressing and grooming, toileting, bathing, and transferring" (Hindmarch et al., 1998). In this study, each of six ADL items was rated as 1 – "performing this activity independently" and 0 – "performing this activity with dependence." The total range of ADL ranged between 0 (worst ADL condition) and 6 (best ADL condition). IADL measured abilities required for complex cognitive thinking and organizational activities. IADL activity items included "managing finances, managing transportation, shopping, meal preparation, house cleaning, home maintenance, managing communication, and managing medications" (M. L. Chan et al., 2021). In this study, each of eight IADL items was rated as 1 – "performing this activity independently" and 0 – "performing this activity with dependence." The total range of IADL ranged between 0 (worst ADL condition) and 8 (best ADL condition) (**Table 1**).

Table 1. Measurements

Variables	T0	ESM	T1	Agent	Instruments	Acronym
Inclusion criteria						
Activities of daily living	✔			PwD	Activities of daily living	ADL
Instrumental activities of daily living	✔			PwD	Instrumental activities of daily living	IADL
Dependent: Well-being						
Positive affectivity	✔	✔	✔	WDC	Positive and Negative Affect Schedule	PANAS
Negative affectivity	✔	✔	✔	WDC	Positive and Negative Affect Schedule	PANAS
Social well-being	✔		✔	WDC	Social well-being	SWB
Depressive symptoms	✔			WDC	Patient Health Questionnaire-9	PHQ
Independent: Stress						
Care stress	✔		✔	WDC	Zarit Burden Interview	ZBI
Work stress	✔		✔	WDC	Occupational Stress Index	OSI
Moderator: Dyadic relationship						
Dyadic relationship	✔		✔	WDC	Unidimensional Relationship Closeness Scale	URCS
Mediator: Meaning-making						
Care meaning 1	✔		✔	WDC	Finding Meaning through Caregiving Scale	FMTC
Care meaning 2	✔		✔	WDC	Positive Aspects of Caregiving	PAC
Work meaning	✔		✔	WDC	Work and Meaning Inventory	WAMI
Life meaning	✔		✔	WDC	Meaning in Life Questionnaire	MLQ
Meaning discrepancy	✔		✔	WDC	Meaning discrepancy	MD
ESM		✔				
Date and time		✔		WDC	Date and time	E1
Location		✔		WDC	Location	E2
Activity of engagement		✔		WDC	Activity of engagement	E3
Perception of stress		✔		WDC	Perception of stress	E4
Positive affectivity		✔		WDC	Positive and Negative Affect Schedule	PANAS
Negative affectivity		✔		WDC	Positive and Negative Affect Schedule	PANAS
Covariates						
Psychiatric symptoms severity	✔			PwD	Neuropsychiatric Inventory Severity	NPI-S
Psychiatric symptoms disturbance	✔			WDC	Neuropsychiatric Inventory Disturbance	NPI-D

5.4.2 Dependent Variables: Well-being

Hedonic well-being commonly involves pleasure, enjoyment, satisfaction, comfort, painlessness, and ease. Specifically, hedonic well-being can be measured in the following categories: positive affect (Diener et al., 1999); positive affect balance (Bradburn, 1969; Ryff, 1989); emotional well-being (Keyes, 2002); hedonic enjoyment (Waterman, 1993); pleasure, joyfulness, enjoyment, fun (Diener, 1984), satisfaction (Cassolato et al., 2010; R. M. Ryan et al., 2008; Waterman, 1993), carefreeness (Huta & Ryan, 2010; R. M. Ryan et al., 2008), low negative affect (Diener et al., 1999; Fowers et al., 2010; Huta & Ryan, 2010). In common practices, hedonic well-being is assessed using several aforementioned dimensions. Positive affect and negative affect are the two major indices of hedonic well-being according to Diener (2003). Therefore, hedonic well-being was measured by the Positive and Negative Affect Schedule (PANAS), which was composed of 10 positive affectivity and ten negativity affectivity (PA and NA) (Brdar, 2014; Thompson, 2007). Positive affectivity included "interested, excited, strong, enthusiastic, proud, alert, inspired, determined, attentive, and active"; while negative affectivity included "distressed, upset, guilty, scared, hostile, irritable, ashamed, nervous, jittery, and afraid." Both PA and NA were measured with the 5-point ratings ranging from 1 (never) to 5 (always) (Thompson, 2007). The score for each subscale ranged from 10 to 50, indicating the worst well-being state to the best state for PA and vice versa for NA. The internal reliabilities of PANAS were satisfactory, ranging from 0.86 to 0.90 for positive affectivity and 0.84 to 0.87 for negative affectivity (Magyar-Moe, 2009). PANAS in this study had a reliability of 0.81 for PA and 0.78 for NA. Their validities were 0.80 for PA and 0.79 for NA (**Table 5**). Improved PANAS was calculated using the changes between baseline and follow-up PANAS scores for each individual.

Social well-being was evaluated by the Chinese version of the Social Well-being Scale (SWB), which consisted of 20 items organized into five dimensions: aspects of "social integration, acceptance, contribution, actualization and coherence" on a 7-point scale (from 1 – "completely disagree" to 7 – "completely agree") (Keyes, 1998). The total score of SWB ranged between 20 to 140, indicating the lowest SWB to the highest SWB. The reliabilities were validated in the following: social integration ($\alpha = .81$), social acceptance ($\alpha = .77$), social contribution ($\alpha = .75$), social actualization ($\alpha = .69$) and social coherence ($\alpha = .64$) (Keyes, 1998). The reliability in this study was 0.85 and the reliability was 0.70.

Caregivers' mental wellness (depressive symptoms) was assessed by the Patient Health Questionnaire-9 (PHQ). The depression scale of PHQ was used as an indicator of the severity of depression (Kroenke et al., 2001). PHQ was conventionally administered in the written form, and in this study was self-reported on the electronic form by the participants. This measurement used nine major clinical diagnostic items evaluating one's depressive levels. Each item was scored from 0 – "none" to 3 – "everyday." The total score ranged from 0 to 27, indicating minimal depressiveness to severe depressiveness. It was suggested minimal depression was scored between 1 to 4, mild depression 5 to 9, moderate depression 10 to 14, and severe depression 15 and above (Kroenke et al., 2001). It has been validated for use in primary care and with confirmed internal consistency reliability (Cronbach's α = 0.89) and inter-rater reliability (Coefficient = 0.94) (Indu et al., 2018; Kroenke et al., 2001; Spitzer et al., 1999). The Cronbach's α of PHQ in this study was 0.80 (Indu et al., 2018), while its validity was 0.84.

5.4.3 Independent Variables: Care and Work Stress

Caregiving burden and stress were measured using the short version of the Zarit Burden Interview (ZBI-4) (Bédard et al., 2001; Zarit, 1980). ZBI consisting of 4 items on the ratings of the 5-point scale (1 – "never" to 5 – "very often") has scores ranging from 4 to 20. Higher scores implied stronger perceptions of burdens. ZBI-4 being quite useful in screening had confirmed internal consistency reliability (Cronbach's α = 0.78) and discriminative ability (AUC range: 0.90 to 0.99) (Higginson et al., 2010). The Cronbach's α of ZBI in this study was 0.78, and its validity was 0.78.

Occupational stress was measured by the Occupational Stress Index (OSI), which was one of the most broadly used inventories to measure work stress (X. Wu et al., 2018). Each of the total 18 items scored on the ratings from 1 – "completely disagree" to 5 – "completely agree", making the total score ranging from 18 to 90, the lightest stress to the heaviest stress. The reliabilities for all sub-scales in OSI ranged between 0.68 to 0.84 (X. Wu et al., 2018). In this study, its reliability was 0.74 and validity 0.62.

5.4.4 Moderating Variable: Dyadic Relationship

The dyadic relationship between the caregiver (CG/WDC) and the care recipient (CR/PwD) was measured using the Unidimensional Relationship Closeness Scale (URCS) at three different time points: prior relationship, the relationship at the time of the baseline survey, and the relationship at the time of the follow-up survey. These URCS assessments were reported by the caregiver and included 12 items scored from 1 – “extremely disagree” to 5 – “extremely agree” (Dibble & Levine, 2011). The total score ranged from 12 to 60, where a higher score indicated a close relationship. The reliabilities were decent for baseline (α = 0.89), ESM (α = 0.85), and followup phases (α = 0.94). The validities were desirable for baseline (0.89), ESM (0.78), and followup phases (0.92).

5.4.5 Mediating variables: Meaning-making and meaning-making choice

The first meaning-making scale, about meaning-making of care, is the Finding Meaning through Caregiving Scale (FMTC), including 43 items on the ratings of 5 points Likert-type scale (from 1 – “strongly agree” to 5 – “strongly disagree”). The items cover three domains: provisional meaning, powerlessness/loss, and ultimate meaning. The scale in this study used the provisional meaning sub-scale had desirable reliability (0.85 to 0.94; a total of 0.91) and validity (0.78 to 0.92; total of 0.88) (Farran et al., 2002).

This study adopted an alternative measurement for care meanings besides FMTC. Positive Aspects of Caregiving (PAC) measured the extent to which the caregiving role was experienced as inspiring and rewarding, yielding positive self-awareness, and enriching one's lived experience (Beach et al., 2000). PAC was measured using 11 items (such as "feel more useful," or "appreciate life more") on a five-point scale each, ranging from 1 ("completely disagree") to 5 ("completely agree"). The PAC scores ranged from 11 to 55. Higher PAC scores indicated a better perception of the caregiving experience. The internal consistency reliability (Cronbach α = .89) and structural validities (0.86) of PAC were desirable as reported (V. W. Q. Lou et al., 2015).

The scale measuring the meaning-making of work was the Work and Meaning Inventory (WAMI) includes ten items on a 5-point Likert scale (from 1 “completely untrue” to 5 “completely true”) that aimed to measure the role of work in individual life consisting of “experiencing positive meaning in work,” “sensing that work was a key avenue for making meaning,” and “perceiving one’s work to benefit some greater good” (M. F. Steger et al., 2012). Its score ranged between 10 to 50, indicating the lowest meaning attached to work versus the highest meaning attached. Its reliability and validity have not been broadly tested while its subscales’ intercorrelated coefficients ranged between 0.65 to 0.78 (M. F. Steger et al., 2012). In this study, its reliability was 0.78 and validity 0.78.

The scale measuring meaning-making of life was the Meaning in Life Inventory (MLQ). MLQ is comprised of 10 items designed to measure two dimensions of meaning in life: 1) Meaning presence, and 2) Search for Meaning (Rose et al., 2017). Respondents answered using a Likert-type scale of 7-point scored from 1 (completely disagree) to 7 (completely agree). The total score ranged between 10 to 70. A higher score indicated a mostly meaningful life. Factorial analysis revealed two critical structural factors with item loadings ranging between 0.67 and 0.82. The internal reliabilities for these two factors were satisfactory (α = 0.81 for "Presence of meaning", and α = 0.79 for "Search for meaning") (Rose et al., 2017). In this study, its reliability was 0.78 and validity 0.79.

Meaning-making Choice was conceptualized as the combination of making meanings of various stressful events, including meaning scores in care, work, and life. The choices of meaning-making include the following categories: 1) care, work, and life, 2) care and work, 3) care and life, 4) work and life, 5) care only, 6) work only, 7) life only, 8) no meaning is made (**Figure 2**). Caregivers with the choices (of 1, 2 and 3) in blue were flourishing caregivers with a positive meaning-making level. Caregivers with the choices (of 5, 6, 7 and 8) in orange were languishing caregivers with a negative meaning-making level. The caregivers with the 4th meaning-making choice in green were neutral caregivers with low care meaning. The flourishing caregivers according to hypotheses had better well-being, while the languishing caregivers had a lower level of well-being.

The aforementioned scales were usually treated as continuous measurement. However, empirical evidence showed that a dichotomization can be interpreted in these measurements. According to interviews in a qualitative study on FMTC, a higher level of meaning expectations is associated with the health conditions (C. K. Chan et al., 2020). The rationale of PAC scores also suggests a higher level of positive meanings attached to the experiences benefits one's well-being and results in better quality of life (Lee & Li, 2021; Quinn & Toms, 2018). In the WAMI's guidebook, a lower score in WAMI is considered an absence from work or employment performance, while a higher score indicates a sense of motivation and enthusiasm in employment (M. F. Steger et al., 2012). Similarly, a higher MLQ score suggests a stronger presence in searching for meanings in one's life

(M. Steger, 2012). Therefore, a dichotomous level can be adopted for all these scales to simplify the categorization. In comparison, data-driven approaches overlook the theoretical development and could be biased when the sample is not appropriately surveyed.

Meaning discrepancies were measured with an adapted version of meaning questions raised by Park (2005): "How much does the [incident] interfere with the way you understood the world to work and the way things happen?" and "how much does the [incident] interfere with your daily goals and the everyday things that are important to you?" We extended the set of the questions to three pairs of questions: 1) "how much does caregiving interfere with the way you understood the world to work and the way things happen?" and "how much does caregiving interfere with your daily goals and the everyday things that are important to you?" 2) "how much does work interfere with the way you understood the world to work and the way things happen?" and "how much does work interfere with your daily goals and the everyday things that are important to you?" and, 3) "how much do personal activities interfere with the way you understood the world to work and the way things happen?" and "how much do personal activities interfere with your daily goals and the everyday things that are important to you?" A 7-point Likert scale was adopted ranging from 1 (completely not influential) to 7 (completely influential). The meaning discrepancy was calculated on the differences between the ratings of each paired questions. Even though it was unknown whether positive or negative discrepancies have different implications, we kept both positive and negative values in the raw data in case there was any difference.

5.4.6 Demographic variables

Demographic data of the care recipient was collected, including gender (1 – male, 2 – female), age (year of birth), residence (1 – home, 2 – hospital, 3 – nursing home, elderly home, care home, etc.), number of family members cohabit with, year of dementia diagnosis, dementia stage (1 – early stage when short-term memory compromises and cognitive functions decline, 2 – medium stage with unstable emotions, irritation, loss of independent living capacity, 3 – late stage with complete loss of self-care capacity, memory, incontinence, cognitive deterioration, and communication disability), the severity and disturbance of Neuropsychiatric Inventory (NPI) with 12 psychiatric symptoms (e.g., hallucination, irritation, depressive mood, anxiety, abnormal behaviors, sleep issues, appetite issues, etc.). NPI severity was rated from 0 (no condition) to 3 (severe condition). NPI disturbance level to caregivers was rated from 0 (no disturbance) to 5 (severe disturbance).

Demographic data of the caregivers was collected, including gender (1 – male, 2 – female), age (year of birth), marital status (1 – single, 2 – married, 3 – partnered, 4 – widowed, 5 – separated, 6 – divorced), education (1 – none, 2 – primary school, 3 – secondary school, 4 – vocational college, 5 – college or university, 6 – master's degree or above), religion (1 – none, 2 – Catholic, 3 – Christian, 4 – Islamic, 5 – Confucius, 6 – Taoism, 7 – others), care length (starting year and month of dementia care), living arrangement (1 – living together with the CR, 0 – living separately with the CR), primary caregiver (1 – yes, 0 – no), other caregivers (1 – couple, 2 – son, 3 – daughter, 4 – son-in-law, 5 – daughter-in-law, 6 – in-home domestic helper, 7 – part-time helper, 8 – friends, 9 – others, 10 – none), financial capacity (1 – completely incapable, 2 – quite incapable, 3 – merely capable, 4 – quite capable, 5 – completely capable), care adaptation (1 – yes, 0 – no), and adaptation length (months length since adaptation if one adapts).

5.4.7 ESM questions

ESM assessment was given 2 to 3 times a day on the specified events (care, work, and personal life) for each participant. In each ESM assessment, the participant's ID instead of the participant's name was required to maximally sustain confidentiality. The data and time of each prompted assessment were automatically collected in the Qualtrics system once the participant completed each assessment. Date and time were automatically recorded once a participant registered the assessment platform for participation. The location and activities of engagement were recorded when the prompt message was received. In addition, participants were asked to manually fill in the type of locations, including "home," "workplace," "outdoors, and" "others to specify." The activities of engagement were recorded including "care for PwD," "care for other older adults," "work," "physical activities (not during caregiving activities)," "recreational activities (not including physical activities or caregiving activities)," and "others to specify." Participants were asked to report the perceived stress from the activities of engagement. The perceived stress was measured with one question using a 5-point Likert scale scored ranging from 1 – "completely no stress" to 5 – "extreme stress." PANAS was as well used in the ESM phase using the same scale. The mean score across 14 days was calculated as the mean ESM PANAS score.

At the end of each ESM assessment, a schedule on care, work and personal life for the next day was provided to fill in so that specific reminder prompts can be sent to corresponding time periods of activities, respectively.

5.4.8 Feasibility, usability, and validity

The evaluation of feasibility and usability was given to participants after all assessments. Following the prior experiences, we used an updated version of the instrument including 17 items to assess subjective experiences relevant to the feasibility and usability of the online version of the ESM protocol (Heinonen et al., 2012; H. Liu & Lou, 2019; Neal et al., 2021). These 17 items included 11 items on feasibility (e.g., “the web-based survey platform worked well”) and six items on usability (e.g., “filling in a web-based survey was an interruption of my daily activities”). Each item was evaluated on a 5-point Likert scale (1 – “completely disagree” to 5 – “completely agree”). The results were demonstrated in the form of mean, standard deviation, and percentages of responses scoring above the mean out of all responses (**Table 2**).

The compliance response rates were recorded as a critical index representing the quality of responses. Even though we configured the “force-respond” function for all questions on the Qualtrics platform to avoid dropouts in the middle of the session, there were chances participants opted out of the ESM assessments. A makeup assessment was sent to those who missed the scheduled assessment on the same day. The compliance response rates were calculated by dividing the number of assessments completed by the total number of reminders delivered (**Figure 4**). We have also recorded hourly response rates, which calculated the percentages of responses for different hours across a day for three separate activities (**Figure 5**).

The ecological validity of the collected data was evaluated to indicate the extent of the representativeness of the real-world environment. To calculate the validity, we used reported locations (e.g., “Where are you right now when you receive the reminder message?”) and activities (e.g., “What were you doing right now when you receive the reminder message?”). Locations include home, the workplace, and outdoors (not including the workplace), and others to specify). Activities included caregiving for the PwD, caregiving for people other than PwD, work, physical activities, leisure activities, and others to specify.

Table 2. Feasibility and usability measures (n = 97)

	Item	M[1] (SD)	%[2]
1	The web-based survey platform is easy to use	3.97 (1.04)	63.6
2	It is easy to carry the smartphone with me	4.28 (0.98)	57.6
3	I carried my smartphone with me every day	4.63 (1.33)	67.7
4	With instruction I understood how the web-based survey platform would work	4.22 (0.97)	63.6
5	It was fun to work with the platform	3.22 (1.46)	68.7
6	It was boring to work with the platform	2.89 (1.07)	42.4
7	The web-based survey platform worked well	4.83 (2.79)	45.5
8	I experienced the prompts as reasonable	3.71 (0.37)	44.4
9	The number of prompts was annoying	2.38 (1.84)	49.5
10	I filled in the web-based survey for 14 consecutive days	4.94 (0.88)	86.9
11	I filled in the web-based survey 2 to 3 times a day	4.77 (1.30)	89.9
12	It was easy to fill in the web-based survey on my smartphone	4.36 (2.29)	49.5
13	The questions were well-displayed on my smartphone	4.17 (1.72)	44.4
14	Filling in a web-based survey was an interruption of my daily activities	3.09 (2.62)	42.4
15	Filling in a web-based survey took too long	1.85 (1.01)	70.7
16	The study took too long	3.54 (1.22)	58.6
17	I understood the questions that were asked	3.84 (2.22)	84.8

[1] Scores were based on a 5-point Likert scale (ranging from 1 - totally disagree to 5 – totally agree)

[2] Percentage of scores above the mean

5.4.9 *Triangulation using qualitative data*

To better support our arguments towards the proposed research questions and hypotheses, the triangulation of qualitative data was also adopted in developing a mixed type of data and providing a comprehensive knowledge of the inquired phenomena (Carter et al., 2014). The method of triangulation refers to the use of multiple sampling methods or data sources to probe a comprehensive understanding of the target experiences (Rooshenas et al., 2019). More specifically, the qualitative data were collected with the convenient sampling method from three of our participants who showed interests in participating in in-depth individual interviews (IDI) with the researcher (Carter et al., 2014). Based on the research questions and hypotheses, three major interview questions were asked to our selected participants: 1) do caregivers make different meaningful choices among care, work, and personal life? 2) do more meanings in life help promote your well-being? 3) does the dyadic relationship help promote your well-being when you have found some meanings in care, work and life? The in-depth interviews aimed to encourage interviewees to elaborate their answers with more detailed facts and arguments. IDI enabled spontaneous, flexible, and responsive conversations between an interviewer and an interviewee (Rooshenas et al., 2019). Therefore, all three participants were asked the same set of the aforementioned questions requesting individual responses. Such an independent qualitative interview design can achieve reliable measurement and trustworthiness.

On the other hand, the focus group discussion (FG) was not adopted this time due to very few participants participating in the interviews and their opinions may greatly influence each other's if they were in the same group. In addition, FGs cost more time and coordination to achieve the interviewer's goal. The merit of FGs can extrapolate enriched data from the group discussion.

5.5 Analytic Method

The analytic plan was applied to answer two major inquiries: 1) to confirm that caregivers making different meaning choices present different states of well-being; and 2) to examine the influences of dyadic relationships on the association between meaning-making choice and well-being.

5.5.1 *Preliminary Analyses*

Prior to the essential analyses, preliminary analyses were conducted to 1) check the statistical power of the data; 2) test the feasibility and usability of the ESM assessments; 3) test the ecological validity of the ESM assessments; and 4) validate the reliability of the revised or Chinese version of the questionnaires, including ZBI, OSI, URCS, FMTC, PAC, WAMI, MLQ, PANAS, SWB, and PHQ.

The power analysis was used to calculate the effect size estimated based on the statistical power of 0.80 and significant level of 0.05 using the current sample size of 100 participants in conducting different analyses (**Table 3**). The proposed meaning-making model included seven covariates (i.e., age, gender, educational attainment, religion, financial status, living status, living arrangement, and caregiving lengtfour), four key variables (dependent variable – positive and negative affectivities, or social well-being; independent variable – caregiving stress, or work stress; mediator variable – meaning-making choice; moderator variable – dyadic relationship) in the structure of a moderated mediation model, and a minimum of 98 research participants would suffice the multiple regression analysis according to G*Power analysis (Faul et al., 2007). In addition, the model analysis requires a minimum of 10 subjects per independent or covariate variable in modeling the analysis (Nunnally & Bernstein, 1994). The sample of 100 participants was statistically sufficient to detect small and medium-sized effects in the moderated mediation model (Preacher et al., 2007).

The ANOVA test compared three sub-groups (k = 3; flourishing, languishing and neutral groups) of meaning-making choices with an averaged sample size for each sub-group (n = 33), making the effect size of 0.317. Bivariate correlations between demographic variables (i.e., age, gender, educational attainment, religion, financial status, living status, living arrangement, and caregiving length), the mediator (meaning-making choice), and the moderator (dyadic relationship) had an estimated effect size of 0.276. The Chi-square test involving categories variables (groups of flourishing, languishing, and neutral caregiver groups) had an estimated effect size of 0.310. The T-test between two independent groups (comparison between flourishing, languishing, and neutral caregiver groups) with equal sample sizes generated an effect size of 0.384, and two groups with unequal sample sizes had 0.511. Finally, regression models had at least a minimal effect size of 0.235.

All these power analyses also took into consideration of Type I and Type II errors. A Type I error indicates a false positive condition whereas a Type II error leads to a false negative conclusion. Avoidance of both errors can be impossible in real statistical practices. Statistically significance usually incurs a Type I error while statistically insignificance manifests a Type II error. The power analyses demonstrated a minimally sufficient sample size for all required statistical analyses, while maintained well-balanced Type I and Type II errors by not having a huge sample size that would inflate the statistical significance. Given a small sample size due to ESM' nature in collecting data, a conservative approach with a lower significance level was sought to increase the Type II error while decrease the Type I error. Even though a false negative conclusion could be ended up, the conclusion is more acceptable according to empirical practices (C. J. Smith, 2012).

Table 3. Power analysis

Analyses	N	k	n	p	Power	df	**Effect Size**
ANOVA		3	33	0.05	0.8		**F = 0.317**
Correlation	100			0.05	0.8		**r = 0.276**
Chi-Square	100			0.05	0.8	2	**W = 0.31**
T-test (equal group)	50			0.05	0.8		**r = 0.384**
T-test (unequal group)	60/40			0.05	0.8		**d = 0.511**
Regression	18/82			0.05	0.8		**F = 0.235**

5.5.2 Research Questions and Hypotheses Testing

Four research questions involving 11 hypotheses were tested using correlational analysis, mediation modeling, and moderated mediation modeling.

SPSS 26 and R 3.6.0 were primarily used for data cleaning, management, and analyses. A preliminary correlational analysis was conducted to examine the relationships between potential covariates and key variables to examine influential covariates and avoid collinearity (e.g., age of PwD and WDC). Distribution, skewness, and outliers of variables were checked meeting the assumptions of a moderated mediation model.

The moderated mediational analysis is an analytic approach to explore whether a mediation model (e.g., a third mediating variable explains the relationship between an independent and a dependent variable) can vary by the levels of a fourth moderating variable (e.g., the mediation pathways are significantly different between the high and low levels of a moderator). The moderated mediation models were analyzed using the “PROCESS” macro for SPSS and R using bootstrap sampling methods (Hayes, 2017). The analytic package “PROCESS” available for both SPSS and R estimates indirect effects (mediating pathways) and conditional indirect effects (moderated mediation effects) using bootstrapping confidence intervals. This resampling strategy as a common measure to bootstrap with repeated sampling for 5000 times yielded statistically unbiased estimates of conditional indirect effects in our proposed moderated mediation model (Hayes & Scharkow, 2013), and a better representation of the targeted population from which our sample was derived (Preacher et al., 2007). As shown in the conceptual framework (**Figure 1**), the relationship as the moderator between the WDC and the PwD, was used in the meaning-making choice model to investigate how the dyadic relationship moderated the relationship between meaning-making and one’s well-being levels (the moderated mediation model involving stressors, meaning-making choices, dyadic relationship and well-being outcome).

Echoing the hypotheses, the conceptual model was identified as shown in **Figure 1**, which consisted of two layers: 1) mediation of meaning-making choice between stressors and well-being as in the meaning-making model; 2) the dyadic relationship moderating the meaning-making choice model. These two types of modeling were tested using the models numbered 4 and 14 specified in Hayes’ PROCESS analysis (Hayes, 2017). The moderated mediation models controlled for caregivers’ age, gender, educational attainment, religious belief, financial status, living arrangement, and years of caregiving history. In addition, the dyadic relationship was presented to moderate the relationship between meaning-making choices and well-being. Both the mediation models and moderated mediation models were analyzed using the significance level of 0.1 due to the small sample size. It was acceptable to have statistical significance for p-values between 0.05 and 0.1 for complex models (Thiese et al., 2016).

5.6 Protocol Compliance, Usability, Feasibility

5.6.1 Compliance Response Rates

The response rates were calculated as shown in **Figure 4**. The average time to complete an ESM assessment was 1.1 min (SD = 0.54). The overall response rate was 93.3% out of the scheduled ESM assessments (n = 3979) daily delivered between 7:00 and 23:59 during the 14-day period. Of the three types of events (i.e., care, work, and personal activities), work had the highest average response rate (96.3%), whereas care had a 95.5% of response rate and personal life had 91.8%. Taking the week's view, response rates were the highest on Thursdays (94.4%) and the lowest on Sundays (92.9%). Specifically, response rates reporting work activities were generally higher on weekdays (between 94.4% and 98.6%) than that on weekends (87.1% on Saturdays and 88.9% on Sundays), partly due to the small number of working hours on weekends (**Figure 4**). Response rates reporting care were the highest on Saturdays (94.7%) and Sundays (95.5%). Response rates of personal activities were between 91.0% and 92.9%. Taking a 24-hour day's view, our findings indicated the time preferences participants responded to finish an assessment. Daily ESM assessments were scheduled based on each individual's preference at certain time points. As shown in **Figure 5**, most responses were made between 19:00 to 21:59 (from 9.9% to 16.8%). For care tasks, 19:00 to 19:59 had the most responses from the participants (25.2%). For personal activities, the preferred time zones of 20:00 to 21:59 were mostly responded (from 17.4% to 37.1%). The peak responsive hours during work were 11:00–13:59 (between 12.6% and 17.9%) (**Figure 5**).

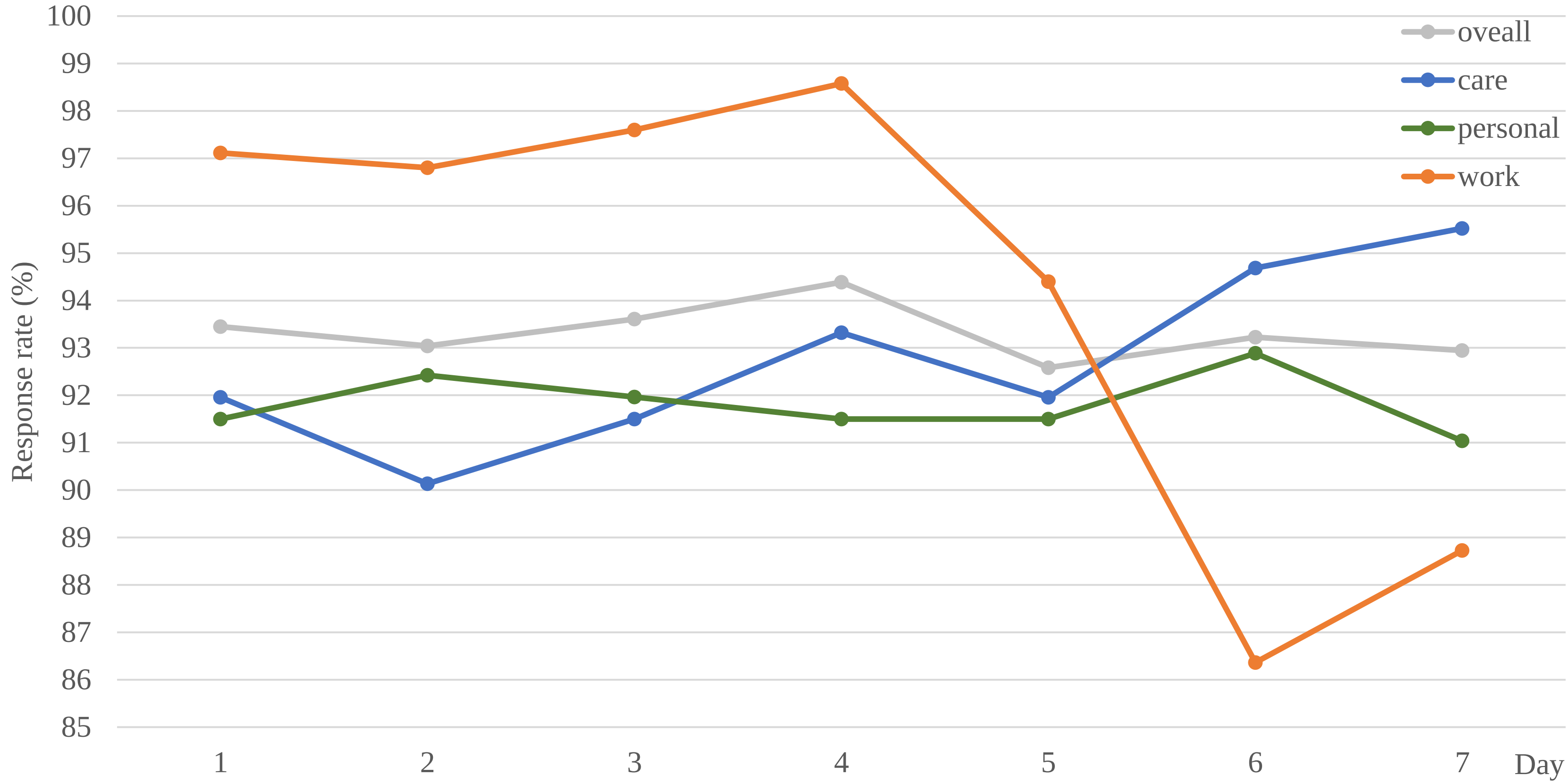

Figure 4. Daily response rates on ESM assessments

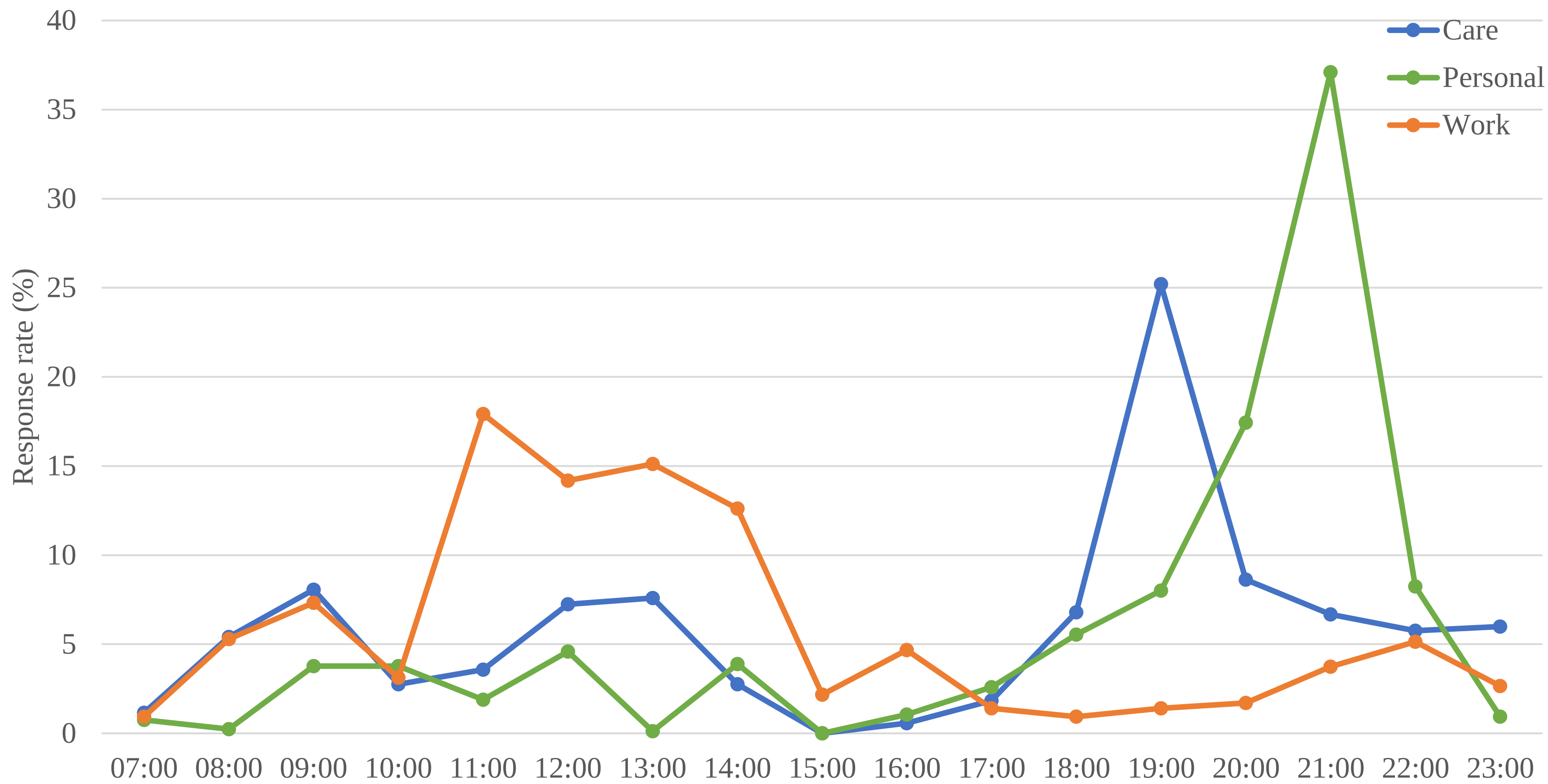

Figure 5. Hourly response preferences on ESM assessments

5.6.2 *Usability and Feasibility*

The evaluation of usability and feasibility was conducted to ensure the user-friendliness of the ESM protocol. Demonstrated in **Table 2**, the results were generated from 99 eligible responses out of all 100 participants. Among reported responses, the majority had positive experiences using the web-based Qualtrics platform. Participants reported positively on “carried the smartphone every day” (M = 4.63, SD = 1.33, 67.7% above the mean) and with researchers’ instruction, they “understood how the web-based survey platform would work” (M = 4.22, SD = 0.97, 63.6%). Most of them “filled in the web-based survey 2 to 3 times a day” (M = 4.77, SD = 1.30, 89.9%) “for 14 consecutive days” (M = 4.94, SD = 0.88, 86.9%). The assessment questions “were well-displayed on the smartphone” (M = 4.17, SD = 1.72, 44.4%) and most participants “understood the questions that were asked” (M = 3.84, SD = 2.22, 84.8%). Fewer of them perceived “filling in a web-based survey was an interruption of daily activities” (M = 3.09, SD = 2.62, 42.4%). Regarding the feasibility of the web-based ESM assessment, majority of the participants responded that the platform was “easy to use” (M = 3.97, SD = 1.04, 63.6%), “worked well” (M = 4.83, SD = 2.79, 45.4%), and “fun to work with” (M = 3.22, SD = 1.46, 68.7%) rather than “boring” (M = 2.89, SD = 1.07, 42.4%). These participants “experienced the prompts as reasonable” (M = 3.71, SD = 0.37, 44.4%) while they found it “easy to carry the smartphone with me” (M = 4.28, SD = 0.98, 57.6%), and “easy to fill in the web-based survey on my smartphone” (M = 4.36, SD = 2.29, 49.5%). In terms of negative experiences, however, many of them reported that “filling in a web-based survey took too long” (M = 3.54, SD = 1.22, 58.6%), and “the number of prompts was annoying” (M = 2.38, SD = 1.84, 49.5%).

5.7 Ecological Validity

Ecological validity was examined by matching the events of engagement and the locations when a participant received a prompt reminder (**Table 4**). Of all activity categories, caregiving was the priority (36.7%) in daily activities, followed by personal leisure activities (35.4%) and work (27.9%). Caregiving was provided majorly at home (86.4%) or at unspecified places including other indoor places, community social service centers, and hospitals (11.8%). Dementia caregivers go to work usually at the workplace (79.2%) or home (11.1%) considering the COVID conditions.

Personal leisure activities took place primarily at home (87.4%) and outdoors (9.0%). Overall, three aspects of activities were broadly taken place at home (65.7%) and workplace (22.8%).

Table 4. Ecological validity

Events	locations[1]				Total[2]
	Home	Workplace	Outdoor	Others	
Care	1178 (86.4%)	24 (1.8%)	1 (0.1%)	161 (11.8%)	1364 (36.7%)
Work	115 (11.1%)	820 (79.2%)	33 (3.2%)	68 (6.6%)	1036 (27.9%)
Personal	1148 (87.4%)	3 (0.2%)	118 (9.0%)	45 (3.4%)	1314 (35.4%)
Total	2441 (65.7%)	847 (22.8%)	152 (4.1%)	274 (7.4%)	3714 (100%)

5.8 Reliability of Instruments

We had a thorough inspection of the internal consistency reliability of mostly relevant instruments (**Table 5**). First, the internal consistency reliabilities of the instrument ranged from 0.42 to 0.94. The reliability for ADL was 0.48 and IADL 0.42. Reliabilities of all health outcomes ranged from 0.50 to 0.85. Positive affectivity and negative affectivity both had high reliability of 0.81 and 0.78. Social well-being had high reliability of 0.85. PHQ had unacceptable reliability of 0.50. Reliabilities for caregiving stress were 0.78 and occupational stress was 0.74. Dyadic relationships generally had high reliabilities at the baseline phase (0.89), ESM phase (0.85), and follow-up phase (0.94). Meaning-making had high reliabilities ranging from 0.77 to 0.89. FMTC's reliability was 0.80, PAC 0.89, WAMI 0.78, and MLQ 0.77. The instruments of NPI severity had a reliability of 0.83, and NPI disturbance had a reliability of 0.71.

Table 5. Reliability and validity

Categories	Instruments	Reliability	Validity
IC			
	ADL	0.48	0.545
	IADL	0.42	0.569
DV: Well-being			
	PA	0.81	0.804
	NA	0.78	0.785

	SWB	0.85	0.696
	PHQ	0.80	0.835
IV: Stress			
	ZBI	0.78	0.777
	OSI	0.74	0.624
Moderator: Dyadic relationship			
	URCS (T0)	0.89	0.890
	URCS (T1)	0.85	0.776
	URCS (T2)	0.94	0.922
	URCS (T0,1,2)	0.91	0.881
Mediator: Meaning-making			
	FMTC	0.80	0.596
	PAC	0.89	0.863
	WAMI	0.78	0.779
	MLQ	0.77	0.793
Covariates			
	NPI severity	0.83	0.695
	NPI disturbance	0.71	0.691

5.9 Correlational Analysis

A preliminary Pearson correlational analysis was conducted to verify any possible potential of having multicollinearity between variables. High correlational coefficients between independent variables may lead to biased estimation in the regression models. As shown in **Figure 6**, all correlational coefficients were provided for independent and outcome variables. Generally, there was no significant multicollinearity between the critical covariate and independent variables involved in our proposed statistical models. However, some variables on the same level are intercorrelated inherently by nature. For instance, on the demographic level, the age of the caregiver and the PwD were highly correlated due to the parent-child relationship. On the mediating level (meaning-making level), there were high correlations between PAC, MLQ, WAMI because they were meaning-making scales. An individual may present a similar level of meaning-making in every aspect of life. On the outcome level, positive and negative affectivity sub-scales of PANAS were found to be highly correlated due to the homogeneity of the sub-scales. PHQ was found to be highly correlated with PANAS sub-scales. To sum, there are no significant interclass correlations that occurred to the critical variables.

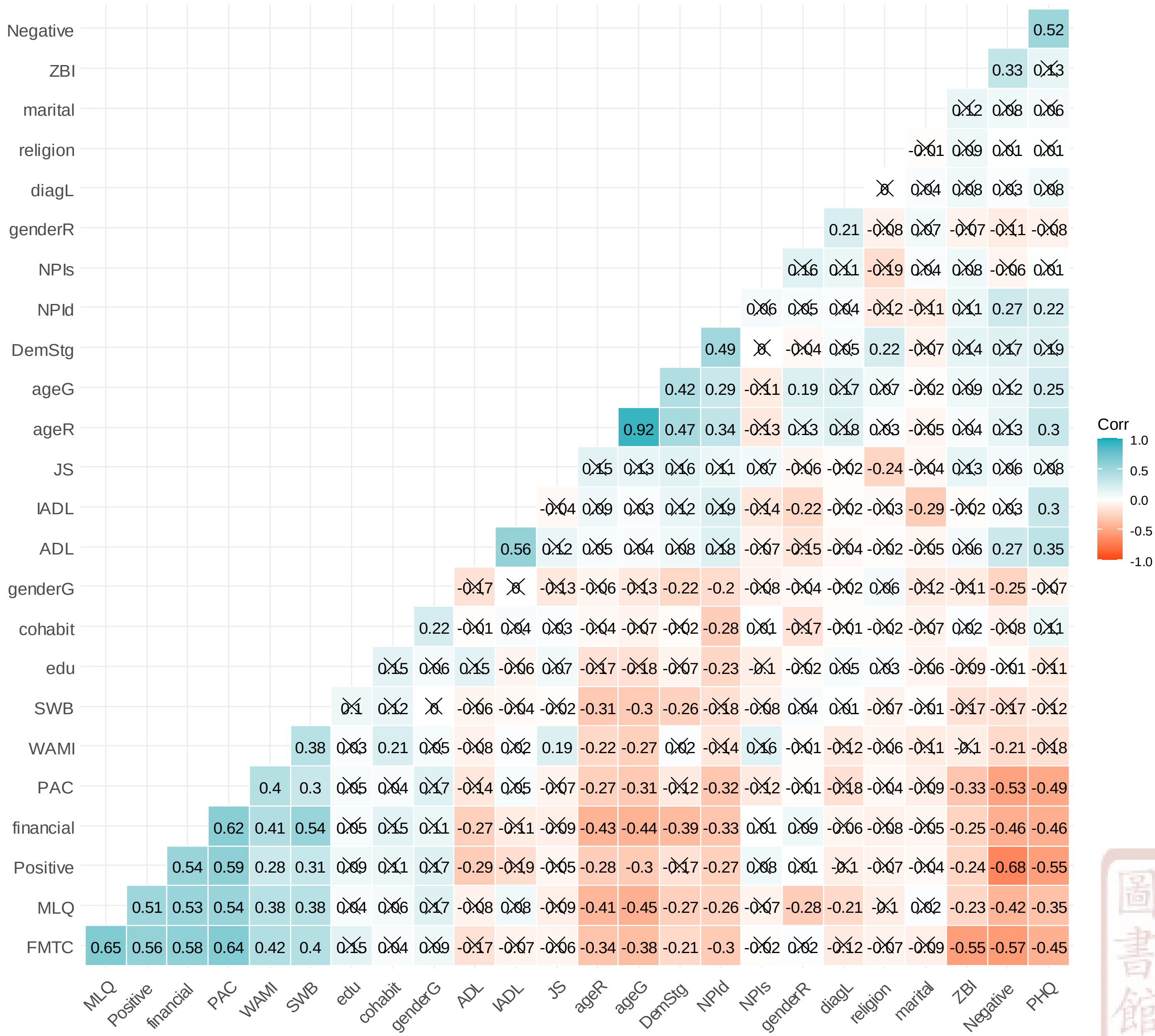

Figure 6. Correlational matrix

5.10 Ethical Consideration

The ethics issues were evaluated with the approval granted by the local Research Ethics Committee in Guangzhou. Recruitment follows the proposed inclusion and exclusion criteria. Participants will be given an information sheet, consent form, and instruction on a digital survey taken on their digital ends, such as cell phones. The participation is on a voluntary basis, on which participants can decide to leave the research anytime. The confidentiality was secured and the consent was informed.

Chapter 6 Results

6.1 Sample Description

The demographic characteristics of the overall sample were presented in **Table 6**. For PwD, the majority of them were females (61%), living with family members (78%), and in their middle dementia stage (48%). PwD on average aged 74.9 year-old (SD = 8.38), were diagnosed for 5.69 years (SD = 4.14), had a medium-high ADL score of 3.70 (SD = 1.15) indicating 3 to 4 ADL difficulties, medium-low IADL score of 2.22 (SD = 1.11) indicating 2 to 3 IADL difficulties, NPI severity score of 15.9 (SD = 6.16). On the other hand, WDCs were primarily females (72%), non-single (i.e., married, partnered; 76%), educated with college and higher diploma (80%), non-religious (90%), and with fair or quite poor financial status (73%). WDCs on average were aged 45.1 year-old (SD = 9.14), and with a moderate low NPI disturbance score of 21.6 (SD = 7.44).

The perceived stressors majorly included caregiving stress (ZBI) and working stress (OSI) as shown in **Table 7**. The overall sample generally scored 11.6 (SD = 3.75) on ZBI, indicating a moderately high caregiving burden; and 54.5 (SD = 8.69) on OSI, indicating moderately high working stress. The meaning-making level was also demonstrated in **Table 8**, which showed a high level of care meaning (M_{FMTC} = 69.0, SD_{FMTC} = 7.96; M_{PAC} = 39.6, SD_{PAC} = 8.39), a moderately high level of work meaning (M_{WAMI} = 32.5, SD_{WAMI} = 6.40), and a moderate level of life meaning (M_{MLQ} = 34.5, SD_{MLQ} = 5,72).

As shown in **Table 8**, on the baseline, hedonic well-being was moderately high (M_{PA} = 26.3, SD_{PA} = 6.58; M_{NA} = 29.8, SD_{NA} = 6.26); social well-being was at a high level (M = 83.7, SD = 9.48); while depressive symptoms PHQ was at a mild level (M = 4.94, SD = 2.31). During the ESM phase, PA was scored 26.5 (SD = 4.41), and NA scored 30.1 (SD = 3.15). According to **Table 9**, the dyadic relationship between the PwD and the WDC was at a moderately higher level (M = 53.9, SD = 8.73).

In addition, the current classification (flourishing caregivers, neutral caregivers, and languishing caregivers) has been cross-checked with the profiles generated from the latent profile analysis (Figure 2a). The latent profile analysis was conducted using the same critical criteria (FMTC, WAMI, and MLQ scores) in a continuous sense, in comparison to the current categorization using the High/Low classification (H/L class). Even though discrepancies in the categorization of working dementia caregivers existed between the theoretically developed classification (the current approach) and the data-driven profile approach, the original classification is sustained as it is theoretically developed and supported.

Table 6. Demographics of the overall sample

Demographics	N (%)	M (SD)	Med [Min, Max]
PwD			
Gender			
Female	61 (61.0%)		
Male	39 (39.0%)		
Age		74.9 (8.38)	76 [56, 89]
Living arrangement			
Living alone	22 (22.0%)		
Living together	78 (78.0%)		
Diagnosis history (in years)		5.69 (4.14)	5 [1, 28]
Dementia stage			
Early	24 (24.0%)		
Middle	48 (48.0%)		
Late	28 (28.0%)		
ADL [6-0]		3.70 (1.15)	4 [1, 6]
IADL [8-0]		2.22 (1.11)	2 [0, 5]
NPI severity [36-0]		15.9 (6.16)	15 [5, 33]
WDC			
Gender			
Female	72 (72.0%)		
Male	28 (28.0%)		
Age		45.1 (9.14)	45.5 [28, 63]
Marital status			
Non-single	76 (76.0%)		
Single	24 (24.0%)		
Education			
Vocational	20 (20.0%)		
College	66 (66.0%)		
Graduate or above	14 (14.0%)		
Religion			
Not Religious	90 (90.0%)		
Religious	10 (10.0%)		
Financial status			
Very poor	8 (8.0%)		
Quite poor	33 (33.0%)		
Fair	40 (40.0%)		
Quite good	11 (11.0%)		
Very good	8 (8.0%)		
NPI disturbance [60-0]		21.6 (7.44)	21 [5, 38]

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range.

Table 7. Stress and meaning-making of the overall sample

	M (SD)	Med [Min, Max]
Stressors		
ZBI [20-4]	11.6 (3.75)	13 [4, 19]
OSI [90-18]	54.5 (8.69)	55.5 [20, 74]
Meaning-making		
FMTC [19-95]	69.0 (7.96)	70 [48, 93]
PAC [11-55]	39.6 (8.39)	40 [18, 52]
WAMI [10-50]	32.5 (6.40)	30.5 [21, 49]
MLQ [10-70]	34.5 (5.72)	36 [21, 45]

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range.

Table 8. Baseline and ESM well-being of the overall sample

	M (SD)	Med [Min, Max]
Well-being (baseline)		
PA [10-50]	26.3 (6.58)	26 [13, 41]
NA [50-10]	29.8 (6.26)	31 [10, 41]
SWB [20-140]	83.7 (9.48)	85 [57, 103]
PHQ [27-0]	4.94 (2.31)	5 [0, 10]
Well-being (ESM)		
PA ESM	26.5 (4.41)	28.4 [17.6, 35.6]
NA ESM	30.1 (3.15)	29.0 [19.4, 37.8]

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range. "PA ESM" and "NA ESM" indicate the positive and negative affectivities measured during the ESM phase.

Table 9. Dyadic relationship of the overall sample

	M (SD)	Med [Min, Max]
Dyadic relation [12-84]	53.9 (8.73)	53.5 [33.0, 76.0]

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range.

Care Meaning	Work Meaning	Life Meaning	
		High	Low
High	High	1 Well-balanced	2 Work to care
	Low	3 Live well to care	5 Filial
Low	High	4 Self-centered	6 Workaholic
	Low	7 Self-carer	8 Hopeless
Classes	Flourishing caregivers: 1, 2, 3		
	Neutral caregivers: 4		
	Languishing caregivers: 5, 6, 7, 8		

H/L class	LPA profiles		
	1	2	3
Flourishing	13	2	28
Neutral	3	0	3
Languishing	8	27	16

Figure 7a. Classes of meaning-making choices (comparing with LPA profiles)

As shown in **Figure 2** and **Table 10**, flourishing caregivers in total took a combined percentage of 43%. Among flourishing caregivers, the meaning-making choice No. 1 took 46.5% of this category, the meaning-making choice No. 2 took 27.9%, while the meaning-making choice No. 3 took 25.6%. Six neutral caregivers who made the meaning-making choice No. 4 took 6% of the overall sample. On the other hand, languishing caregivers took a combined percentage of 51%. Among these languishing caregivers, 11.8% of them picked the meaning-making choice No. 5, 19.6% of the meaning-making choice No. 6, 9.8% of the meaning-making choice No. 7, and 58.8% of the meaning-making choice No. 8.

As shown in **Figure 2** and **Table 11**, caregivers with high work meanings in total took a combined percentage of 50%. Among caregivers with high work meanings, the meaning-making choice No. 1 took 40% of this category, the meaning-making choice No. 2 took 24%, the meaning-making choice No. 4 took 11%, and the meaning-making choice No. 6 took 20%. On the other hand, caregivers with low work meanings took a combined percentage of 50%. Among these caregivers, 24% of them picked the meaning-making choice No. 3, 10% of the meaning-making choice No. 5, 6% of the meaning-making choice No. 7, and 10% of the meaning-making choice No. 8.

Table 10. Meaning-making choices and classes of care meanings

MMC		Overall (N = 100)	Flourishing (N = 43)	Neutral (N = 6)	Languishing (N = 51)
1	↑C↑W↑P	20 (20%)	46.5%		
2	↑C↑W↓P	12 (12%)	27.9%		
3	↑C↓W↑P	11 (11%)	25.6%		
4	↓C↑W↑P	6 (6%)		100%	
5	↑C↓W↓P	6 (6%)			11.8%
6	↓C↑W↓P	10 (10%)			19.6%
7	↓C↓W↑P	5 (5%)			9.8%
8	↓C↓W↓P	30 (30%)			58.8%

Table 11. Meaning-making choices and classes of work meanings

	MMC	Overall (N = 100)	High Work Meaning (N = 50)	Low Work Meaning (N = 50)
1	↑C↑W↑P	20 (20%)	40%	
2	↑C↑W↓P	12 (12%)	24%	
4	↓C↑W↑P	11 (11%)	16%	
6	↓C↑W↓P	6 (6%)	20%	
3	↑C↓W↑P	6 (6%)		24%
5	↑C↓W↓P	10 (10%)		10%
7	↓C↓W↑P	5 (5%)		6%
8	↓C↓W↓P	30 (30%)		10%

6.2 Descriptives on MMC classes

The demographic descriptives by classes of care meaning levels were displayed in **Table 12**. Between classes of care meaning levels, neutral group on average had the youngest PwD (M = 67.2, SD = 5.95, p = 0.006), and the youngest WDC (M = 35.7, SD = 2.07, p = 0.002). Flourishing caregivers had the highest percentages in living together with family members (83.7%, p = 0.091), better financial status (37.3% reporting "quite good" and "very good" versus 5.9% among neutral and languishing caregivers, p = 0.006), and the lowest disturbance by NPI symptoms (M = 20.1, SD = 7.31, p = 0.031). As shown in **Table 13**, between classes of the meaning-making choice No. 1 to 4 (flourishing and neutral caregivers) and the meaning-making choice No. 5-8 (languishing caregivers), the former class on average had younger PwD (M = 72.6, SD = 7.76, p = 0.006), and the youngest WDC (M = 42.1, SD = 7.98, p < 0.001). Caregivers of the former group in general were primarily females (75.5%, p = 0.007), had a better financial status (77.5% reporting fair or better status; p < 0.001).

The demographic descriptives by classes of work meaning levels were displayed in **Table 14**. Between classes of work meaning levels, high work meaning group on average had the youngest PwD (M = 72.6, SD = 8.27, p = 0.006), and WDC (M = 42.3, SD = 8.41, p = 0.002). High work meaning group had the highest percentages of PwD living together with family members (86.0%, p = 0.091), better financial status (72.0% reporting "fair," "quite good," and "very good" versus 46% among the low work meaning group, p = 0.006), and lower disturbance by NPI symptoms (M = 20.1, SD = 7.31, p = 0.031).

Table 12. Descriptives by classes of care meanings (flourishing – the meaning-making choice No. 1-3, neutral – the meaning-making choice No. 4, and languishing caregivers – the meaning-making choice No. 5-8)

	Flourishing (N = 43)	**Neutral (N = 6)**	**Languishing (N = 51)**	***p***
PwD				
Gender				0.682
Female	31 (72.1%)	6 (100%)	24 (47.1%)	
Male	12 (27.9%)	0 (0%)	27 (52.9%)	
Age				**0.006**
Mean (SD)	73.3 (7.74)	**67.2 (5.95)**	77.1 (8.42)	
Median [Min, Max]	73 [62, 89]	68.5 [56, 72]	78 [58, 89]	
Living arrangement				**0.091**
Living alone	7 (16.3%)	0 (0%)	15 (29.4%)	
Living together	**36 (83.7%)**	6 (100%)	36 (70.6%)	
Diagnosis history (in years)				0.374
Mean (SD)	5.60 (3.89)	4 (2.76)	5.96 (4.47)	
Median [Min, Max]	5 [1, 21]	3 [2, 9]	5 [1, 28]	
Dementia stage				0.640
Early	14 (32.6%)	1 (16.7%)	9 (17.6%)	
Middle	14 (32.6%)	5 (83.3%)	29 (56.9%)	
Late	15 (34.9%)	0 (0%)	13 (25.5%)	
ADL [6-0]				0.490
Mean (SD)	3.74 (1.16)	4.17 (0.753)	3.61 (1.18)	
Median [Min, Max]	4 [1, 6]	4 [3, 5]	4 [1, 6]	
IADL [8-0]				0.858
Mean (SD)	2.33 (1.21)	2.83 (0.983)	26 (11)	
Median [Min, Max]	2 [0, 5]	3 [1, 4]	2 [0, 4]	
NPI severity [36-0]				0.640
Mean (SD)	16 (5.70)	13.8 (5.27)	16 (6.68)	
Median [Min, Max]	15 [8, 33]	13.5 [8, 21]	15 [5, 33]	
WDC				
Gender				0.824
Female	33 (76.7%)	3 (50%)	36 (70.6%)	
Male	10 (23.3%)	3 (50%)	15 (29.4%)	
Age				**0.002**
Mean (SD)	43 (89.0)	**35.7 (27.0)**	48 (9.33)	
Median [Min, Max]	40 [31, 58]	36 [32, 38]	49 [28, 63]	
Marital status				0.242

Non-single	31 (72.1%)	5 (83.3%)	40 (78.4%)	
Single	12 (27.9%)	1 (16.7%)	11 (21.6%)	
Education				0.496
Vocational	8 (18.6%)	0 (0%)	12 (23.5%)	
College	25 (58.1%)	6 (100%)	35 (68.6%)	
Graduate or above	10 (23.3%)	0 (0%)	4 (7.8%)	
Religion				0.739
Not Religious	42 (97.7%)	5 (83.3%)	43 (84.3%)	
Religious	1 (2.3%)	1 (16.7%)	8 (15.7%)	
Financial status				**0.006**
Very poor	0 (0%)	0 (0%)	8 (15.7%)	
Quite poor	10 (23.3%)	1 (16.7%)	22 (43.1%)	
Fair	17 (39.5%)	5 (83.3%)	18 (35.3%)	
Quite good	**10 (23.3%)**	0 (0%)	1 (2%)	
Very good	6 (14%)	0 (0%)	2 (3.9%)	
NPI disturbance [60-0]				**0.013**
Mean (SD)	**20.1 (7.31)**	23.7 (8.57)	22.6 (7.32)	
Median [Min, Max]	19 [7, 37]	23.5 [14, 33]	24 [5, 38]	

Table 13. Descriptives by classes of care meanings (flourishing and neutral – the meaning-making choice No. 1- 4, and languishing caregivers – the meaning-making choice No. 5-8)

	Flourishing & Neutral (N = 49)	**Languishing (N = 51)**	***p***
PwD			
Gender			**0.007**
Female	**37 (75.5%)**	24 (47.1%)	
Male	12 (24.5%)	27 (52.9%)	
Age			**0.006**
Mean (SD)	**72.6 (7.76)**	77.1 (8.42)	
Median [Min, Max]	72 [56, 89]	78 [58, 89]	
Living arrangement			0.113
Living alone	7 (14.3%)	15 (29.4%)	
Living together	42 (85.7%)	36 (70.6%)	
Diagnosis history (in years)			0.506
Mean (SD)	5.41 (3.79)	5.96 (4.47)	
Median [Min, Max]	5 [1, 21]	5 [1, 28]	
Dementia stage			0.158
Early	15 (30.6%)	9 (17.6%)	
Middle	19 (38.8%)	29 (56.9%)	
Late	15 (30.6%)	13 (25.5%)	
ADL [6-0]			0.416
Mean (SD)	3.80 (1.12)	3.61 (1.18)	
Median [Min, Max]	4 [1, 6]	4 [1, 6]	
IADL [8-0]			0.139
Mean (SD)	2.39 (1.19)	2.06 (1.01)	
Median [Min, Max]	2 [0, 5]	2 [0, 4]	
NPI severity [36-0]			0.855
Mean (SD)	15.7 (5.64)	16.0 (6.68)	
Median [Min, Max]	15 [8, 33]	15 [5, 33]	
WDC			
Gender			0.922
Female	36 (73.5%)	36 (70.6%)	
Male	13 (26.5%)	15 (29.4%)	
Age			**< 0.001**
Mean (SD)	**42.1 (7.98)**	48.0 (9.33)	
Median [Min, Max]	39 [31, 58]	49 [28, 63]	
Marital status			0.729
Non-single	36 (73.5%)	40 (78.4%)	

Single	13 (26.5%)	11 (21.6%)	
Education			0.167
Vocational	8 (16.3%)	12 (23.5%)	
College	31 (63.3%)	35 (68.6%)	
Graduate or above	10 (20.4%)	4 (7.8%)	
Religion			0.110
Not Religious	47 (95.9%)	43 (84.3%)	
Religious	2 (4.1%)	8 (15.7%)	
Financial status			**< 0.001**
Very poor	0 (0%)	8 (15.7%)	
Quite poor	11 (22.4%)	**22 (43.1%)**	
Fair	**22 (44.9%)**	18 (35.3%)	
Quite good	10 (20.4%)	1 (2.0%)	
Very good	6 (12.2%)	2 (3.9%)	
NPI disturbance [60-0]			0.152
Mean (SD)	20.5 (7.48)	22.6 (7.32)	
Median [Min, Max]	19 [7, 37]	24 [5, 38]	
DR [12-84]			**< 0.001**
Mean (SD)	**56.2 (6.40)**	51.7 (10.1)	
Median [Min, Max]	56.0 [44.0, 70.0]	52.0 [33.0, 76.0]	

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range.

Table 14. Descriptives by classes of work meanings (low work meaning – the meaning-making choice No. 3, 5, 7 and 8, high work meaning – the meaning-making choice No. 1, 2, 4 and 6)

	High Work Meaning (N = 50)	**Low Work Meaning (N = 50)**	***p***
PwD			
Gender			0.682
Female	29 (58.0%)	32 (64.0%)	
Male	21 (42.0%)	18 (36.0%)	
Age			**0.006**
Mean (SD)	**72.6 (8.27)**	77.1 (7.94)	
Median [Min, Max]	71.5 [56, 89]	78 [58, 89]	
Living arrangement			**0.091**
Living alone	7 (14.0%)	15 (30.0%)	
Living together	**43 (86.0%)**	35 (70.0%)	
Diagnosis history (in years)			0.374
Mean (SD)	5.32 (3.61)	6.06 (4.61)	
Median [Min, Max]	4.50 [1, 21]	5 [1, 28]	
Dementia stage			0.640
Early	14 (28.0%)	10 (20.0%)	
Middle	23 (46.0%)	25 (50.0%)	
Late	13 (26.0%)	15 (30.0%)	
ADL [6-0]			0.490
Mean (SD)	3.62 (1.23)	3.78 (1.07)	
Median [Min, Max]	4 [1, 6]	4 [1, 6]	
IADL [8-0]			0.858
Mean (SD)	2.20 (1.11)	2.24 (1.12)	
Median [Min, Max]	2 [0, 5]	2 [0, 5]	
NPI severity [36-0]			0.640
Mean (SD)	16.1 (5.91)	15.6 (6.45)	
Median [Min, Max]	16 [8, 33]	15 [5, 33]	
WDC			
Gender			0.824
Female	35 (70.0%)	37 (74.0%)	
Male	15 (30.0%)	13 (26.0%)	
Age			**0.002**
Mean (SD)	**42.3 (8.41)**	47.9 (9.06)	
Median [Min, Max]	38.5 [31, 59]	49 [28, 63]	
Marital status			0.242
Non-single	35 (70.0%)	41 (82.0%)	

Single	15 (30.0%)	9 (18.0%)	
Education			0.496
Vocational	9 (18.0%)	11 (22.0%)	
College	32 (64.0%)	34 (68.0%)	
Graduate or above	9 (18.0%)	5 (10.0%)	
Religion			0.739
Not Religious	46 (92.0%)	44 (88.0%)	
Religious	4 (8.0%)	6 (12.0%)	
Financial status			**0.006**
Very poor	0 (0%)	8 (16.0%)	
Quite poor	14 (28.0%)	19 (38.0%)	
Fair	**22 (44.0%)**	18 (36.0%)	
Quite good	7 (14.0%)	4 (8.0%)	
Very good	7 (14.0%)	1 (2.0%)	
NPI disturbance [60-0]			**0.013**
Mean (SD)	**19.8 (7.44)**	23.4 (7.03)	
Median [Min, Max]	19 [6, 38]	24 [5, 37]	

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range.

6.3 Stressors and meaning-making levels by the meaning-making choice classes

The stressors between classes of care meaning levels were displayed in **Table 15**. There was no significant difference in stressors between classes of flourishers, neutralers, and languishers. The flourishing caregivers had the highest meaning-making level ($M_{FMTC} = 73.3$, $SD_{FMTC} = 2.08$, $p < 0.001$; $M_{PAC} = 42.6$, $SD_{PAC} = 6.80$, $p = 0.001$). Neutral caregivers had the highest work meaning-making level ($M = 37.3$, $SD = 7.74$, $p < 0.001$) , and the highest life meaning-making level ($M = 40.3$, $SD = 2.58$, $P < 0.001$), respectively followed by flourishing caregivers ($M_{WAMI} = 34.5$, $SD_{WAMI} = 6.21$, $p < 0.001$; $M_{MLQ} = 37.1$, $SD_{MLQ} = 3.06$, $p < 0.001$). As shown in **Table 16**, between classes of the meaning-making choice No. 1-4 (flourishing and neutral caregivers) and the meaning-making choice No. 5-8 (languishing caregivers), the former class on average had higher levels of care meanings ($M_{FMTC} = 72.2$, $SD_{FMTC} = 3.84$, $p < 0.001$; $M_{PAC} = 42.2$, $SD_{PAC} = 6.75$, $p = 0.002$), work meanings ($M = 34.9$, $SD = 6.39$, $p < 0.001$), and life meanings ($M = 37.5$, $SD = 3.16$, $p < 0.001$). There were no significant differences on stressors between two classes.

The stressors between classes of work meaning levels were displayed in **Table 17**. There was no significant difference in stressors between classes of high and low work meaning levels. The high work meaning caregivers had the highest meaning-making level ($M_{FMTC} = 72.3$, $SD_{FMTC} = 6.87$, $p < 0.001$; $M_{PAC} = 42.6$, $SD_{PAC} = 7.33$, $p = 0.002$), the highest work meaning-making level ($M = 37.4$, $SD = 5.62$, $p < 0.001$) , and the highest life meaning-making level ($M = 36.6$, $SD = 4.20$, $p < 0.001$).

Table 15. Stressors and meaning-making by classes of care meanings (flourishing – the meaning-making choice No. 1-3, neutral – the meaning-making choice No. 4, and languishing caregivers – the meaning-making choice No. 5-8)

	Flourishing (N = 43)	**Neutral (N = 6)**	**Languishing (N = 51)**	***p***
Stressors				
ZBI [20-4]				0.143
Mean (SD)	10.8 (33)	12.8 (42)	12.2 (4.17)	
Median [Min, Max]	10 [5, 19]	14.5 [6, 16]	14 [4, 18]	
OSI [90-18]				0.336
Mean (SD)	55.3 (8.16)	51 (4.98)	54.3 (9.44)	
Median [Min, Max]	56 [33, 68]	50.5 [46, 60]	56 [20, 74]	
Meaning-making				
FMTC [19-95]				**< 0.001**
Mean (SD)	**73.3 (2.08)**	64 (3.58)	65.9 (9.58)	
Median [Min, Max]	73 [70, 77]	63.5 [59, 69]	63 [48, 93]	
PAC [11-55]				**< 0.001**
Mean (SD)	**42.6 (6.80)**	39.2 (6.05)	37.1 (9.10)	
Median [Min, Max]	44 [23, 52]	40.5 [29, 46]	36 [18, 52]	
WAMI [10-50]				**< 0.001**
Mean (SD)	34.5 (6.21)	**37.3 (7.74)**	30.3 (5.62)	
Median [Min, Max]	33 [27, 48]	34 [31, 48]	28 [21, 49]	
MLQ [10-70]				**< 0.001**
Mean (SD)	37.1 (3.06)	**40.3 (2.58)**	31.5 (6.10)	
Median [Min, Max]	37 [29, 43]	40.5 [36, 44]	33 [21, 45]	

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range.

Table 16. Stressors and meaning-making by classes of care meanings (flourishing and neutral – the meaning-making choice No. 1- 4, and languishing caregivers – the meaning-making choice No. 5-8)

	Flourishing & Neutral (N = 49)	**Languishing (N = 51)**	***p***
Stressors			
ZBI [20-4]			0.136
Mean (SD)	11.1 (3.18)	12.2 (4.17)	
Median [Min, Max]	10 [5, 19]	14 [4, 18]	
OSI [90-18]			0.792
Mean (SD)	54.8 (7.93)	54.3 (9.44)	
Median [Min, Max]	55 [33, 68]	56 [20, 74]	
Meaning-making			
FMTC [19-95]			**< 0.001**
Mean (SD)	**72.2 (3.84)**	65.9 (9.58)	
Median [Min, Max]	73 [59, 77]	63 [48, 93]	
PAC [11-55]			**0.002**
Mean (SD)	**42.2 (6.75)**	37.1 (9.10)	
Median [Min, Max]	43 [23, 52]	36 [18, 52]	
WAMI [10-50]			**< 0.001**
Mean (SD)	**34.9 (6.39)**	30.3 (5.62)	
Median [Min, Max]	33 [27, 48]	28 [21, 49]	
MLQ [10-70]			**< 0.001**
Mean (SD)	**37.5 (3.16)**	31.5 (6.10)	
Median [Min, Max]	38 [29, 44]	33 [21, 45]	

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range.

Table 17. Stressors and meaning-making by classes of work meanings (low work meaning – the meaning-making choice No. 3, 5, 7 and 8, high work meaning – the meaning-making choice No. 1, 2, 4 and 6)

	High Work Meaning (N = 50)	**Low Work Meaning (N = 50)**	***p***
Stressors			
ZBI [20-4]			0.136
Mean (SD)	11.1 (3.78)	12.2 (3.66)	
Median [Min, Max]	11 [4, 19]	13.5 [6, 18]	
OSI [90-18]			0.792
Mean (SD)	55.4 (8.19)	53.7 (9.17)	
Median [Min, Max]	57 [33, 68]	54 [20, 74]	
Meaning-making			
FMTC [19-95]			**< 0.001**
Mean (SD)	**72.3 (6.87)**	65.7 (7.63)	
Median [Min, Max]	72.5 [59, 93]	64 [48, 79]	
PAC [11-55]			**0.002**
Mean (SD)	**42.6 (7.33)**	36.6 (8.36)	
Median [Min, Max]	44 [23, 52]	37.5 [18, 51]	
WAMI [10-50]			**< 0.001**
Mean (SD)	**37.4 (5.62)**	27.7 (1.66)	
Median [Min, Max]	36 [31, 49]	28 [21, 30]	
MLQ [10-70]			**< 0.001**
Mean (SD)	**36.6 (4.20)**	32.4 (6.29)	
Median [Min, Max]	37 [23, 45]	33 [21, 43]	

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range.

6.4 Well-being by meaning-making choices

As shown in **Table 18**, the findings indicated that the baseline PANAS was generally better for the meaning-making choice No. 1 to 4 (PA ranging from 26.25 to 31.60; NA ranging from 24.50 to 31.63) than the meaning-making choice No. 6, 7 and 8 (PA from 21.00 to 23.33; NA from 30.33 to 33.27) (**Figure 8**). However, the meaning-making choice No. 5 had the highest positive affectivity (M = 32.20) and the second-lowest negative affectivity (M = 26.20).

In comparison, the ESM well-being demonstrated more expected outcomes, in which the meaning-making choice No. 1 to 4 had the highest positive affectivity (ranging from 28.01 to 29.63) and relatively low negative affectivity (ranging from 21.57 to 29.21), whereas the meaning-making choice No. 5 to 8 had the lowest positive affectivity (ranging from 22.26 to 29.09) and relatively higher negative affectivity (ranging from 28.55 to 32.96) (**Table 19** and **Figure 9**). As expected, the meaning-making choice No. 1 to 4 had higher levels of social well-being (ranging from 86.50 to 87.55) than the meaning-making choice No. 5 to 8 (ranging from 76.83 to 86.90) (**Table 19** and **Figure 10**). In terms of depressive symptoms, the meaning-making choice No. 1 to 4 had lower scores (ranging from 2.95 to 5.25) than the meaning-making choice No. 6 to 8 (ranging from 5.70 to 7.00). The meaning-making choice No. 5 had the lowest depressive symptoms score of 3.40 (**Table 19** and **Figure 11**).

Table 18. Outcomes by classes (baseline)

	PHQ	SWB	PA	NA
1 ↑C↑W↑P	2.95	87.55	31.60	24.50
2 ↑C↑W↓P	5.17	86.50	26.25	29.58
3 ↑C↓W↑P	4.58	87.00	28.50	28.58
4 ↓C↑W↑P	5.25	87.13	29.13	31.63
5 ↑C↓W↓P	3.40	82.00	32.20	26.20
6 ↓C↑W↓P	7.00	86.90	21.00	32.30
7 ↓C↓W↑P	6.00	85.33	23.33	30.33
8 ↓C↓W↓P	5.70	76.83	22.33	33.27

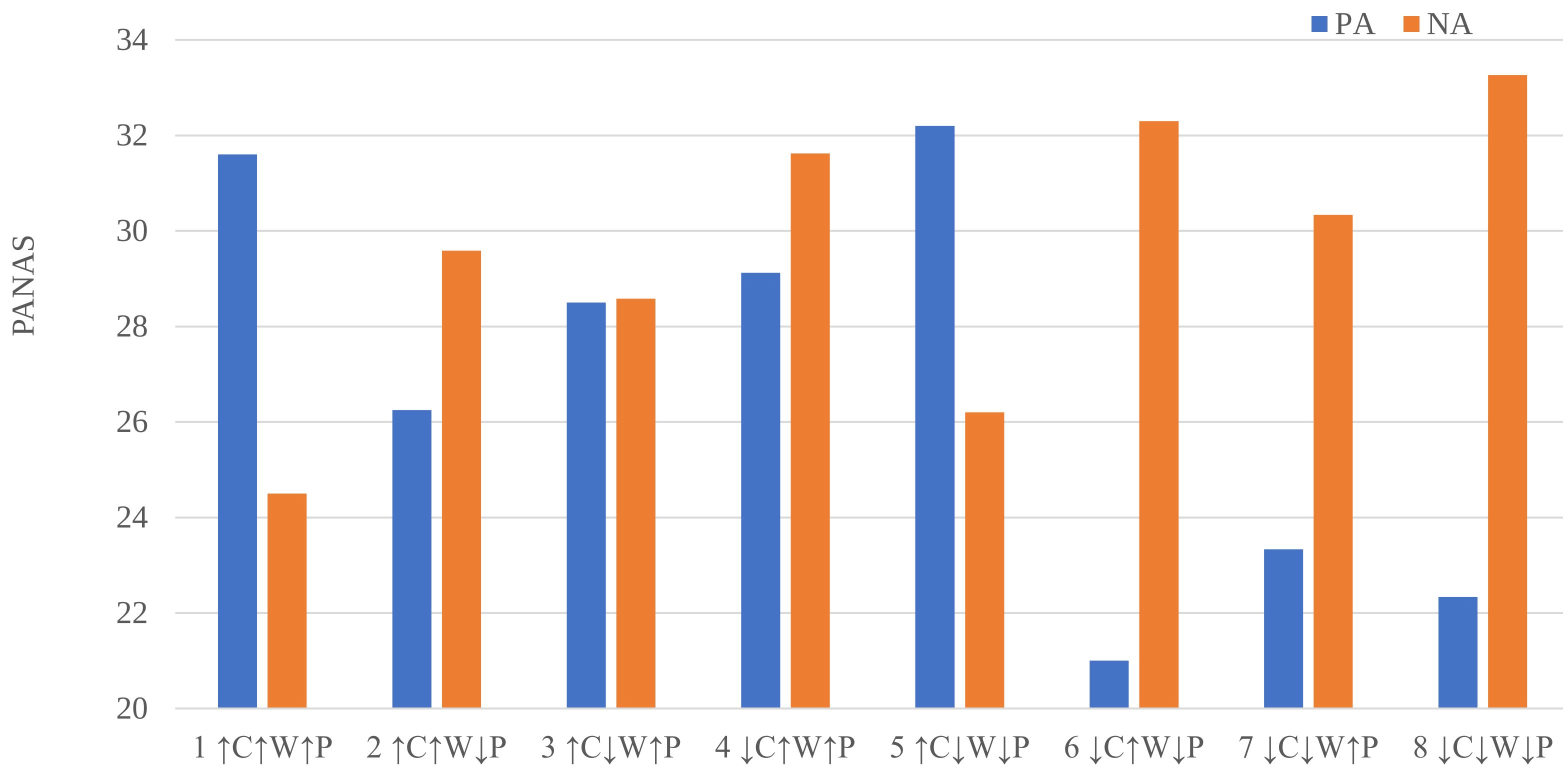

Figure 8. Baseline PANAS by MMC

Table 19. Outcomes by classes (ESM)

MMC	E-PA	E-NA
1 ↑C↑W↑P	29.87	27.34
2 ↑C↑W↓P	29.46	28.89
3 ↑C↓W↑P	29.86	28.38
4 ↓C↑W↑P	29.25	28.21
5 ↑C↓W↓P	29.21	28.55
6 ↓C↑W↓P	22.52	31.84
7 ↓C↓W↑P	28.24	29.16
8 ↓C↓W↓P	21.57	32.96

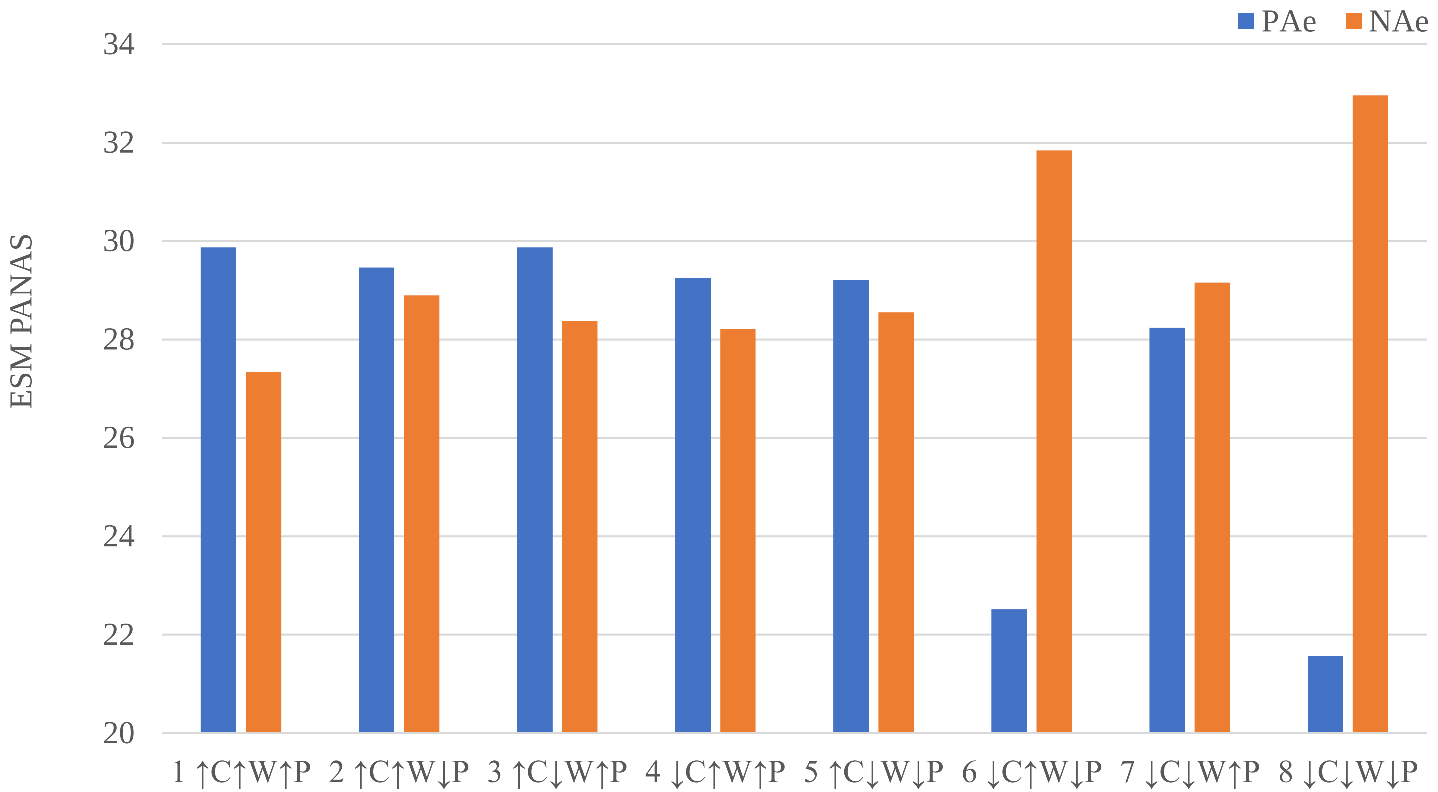

Figure 9. ESM PANAS by MMC

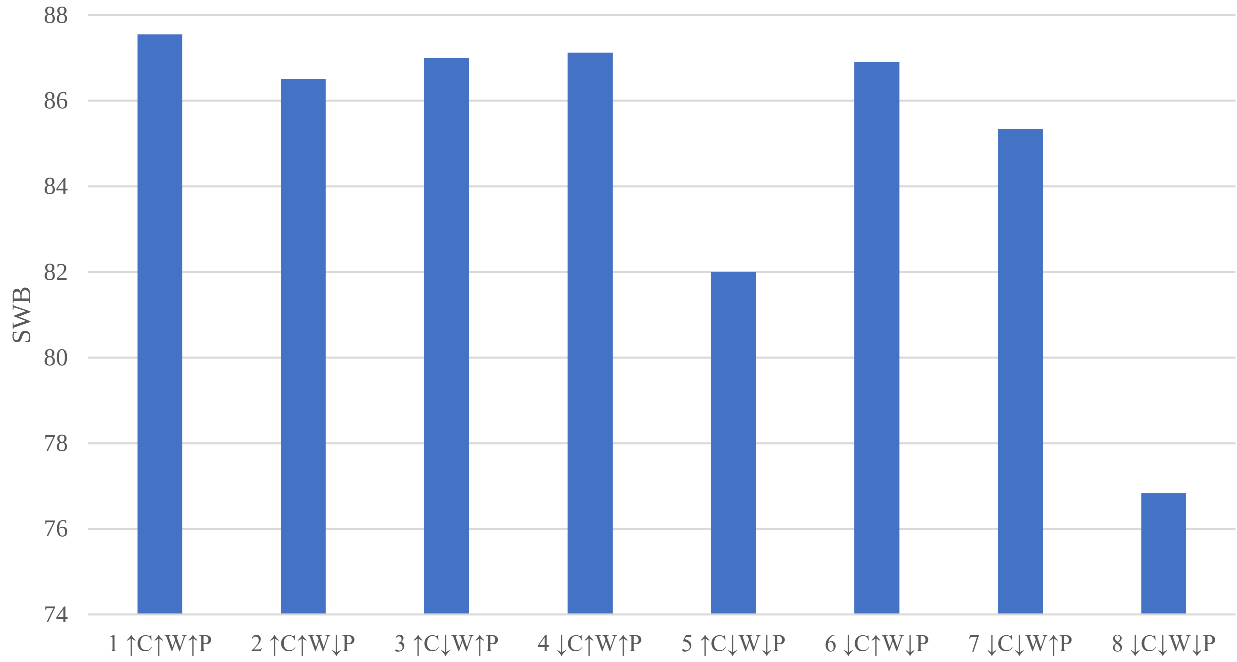

Figure 10. SWB by MMC

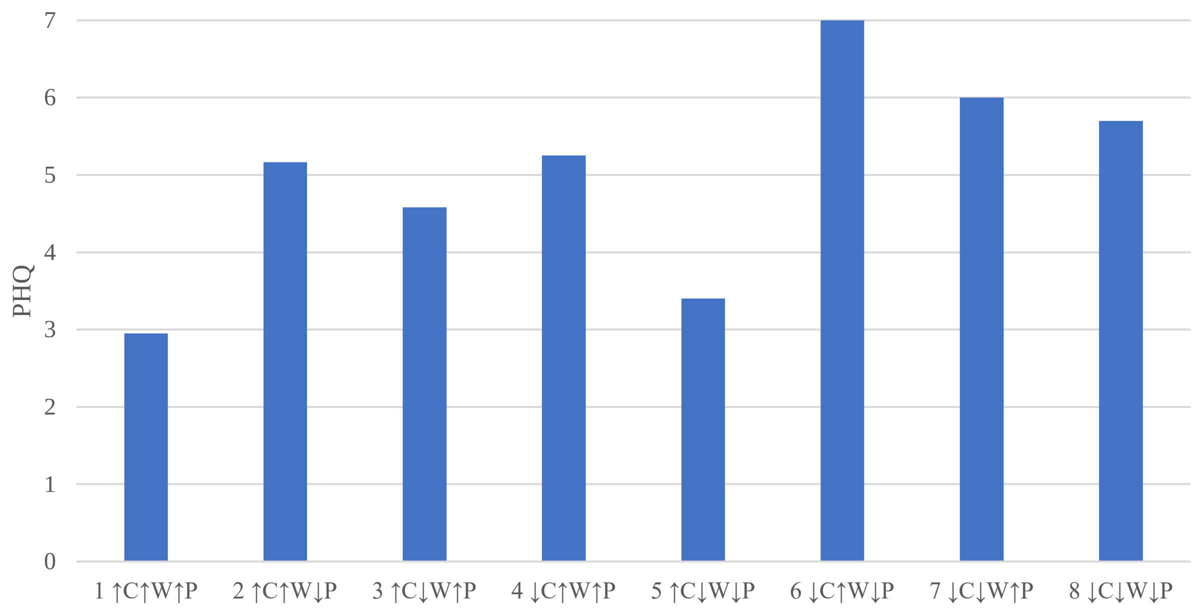

Figure 11. PHQ by MMC

6.5 Well-being by MMC classes

Our findings indicated that flourishing caregivers generally had better well-being (**Table 20**). In terms of positive affectivity, neutral caregivers with the highest average score ($M = 29.2$, $SD = 6.97$) and flourishing caregiver with the second highest ($M = 27.9$, $SD = 5.42$) were both significantly higher than that of languishing caregivers ($M = 24.7$, $SD = 7.10$, $p < 0.001$). Flourishing caregivers had the lowest negative affectivity ($M = 28.0$, $SD = 5.97$, $p < 0.001$) in comparison with neutral caregivers ($M = 33.8$, $SD = 6.08$) and languishing caregivers ($M = 30.9$, $SD = 6.14$). In terms of social well-being, neutral caregivers had the highest social well-being ($M = 87.3$, $SD = 10.1$) followed by flourishing caregivers ($M = 86.1$, $SD = 8.28$), both of which were significantly higher than languishing caregivers ($M = 81.3$, $SD = 9.87$, $p < 0.001$). Depressive symptoms had no differences between classes of care meanings ($p = 0.227$). During the ESM phase, flourishing caregivers in general had the highest positive affectivity ($M = 29.0$, $SD = 2.82$, $p < 0.001$) and the lowest negative affectivity ($M = 28.7$, $SD = 1.93$, $p = 0.003$). As shown in **Table 21**, between classes of the meaning-making choice No. 1 to 4 (flourishing and neutral caregivers) and the meaning-making choice No. 5 to 8 (languishing caregivers), the former class on average had better status in positive affectivity ($M = 28.1$, $SD = 5.56$, $p = 0.009$), negative affectivity ($M = 28.7$, $SD = 6.23$, $p = 0.072$), social well-being ($M = 86.2$, $SD = 8.42$, $p = 0.008$). There was no significant difference on depressive symptoms between two classes ($p = 0.543$). During the ESM phase, the former class had a higher positive affectivity ($M = 28.8$, $SD = 3.40$, $p < 0.001$), and a lower negative affectivity ($M = 28.8$, $SD = 2.27$, $p = 0.003$).

Between high and low work meaning classes, high work meaning group had a higher level of positive affectivity ($M = 27.8$, $SD = 6.78$, $p = 0.009$) than low work meaning class ($M = 24.9$, $SD = 6.10$) (**Table 22**). By the same token, high work meaning class had a lower level of negative affectivity ($M = 28.4$, $SD = 7.11$, $p = 0.072$) than low work meaning group ($M = 31.3$, $SD = 4.94$). Social well-being was scored higher in high work meaning group ($M = 87.1$, $SD = 7.92$) than the low work meaning group ($M = 80.3$, $SD = 9.75$). There was no significant difference between two groups on depressive symptoms. During the ESM phase, high work meaning class reported a higher positive affectivity ($M = 27.9$, $SD = 4.20$) and a lower negative positivity ($M = 29.1$, $SD = 2.72$) than low work meaning class, respectively ($M_{PA} = 25.0$, $SD_{PA} = 4.15$, $p < 0.001$; $M_{NA} = 31.0$, $SD_{NA} = 3.30$, $p = 0.003$).

Table 20. Well-being by classes of care meanings (flourishing – the meaning-making choice No. 1-3, neutral – the meaning-making choice No. 4, and languishing caregivers – the meaning-making choice No. 5-8)

Well-being	Flourishing (N = 43)	Neutral (N = 6)	Languishing (N = 51)	*p*
Baseline				
PA [10-50]				**< 0.001**
Mean (SD)	27.9 (5.42)	**29.2 (6.97)**	24.7 (7.10)	
Median [Min, Max]	27 [15, 36]	28.5 [22, 41]	24 [13, 40]	
NA [50-10]				**< 0.001**
Mean (SD)	**28.0 (5.97)**	33.8 (6.08)	30.9 (6.14)	
Median [Min, Max]	28 [14, 39]	35 [26, 41]	32 [10, 39]	
SWB [20-140]				**< 0.001**
Mean (SD)	86.1 (8.28)	**87.3 (10.1)**	81.3 (9.87)	
Median [Min, Max]	87 [69, 103]	93 [72, 95]	83 [57, 103]	
PHQ [27-0]				0.227
Mean (SD)	4.63 (2.39)	6 (1.26)	5.08 (2.31)	
Median [Min, Max]	4 [0, 10]	6.50 [4, 7]	5 [0, 10]	
ESM				
PA ESM [10-50]				**< 0.001**
Mean (SD)	**29.0 (2.82)**	27.4 (6.45)	24.2 (4.12)	
Median [Min, Max]	29.6 [20.3, 32.4]	29.3 [19.2, 35.6]	22.4 [17.6, 31.8]	
NA ESM [10-50]				**0.003**
Mean (SD)	**28.7 (1.93)**	29.7 (4.15)	31.3 (3.42)	
Median [Min, Max]	28.4 [23.4, 33.4]	28.2 [25.7, 36.8]	31.8 [19.4, 37.8]	

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range. "PA ESM" and "NA ESM" indicate the positive and negative affectivities measured during the ESM phase.

Table 21. Well-being by classes of care meanings (flourishing and neutral – the meaning-making choice No. 1- 4, and languishing caregivers – the meaning-making choice No. 5-8)

Well-being	**Flourishing & Neutral (N = 49)**	**Languishing (N = 51)**	***p***
Baseline			
PA [10-50]			**0.009**
Mean (SD)	**28.1 (5.56)**	24.7 (7.10)	
Median [Min, Max]	27 [15, 41]	24 [13, 40]	
NA [50-10]			**0.072**
Mean (SD)	**28.7 (6.23)**	30.9 (6.14)	
Median [Min, Max]	28 [14, 41]	32 [10, 39]	
SWB [20-140]			**0.008**
Mean (SD)	**86.2 (8.42)**	81.3 (9.87)	
Median [Min, Max]	88 [69, 103]	83 [57, 103]	
PHQ [27-0]			0.543
Mean (SD)	4.80 (2.32)	5.08 (2.31)	
Median [Min, Max]	5 [0, 10]	5 [0, 10]	
ESM			
PA ESM [10-50]			**< 0.001**
Mean (SD)	**28.8 (3.40)**	24.2 (4.12)	
Median [Min, Max]	29.6 [19.2, 35.6]	22.4 [17.6, 31.8]	
NA ESM [10-50]			**0.003**
Mean (SD)	**28.8 (2.27)**	31.3 (3.42)	
Median [Min, Max]	28.4 [23.4, 36.8]	31.8 [19.4, 37.8]	

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range. "PA ESM" and "NA ESM" indicate the positive and negative affectivities measured during the ESM phase.

Table 22. Well-being by classes of work meanings (low work meaning – the meaning-making choice No. 3, 5, 7 and 8, high work meaning – the meaning-making choice No. 1, 2, 4 and 6)

Well-being	**High Work Meaning (N = 50)**	**Low Work Meaning (N = 50)**	***p***
Baseline			
PA [10-50]			**0.009**
Mean (SD)	**27.8 (6.78)**	24.9 (6.10)	
Median [Min, Max]	27 [14, 41]	24.5 [13, 36]	
NA [50-10]			**0.072**
Mean (SD)	**28.4 (7.11)**	31.3 (4.94)	
Median [Min, Max]	28.5 [10, 41]	32 [17, 39]	
SWB [20-140]			**0.008**
Mean (SD)	**87.1 (7.92)**	80.3 (9.75)	
Median [Min, Max]	88 [71, 103]	80.5 [57, 103]	
PHQ [27-0]			0.543
Mean (SD)	4.66 (2.52)	5.22 (2.06)	
Median [Min, Max]	4.50 [0, 10]	5 [2, 10]	
ESM			
PA ESM [10-50]			**< 0.001**
Mean (SD)	**27.9 (4.20)**	25.0 (4.15)	
Median [Min, Max]	29.6 [19.2, 35.6]	23.5 [17.6, 31.6]	
NA ESM [10-50]			**0.003**
Mean (SD)	**29.1 (2.72)**	31.0 (3.30)	
Median [Min, Max]	28.5 [25.6, 37.1]	31.4 [19.4, 37.8]	

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range.

6.6 Hypotheses Testing Meaning-making Choices and Well-being

6.6.1 H1.1: participants with MMC 1 to 3 are flourishing caregivers

The first class, flourishing caregivers, included the meaning-making choices of 1 to 3 referring to high meaning-making levels of at least two aspects. For instance, high meaning-making levels of care and work, or high meaning-making levels of care and life. Also, high meaning-making levels of all three aspects are counted as the first category.

According to **Table 23**, a total number of 43 participants who made the meaning-making choice No. 1 to 3 were flourishing caregivers reporting higher positive affectivity and lower negative affectivity. Among identified flourishing caregivers, participants who made the meaning-making choice No. 1 took the share of 46.5%, the meaning-making choice No. 2 took the share of 27.9%, and the meaning-making choice No. 3 took the share of 25.6%. Overall, these flourishing caregivers took 43% of the total population.

In comparison to the rest ($n = 57$), flourishing caregivers on the baseline reported a higher level of positive affectivity ($M = 27.9$, $SD = 5.42$, $p < 0.001$), and social well-being ($M = 86.1$, $SD = 8.28$, $p < 0.001$); and a lower level of negative affectivity ($M = 28.0$, $SD = 5.97$, $p < 0.001$) (**Table 24**). During the ESM period, flourishing caregivers reported a higher level of positive affectivity ($M = 29.0$, $SD = 2.82$, $p < 0.001$) and a lower level of negative affectivity ($M = 28.7$, $SD = 1.93$, $p = 0.003$).

Table 23. MMC 1 to 3 and classes

	MMC	Flourishing (N = 43)	Neutral (N = 6)	Languishing (N = 51)
1	**↑C↑W↑P**	**20 (46.5%)**		
2	**↑C↑W↓P**	**13 (30.2%)**		
3	**↑C↓W↑P**	**10 (23.3%)**		
4	↓C↑W↑P		6 (100%)	
5	↑C↓W↓P			5 (9.8%)
6	↓C↑W↓P			11 (21.6%)
7	↓C↓W↑P			5 (9.8%)
8	↓C↓W↓P			30 (58.8%)

Table 24. Well-being of MMC 1 to 3

Well-being	Flourishing (N = 43)	Others (N = 57)	*p*
Baseline			
PA [10-50]	**27.9 (5.42)**	24.9 (7.02)	**< 0.001**
NA [50-10]	**28.0 (5.97)**	31.3 (6.13)	**< 0.001**
SWB [20-140]	**86.1 (8.28)**	81.9 (9.91)	**< 0.001**
PHQ [27-0]	4.63 (2.39)	5.13 (2.11)	0.227
ESM			
PA ESM [10-50]	**29.0 (2.82)**	24.4 (4.75)	**< 0.001**
NA ESM [10-50]	**28.7 (1.93)**	31.5 (3.65)	**0.003**

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range. "PA ESM" and "NA ESM" indicate the positive and negative affectivities measured during the ESM phase.

6.6.2 H1.2: participants with MMC 4 are neutral caregivers

The second class, neutral caregivers, included the meaning-making choices of 4, referring to high meaning-making levels of work and life and a low meaning-making level of care.

According to **Table 25**, a total number of 6 neutral participants who made the meaning-making choice No. 4 reported higher positive affectivity and lower negative affectivity. Overall, these neutral caregivers took 6% of the total population. In comparison to the rest ($n = 94$), neutral caregivers on the baseline reported a higher level of positive affectivity ($M = 29.2$, $SD = 6.97$, $p < 0.001$), and social well-being ($M = 87.3$, $SD = 10.1$, $p < 0.001$); and a higher level of negative affectivity ($M = 33.8$, $SD = 6.08$, $p < 0.001$). During the ESM period, neutral caregivers reported a higher level of positive affectivity ($M = 27.4$, $SD = 6.45$, $p < 0.001$) and a lower level of negative affectivity ($M = 29.7$, $SD = 4.15$, $p = 0.003$) (**Table 26**). The between-group comparison on the baseline lacked sufficient explanatory power due to its limited sample size ($n = 6$), leading to a questionable conclusion. On the other hand, ESM data collecting 38 data points for each individual had a total number of 228 data points to sufficiently make the comparison.

Table 25. MMC 4 and classes

	MMC	Flourishing (N = 43)	Neutral (N = 6)	Languishing (N = 51)
1	↑C↑W↑P	20 (46.5%)		
2	↑C↑W↓P	13 (30.2%)		
3	↑C↓W↑P	10 (23.3%)		
4	**↓C↑W↑P**		**6 (100%)**	
5	↑C↓W↓P			5 (9.8%)
6	↓C↑W↓P			11 (21.6%)
7	↓C↓W↑P			5 (9.8%)
8	↓C↓W↓P			30 (58.8%)

Table 26. Well-being of MMC 4

Well-being	Neutral (N = 6)	Others (N = 94)	*p*
Baseline			
PA [10-50]	**29.2 (6.97)**	25.9 (6.12)	**< 0.001**
NA [50-10]	33.8 (6.08)	**29.5 (6.22)**	**< 0.001**
SWB [20-140]	**87.3 (10.1)**	83.5 (9.17)	**< 0.001**
PHQ [27-0]	6 (1.26)	4.92 (2.32)	0.227
ESM			
PA ESM [10-50]	**27.4 (6.45)**	26.4 (3.25)	**< 0.001**
NA ESM [10-50]	**29.7 (4.15)**	37.5 (3.35)	**0.003**

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range. "PA ESM" and "NA ESM" indicate the positive and negative affectivities measured during the ESM phase.

6.6.3 H1.3: participants with MMC 5 to 8 are languishing caregivers

The third class, languishing caregivers, included the meaning-making choices of 5 to 8, referring to low meaning-making levels of at least two aspects. For instance, low meaning-making levels of care and work, or low meaning-making levels of care and life. Also, low meaning-making levels of all three aspects are counted as the third category.

According to **Table 27**, among the total number of 51 languishing participants who made the meaning-making choice No. 5 to 8, 11.8% ($n = 6$) of them made the meaning-making choice No. 5, 19.6% ($n = 10$) made the meaning-making choice No. 6, 9.8% ($n = 5$) made the meaning-making choice No. 7, and 58.8% ($n = 30$) made the meaning-making choice No. 8. Languishing caregivers took 51% of the total population.

In comparison to the rest ($n = 49$), languishing caregivers on the baseline reported a lower level of positive affectivity ($M = 24.7$, $SD = 7.10$, $p < 0.001$), and social well-being ($M = 81.3$, $SD = 9.87$, $p < 0.001$); and a higher level of negative affectivity ($M = 30.9$, $SD = 6.14$, $p < 0.001$). During the ESM period, languishing caregivers reported a lower level of positive affectivity ($M = 24.2$, $SD = 4.12$, $p < 0.001$) and a higher level of negative affectivity ($M = 31.3$, $SD = 3.42$, $p = 0.003$) (**Table 28**).

Table 27. MMC 5 to 8 and classes

	MMC	Flourishing (N = 43)	Neutral (N = 6)	Languishing (N = 51)
1	↑C↑W↑P	20 (46.5%)		
2	↑C↑W↓P	13 (30.2%)		
3	↑C↓W↑P	10 (23.3%)		
4	↓C↑W↑P		6 (100%)	
5	**↑C↓W↓P**			**5 (9.8%)**
6	**↓C↑W↓P**			**11 (21.6%)**
7	**↓C↓W↑P**			**5 (9.8%)**
8	**↓C↓W↓P**			**30 (58.8%)**

Table 28. Well-being of MMC 5 to 8

Well-being	Languishing (N = 51)	Others (N = 49)	*p*
Baseline			
PA [10-50]	24.7 (7.10)	**28.1 (6.11)**	**< 0.001**
NA [50-10]	30.9 (6.14)	**28.6 (6.23)**	**< 0.001**
SWB [20-140]	81.3 (9.87)	**86.2 (9.33)**	**< 0.001**
PHQ [27-0]	5.08 (2.31)	4.79 (2.21)	0.227
ESM			
PA ESM [10-50]	24.2 (4.12)	**28.5 (4.66)**	**< 0.001**
NA ESM [10-50]	31.3 (3.42)	**29.6 (3.08)**	**0.003**

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range. "PA ESM" and "NA ESM" indicate the positive and negative affectivities measured during the ESM phase.

6.6.4 H2.1: participants with MMC 1 to 4 have higher PA than MMC 5 to 8

Classes of flourishing caregivers and neutral caregivers were combined to respond to this hypothesis.

According to **Table 29**, among the total number of 43 flourishing and 6 neutral participants who made the meaning-making choice No. 1 to 4, they reported higher positive affectivity and lower negative affectivity. Among identified 49 caregivers, participants who made the meaning-making choice No. 1 took the share of 40.8%, the meaning-making choice No. 2 took the share of 24.5%, the meaning-making choice No. 3 took the share of 22.4%, and the meaning-making choice No. 4 took the share of 12.2%. Overall, these flourishing caregivers took 49% of the total population.

In comparison to the rest ($n = 51$), flourishing and neutral caregivers on the baseline reported a higher level of positive affectivity ($M = 28.1$, $SD = 5.56$, $p = 0.009$), and social well-being ($M = 86.2$, $SD = 8.42$, $p = 0.008$); and a lower level of negative affectivity ($M = 28.7$, $SD = 6.23$, $p = 0.072$). During the ESM period, flourishing caregivers reported a higher level of positive affectivity ($M = 28.8$, $SD = 3.40$, $p < 0.001$) and a higher level of negative affectivity ($M = 28.8$, $SD = 2.27$, $p = 0.003$) (**Table 30**).

Qualitative interview indicated that higher meaning levels in multiple dimensions of care, work and personal life resulted in better well-being. In other words, flourishing caregivers with meaningful daily activities, taking care and work for instance, reported better well-being (**Appendix V**). According to the narratives, higher levels of care, work, or personal life lead to better well-being among Chinese working dementia caregivers.

Table 29. MMC 1 to 4 and classes

	MMC	Flourishing+Neutral (N = 49)	Languishing (N = 51)
1	**↑C↑W↑P**	**20 (40.8%)**	
2	**↑C↑W↓P**	**13 (26.5%)**	
3	**↑C↓W↑P**	**11 (22.4%)**	
4	**↓C↑W↑P**	**6 (12.2%)**	
5	↑C↓W↓P		5 (9.8%)
6	↓C↑W↓P		11 (21.6%)
7	↓C↓W↑P		5 (9.8%)
8	↓C↓W↓P		30 (58.8%)

Table 30. Well-being of MMC 1 to 4

Well-being	Overall (N = 100)	Flourishing & Neutral (N = 49)	Languishing (N = 51)	*p*
Baseline				
PA [10-50]	26.3 (6.58)	**28.1 (5.56)**	24.7 (7.10)	**0.009**
NA [50-10]	29.8 (6.26)	**28.7 (6.23)**	30.9 (6.14)	**0.072**
SWB [20-140]	83.7 (9.48)	**86.2 (8.42)**	81.3 (9.87)	**0.008**
PHQ [27-0]	4.94 (2.31)	4.80 (2.32)	5.08 (2.31)	0.543
ESM				
PA ESM [10-50]	26.5 (4.41)	**28.8 (3.40)**	24.2 (4.12)	**< 0.001**
NA ESM [10-50]	30.1 (3.15)	**28.8 (2.27)**	31.3 (3.42)	**0.003**

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range. "PA ESM" and "NA ESM" indicate the positive and negative affectivities measured during the ESM phase.

6.6.5 H2.2: participants with MMC 1 to 3 have lower NA than MMC 4 to 8

According to **Table 31**, caregivers with the meaning-making choice No. 1 to 3 on the baseline reported a lower level of negative affectivity (M = 28.0, SD = 5.97, p < 0.001) in comparison with others (M = 31.3, SD = 6.13) (**Table 24**). During the ESM period, flourishing caregivers reported a lower level of negative affectivity (M = 28.7, SD = 1.93, p = 0.003) than the others (M = 31.5, SD = 3.65).

Qualitative interviews indicated that no meaning assigned to daily living events resulted in worse well-being. In other words, languishing caregivers holding meaningless attitudes towards daily activities, taking care and work for instance, reported worse well-being than their counterparts with higher meaning levels (**Appendix V**).

Table 31. NA of MMC 1 to 3

Well-being	**Flourishing (N = 43)**	**Others (N = 57)**	***p***
Baseline NA	**28.0 (5.97)**	31.3 (6.13)	**< 0.001**
ESM NA	**28.7 (1.93)**	31.5 (3.65)	**0.003**

6.6.6 H2.3: participants with MMC of 1, 2, 4 and 6 have higher SWB than MMC 3, 5, 7 and 8

According to **Table 32**, among the total number of 50 participants who made the meaning-making choice No. 1, 2, 4 and 6, all of them reported high work meanings. Among identified 50 caregivers, participants who made the meaning-making choice No. 1 took the share of 40%, the meaning-making choice No. 2 took the share of 24%, the meaning-making choice No. 3 took the share of 16%, and the meaning-making choice No. 4 took the share of 20%. Overall, these flourishing caregivers took up 50% of the total population.

In comparison to the rest (n = 50), caregivers with a high work meaning on the baseline reported a higher level of positive affectivity (M = 27.8, SD = 6.78, p = 0.009), and social well-being (M = 87.1, SD = 7.92, p = 0.008); and a lower level of negative affectivity (M = 28.4, SD = 7.11, p < 0.072). During the ESM period, languishing caregivers reported a higher level of positive affectivity (M = 27.9, SD = 4.20, p < 0.001) and a lower level of negative affectivity (M = 29.1, SD = 2.72, p = 0.003) (**Table 33**).

Table 32. MMC 1, 2, 4 and 6 and classes

	MMC	High Work Meaning (N = 50)	Low Work Meaning (N = 50)
1	**↑C↑W↑P**	**20 (40%)**	
2	**↑C↑W↓P**	**13 (26%)**	
4	**↓C↑W↑P**	**6 (12%)**	
6	**↓C↑W↓P**	**11 (22%)**	
3	↑C↓W↑P		10 (20%)
5	↑C↓W↓P		5 (10%)
7	↓C↓W↑P		5 (10%)
8	↓C↓W↓P		30 (100%)

Table 33. Well-being of MMC 1, 2, 4 and 6

Well-being	High Work Meaning (N = 50)	Low Work Meaning (N = 50)	*p*
Baseline			
PA [10-50]			**0.009**
Mean (SD)	**27.8 (6.78)**	24.9 (6.10)	
Median [Min, Max]	27 [14, 41]	24.5 [13, 36]	
NA [50-10]			**0.072**
Mean (SD)	**28.4 (7.11)**	31.3 (4.94)	
Median [Min, Max]	28.5 [10, 41]	32 [17, 39]	
SWB [20-140]			**0.008**
Mean (SD)	**87.1 (7.92)**	80.3 (9.75)	
Median [Min, Max]	88 [71, 103]	80.5 [57, 103]	
PHQ [27-0]			0.543
Mean (SD)	4.66 (2.52)	5.22 (2.06)	
Median [Min, Max]	4.50 [0, 10]	5 [2, 10]	
ESM			
PA ESM [10-50]			**< 0.001**
Mean (SD)	**27.9 (4.20)**	25.0 (4.15)	
Median [Min, Max]	29.6 [19.2, 35.6]	23.5 [17.6, 31.6]	
NA ESM [10-50]			**0.003**
Mean (SD)	**29.1 (2.72)**	31.0 (3.30)	
Median [Min, Max]	28.5 [25.6, 37.1]	31.4 [19.4, 37.8]	

Note: square brackets after each variable's name denoted in "[]" indicate the lowest/worst condition to the highest/best condition in its range. "PA ESM" and "NA ESM" indicate the positive and negative affectivities measured during the ESM phase.

6.6.7 H3.1: MMC 1 to 4 predict improved PA than MMC 5 to 8

As shown in **Table 34**, in the path A of the model, one's meaning-making choice to be flourishing was predicted by non-religious belief marginally (B = 0.496, p = 0.098) and better financial status (B = -0.350, p = 0.001). The path A was statistically robust (R^2 = 0.275, p < 0.001), while caregiving stress significantly predicted meaning-making choice of 1 to 4 (B = 0.046, p = 0.048). In the path B of the model, one's positive affectivity was predicted by worse financial status (B = -0.199, p = 0.072), living arrangement with family members (B = 0.437, p = 0.060). After controlling for several covariates (age, gender, education, religion, financial status, living arrangement, and length of care), the meaning-making choice of 1 to 4 insignificantly predicted improved positive affectivity (B = 0.172, p = 0.114) while the caregiving stress insignificantly predicted improved positive affectivity (B = -0.036, p = 0.160). The mediation model was not statistically robust (R^2 = 0.111, p = 0.158).

As shown in **Table 35** using ESM PANAS, in comparison, in the path A of the model, one's meaning-making choice to be flourishing was predicted by better financial status (B = -0.276, p = 0.213), and less caregiving burden (B = 0.046, p = 0.048). In the path B of the model, one's positive affectivity was predicted by younger in age (B = -0.087, p = 0.048), better financial status (B = 1.524, p < 0.001). After controlling for several covariates (age, gender, education, religion, financial status, living arrangement, and length of care), the meaning-making choice of 1 to 4 significantly predicted improved ESM positive affectivity (B = -1.483, p = 0.001) while the caregiving stress significantly predicted ESM improved positive affectivity (B = -0.306, p = 0.057). The mediation model was statistically robust for both path A (R^2 = 0.275, p < 0.001) and path B (R^2 = 0.461, p < 0.001).

Qualitative interviews indicated that higher meaning levels in multiple dimensions of care, work and personal life resulted in improved well-being (**Appendix V**). Narratives implied that by making sense of their caregiving experiences as well as other aspects of daily life, WDCs can have enhanced experiences with more positive aspects and attitudes towards the stressful experiences.

Table 34. Mediation model in prediction of PA

	B	S.E.	t	p	LLCI	ULCI
ZBI→MC						
constant	2.775	0.896	3.096	**0.003**	0.995	4.556
Age	0.006	0.011	0.537	0.593	-0.016	0.028
Gender	0.302	0.201	1.503	0.136	-0.097	0.700
Education attainment	-0.224	0.154	-1.453	0.150	-0.529	0.082
Religion	0.496	0.296	1.672	**0.098**	-0.093	1.085
Financial status	-0.350	0.100	-3.517	**0.001**	-0.548	-0.152
Living arrangement	-0.276	0.220	-1.255	0.213	-0.713	0.161
Caregiving length	-0.001	0.003	-0.213	0.832	-0.007	0.006
Caregiving stress	0.046	0.022	2.091	**0.048**	-0.026	0.071
MC→PA						
constant	1.933	0.973	1.987	0.050	0.001	3.865
Age	-0.003	0.011	-0.300	0.765	-0.026	0.019
Gender	0.016	0.210	0.074	0.941	-0.401	0.432
Education attainment	0.084	0.161	0.521	0.604	-0.235	0.403
Religion	-0.373	0.311	-1.199	0.234	-0.990	0.245
Financial status	-0.199	0.110	-1.819	**0.072**	-0.417	0.018
Living arrangement	0.437	0.229	1.907	**0.060**	-0.018	0.891
Caregiving length	0.003	0.003	0.903	0.369	-0.004	0.010
Caregiving stress	-0.036	0.025	-1.417	0.160	-0.086	0.014
MC	0.172	0.108	1.594	0.114	-0.042	0.387
	R^2	**MSE**	**F**	**p**		
ZBI→MC	0.275	0.744	4.307	**< 0.001**		
MC→PA	0.131	0.792	1.505	0.158		

Table 35. Mediation model in prediction of ESM PA

	B	S.E.	t	p	LLCI	ULCI
ZBI→MC						
constant	2.775	0.896	3.096	**0.003**	0.995	4.556
Age	0.006	0.011	0.537	0.593	-0.016	0.028
Gender	0.302	0.201	1.503	0.136	-0.097	0.700
Education attainment	-0.224	0.154	-1.453	0.150	-0.529	0.082
Religion	0.496	0.296	1.672	0.098	-0.093	1.085
Financial status	-0.350	0.100	-3.517	**0.001**	-0.548	-0.152
Living arrangement	-0.276	0.220	-1.255	0.213	-0.713	0.161
Caregiving length	-0.001	0.003	-0.213	0.832	-0.007	0.006
Caregiving stress	0.046	0.022	2.091	**0.048**	-0.026	0.071
MC→PA (ESM)						
constant	28.898	3.711	7.788	< **0.001**	21.527	36.270
Age	-0.087	0.043	-2.002	**0.048**	-0.174	-0.001
Gender	-0.206	0.800	-0.258	0.797	-1.795	1.382
Education attainment	0.018	0.613	0.029	0.977	-1.199	1.235
Religion	0.742	1.185	0.626	0.533	-1.612	3.096
Financial status	1.524	0.418	3.648	< **0.001**	0.694	2.355
Living arrangement	0.638	0.873	0.731	0.467	-1.097	2.373
Caregiving length	-0.011	0.013	-0.845	0.400	-0.037	0.015
Caregiving stress	-0.306	0.157	-1.949	**0.057**	-0.698	0.086
MC	-1.483	0.413	-3.594	**0.001**	-2.303	-0.663
	R^2	**MSE**	**F**	**p**		
ZBI→MC	0.275	0.744	4.307	< **0.001**		
MC→PA (ESM)	0.461	11.535	8.563	< **0.001**		

6.6.8 H3.2: MMC 1 to 3 predict declined NA than MMC 4 to 8

As shown in **Table 36**, in the path A of the model, one's meaning-making choice of 1 to 3 was predicted by non-religious belief marginally (B = 0.496, p = 0.098), and better financial status (B = -0.350, p = 0.001). The path A was statistically robust (R^2 = 0.275, p < 0.001), while caregiving stress significantly predicted meaning-making choice of 5 to 8 (B = 0.022, p = 0.368). In the path B of the model, one's negative affectivity was predicted by non-religious belief (B = -0.798, p = 0.034). After controlling for several covariates (age, gender, education, religion, financial status, living arrangement, and length of care), the meaning-making choice of 1 to 3 insignificantly predicted worsened negative affectivity (B = 0.412, p = 0.527) while the caregiving stress insignificantly predicted worsened negative affectivity (B = -0.044, p = 0.149). The mediation model was not statistically robust (R^2 = 0.145, p = 0.103).

As shown in **Table 37** using ESM PANAS, in comparison, in the path A of the model, one's meaning-making choice of 1 to 3 was predicted by non-religious belief marginally (B = 0.496, p = 0.098), better financial status (B = -0.350, p = 0.001), and more caregiving stress (B = 0.046, p = 0.048). In the path B of the model, one's ESM negative affectivity was predicted by worse financial status (B = -0.931, p = 0.004) and longer caregiving length (B = 0.021, p = 0.004). After controlling for several covariates (age, gender, education, religion, financial status, living arrangement, and length of care), the meaning-making choice of 1 to 3 significantly predicted worsened ESM negative affectivity (B = 0.592, p = 0.060) while the caregiving stress significantly predicted worsened ESM negative affectivity (B = 0.444, p = 0.004). The mediation model was statistically robust for both path A (R^2 = 0.275, p < 0.001) and path B (R^2 = 0.400, p < 0.001).

Qualitative interview indicated that lower meaning levels led to compromised well-being (**Appendix V**). Narratives implied that by making sense of their caregiving experiences as well as other aspects of daily life, WDCs can have enhanced experiences with more positive aspects and attitudes towards the stressful experiences.

Table 36. Mediation model in prediction of NA

	B	S.E.	t	p	LLCI	ULCI
ZBI→MC						
constant	2.775	0.896	3.096	**0.003**	0.995	4.556
Age	0.006	0.011	0.537	0.593	-0.016	0.028
Gender	0.302	0.201	1.503	0.136	-0.097	0.700
Education attainment	-0.224	0.154	-1.453	0.150	-0.529	0.082
Religion	0.496	0.296	1.672	**0.098**	-0.093	1.085
Financial status	-0.350	0.100	-3.517	**0.001**	-0.548	-0.152
Living arrangement	-0.276	0.220	-1.255	0.213	-0.713	0.161
Caregiving length	-0.001	0.003	-0.213	0.832	-0.007	0.006
Caregiving stress	0.046	0.022	2.091	**0.048**	-0.026	0.071
MC→NA						
constant	2.636	1.162	2.269	**0.026**	0.328	4.944
Age	-0.006	0.014	-0.466	0.642	-0.033	0.021
Gender	-0.110	0.250	-0.441	0.660	-0.608	0.387
Education attainment	-0.067	0.192	-0.347	0.730	-0.448	0.315
Religion	-0.798	0.371	-2.150	**0.034**	-1.535	-0.060
Financial status	-0.119	0.131	-0.910	0.365	-0.379	0.141
Living arrangement	0.112	0.273	0.408	0.684	-0.432	0.655
Caregiving length	0.003	0.004	0.834	0.406	-0.005	0.012
Caregiving stress	-0.044	0.030	-1.456	0.149	-0.104	0.016
MC	0.412	0.649	0.635	0.527	-0.878	1.702
	R^2	**MSE**	**F**	**p**		
ZBI→MC	0.275	0.744	4.307	**< 0.001**		
MC→NA	0.145	1.131	1.691	0.103		

Table 37. Mediation model in prediction of ESM NA

	B	S.E.	t	p	LLCI	ULCI
ZBI→MC						
constant	2.775	0.896	3.096	**0.003**	0.995	4.556
Age	0.006	0.011	0.537	0.593	-0.016	0.028
Gender	0.302	0.201	1.503	0.136	-0.097	0.700
Education attainment	-0.224	0.154	-1.453	0.150	-0.529	0.082
Religion	0.496	0.296	1.672	**0.098**	-0.093	1.085
Financial status	-0.350	0.100	-3.517	**0.001**	-0.548	-0.152
Living arrangement	-0.276	0.220	-1.255	0.213	-0.713	0.161
Caregiving length	-0.001	0.003	-0.213	0.832	-0.007	0.006
Caregiving stress	0.046	0.022	2.091	**0.048**	-0.026	0.071
MC→NA (ESM)						
constant	30.377	2.793	10.875	< **0.001**	24.827	35.926
Age	0.031	0.033	0.933	0.353	-0.035	0.096
Gender	-0.943	0.602	-1.566	0.121	-2.139	0.253
Education attainment	-0.017	0.461	-0.037	0.971	-0.933	0.899
Religion	1.223	0.892	1.371	0.174	-0.549	2.995
Financial status	-0.931	0.315	-2.958	**0.004**	-1.556	-0.306
Living arrangement	-1.034	0.657	-1.573	0.119	-2.340	0.272
Caregiving length	0.021	0.010	2.151	**0.034**	0.002	0.041
Caregiving stress	0.444	0.152	2.924	**0.004**	0.142	0.746
MC	0.592	0.258	2.295	**0.030**	-0.025	1.209
	R^2	**MSE**	**F**	**p**		
ZBI→MC	0.275	0.744	4.307	< **0.001**		
MC→NA (ESM)	0.400	6.537	6.672	< **0.001**		

6.6.9 H3.3: MMC 1, 2, 4 and 6 predict improved SWB than MMC 3, 5, 7 and 8

As shown in **Table 38**, in path A of the model, one's meaning-making choice of 1, 2, 4 and 6 was predicted by more employment stress (B = -0.375, $p < 0.001$). In the path B of the model, one's social well-being was predicted by younger in age (B = -0.278, p = 0.050), better financial status (B = 4.760, $p < 0.001$). After controlling for several covariates (age, gender, education, religion, financial status, living arrangement, and length of care), the meaning-making choice of 1, 2, 4 and 6 significantly predicted improved social well-being (B = 2.423, p = 0.047) while the employment stress insignificantly predicted improved social well-being (B = 0.017, p = 0.899). The mediation model was statistically robust ($R^2 = 0.401$, $p < 0.001$).

Qualitative interviews indicated that higher meaning levels in work resulted in improved social well-being (**Appendix V**). Narratives implied that considering work as a way to earn more money or achieve self-actualization, WDCs can have enhanced work experiences and motivations with more positive aspects and attitudes towards work stress.

Table 38. Mediation model in prediction of SWB

	B	S.E.	t	p	LLCI	ULCI
OSI→MC						
constant	3.273	0.958	3.416	**0.001**	1.370	5.176
Age	0.006	0.011	0.573	0.568	-0.016	0.029
Gender	0.281	0.202	1.393	0.167	-0.120	0.682
Education attainment	-0.227	0.155	-1.461	0.148	-0.535	0.082
Religion	0.453	0.310	1.462	0.147	-0.162	1.068
Financial status	-0.003	0.011	-0.312	0.756	-0.025	0.018
Living arrangement	-0.244	0.219	-1.115	0.268	-0.679	0.191
Caregiving length	-0.001	0.003	-0.211	0.834	-0.007	0.006
Employment stress	-0.375	0.097	-3.885	< **0.001**	-0.567	-0.183
MC→SWB						
constant	106.446	12.677	8.397	< **0.001**	81.262	131.631
Age	-0.278	0.140	-1.983	**0.050**	-0.556	0.001
Gender	-1.398	2.540	-0.550	0.583	-6.445	3.649
Education attainment	0.278	1.956	0.142	0.887	-3.609	4.165
Religion	-6.425	3.903	-1.646	0.103	-14.178	1.328
Financial status	4.760	1.299	3.664	< **0.001**	2.180	7.341
Living arrangement	-2.385	2.746	-0.868	0.388	-7.840	3.071
Caregiving length	-0.032	0.042	-0.777	0.439	-0.115	0.050
Employment stress	0.017	0.133	0.127	0.899	-0.248	0.281
MC	2.423	1.306	1.855	**0.047**	-5.017	0.172
	R^2	**MSE**	**F**	**p**		
OSI→MC	0.269	0.750	4.183	< **0.001**		
MC→SWB	0.401	116.392	6.701	< **0.001**		

6.6.10 *H4.1: among participants who have improved DR, the ones with MMC 1 to 4 are more likely to have improved PA than MMC 5 to 8*

As shown in **Table 39**, in path A of the model, one's meaning-making choice to be flourishing was predicted by non-religious belief marginally (B = 0.496, $p = 0.098$) and better financial status (B = -0.350, $p = 0.001$). Even though the path A was statistically robust ($R^2 = 0.275$, $p < 0.001$), caregiving stress did not significantly predict meaning-making choice of 1 to 4 (B = 0.022, $p = 0.368$). In the path B of the model, one's positive affectivity was predicted by worse financial status (B = -0.181, $p = 0.095$), living arrangement with family members (B = 0.430, $p = 0.056$). After controlling for several covariates (age, gender, education, religion, financial status, living arrangement, and length of care), the meaning-making choice of 1 to 4 significantly predicted improved positive affectivity (B = 0.177, $p = 0.091$) while the caregiving stress insignificantly predicted improved positive affectivity (B = -0.032, $p = 0.207$). The interaction terms between meaning-making choice and dyadic relationship were significant (B = 0.119, $p = 0.088$). More specifically, the meaning-making choices were significantly moderated by a lower level ($p = 0.016$), and a moderate level of dyadic relationship ($p = 0.076$). Even though the higher level was not a significant moderator ($p = 0.\ 927$), the **Figure 12** showed that people with the meaning-making choice of 1 to 4, in general, reported a higher level of positive affectivity than the ones with the meaning-making choice of 5 to 8 regardless of dyadic relationship levels. From the perspective of dyadic relationships, a higher level of relationship improved the positive affectivity of the ones with the meaning-making choice of 5 to 8; yet compromised the positive affectivity of the ones with the meaning-making choice of 1 to 4 as the relationship was at a better level. The moderated mediation model was statistically robust for both path A ($R^2 = 0.275$, $p < 0.001$) and path B ($R^2 = 0.226$, $p = 0.015$).

Table 39. Moderated mediation model in prediction of PA

	B	S.E.	t	p	LLCI	ULCI
ZBI→MC						
constant	0.695	0.896	0.776	0.440	-1.085	2.476
Age	0.006	0.011	0.537	0.593	-0.016	0.028
Gender	0.302	0.201	1.503	0.136	-0.097	0.700
Education attainment	-0.224	0.154	-1.453	0.150	-0.529	0.082
Religion	0.496	0.296	1.672	**0.098**	-0.093	1.085
Financial status	-0.350	0.100	-3.517	**0.001**	-0.548	-0.152
Living arrangement	-0.276	0.220	-1.255	0.213	-0.713	0.161
Caregiving length	-0.001	0.003	-0.213	0.832	-0.007	0.006
Caregiving burden	0.022	0.024	0.905	0.368	-0.026	0.071
MC×DR→PA						
constant	2.434	0.902	2.698	**0.008**	0.641	4.226
Age	-0.008	0.011	-0.748	0.457	-0.031	0.014
Gender	-0.001	0.200	-0.005	0.996	-0.399	0.397
Education attainment	0.089	0.154	0.582	0.562	-0.216	0.394
Religion	-0.207	0.304	-0.682	0.497	-0.811	0.397
Financial status	-0.181	0.107	-1.689	**0.095**	-0.394	0.032
Living arrangement	0.430	0.222	1.935	**0.056**	-0.012	0.872
Caregiving length	0.003	0.003	0.769	0.444	-0.004	0.009
Caregiving stress	-0.032	0.025	-1.271	0.207	-0.081	0.018
MC	0.177	0.103	1.706	**0.091**	-0.029	0.382
DR	0.005	0.008	0.615	0.540	-0.011	0.021
MC×DR	0.119	0.069	1.725	**0.088**	-0.018	0.257
MC×DR-H	-2.908	1.182	-2.460	**0.016**	-5.257	-0.558
MC×DR-M	-1.506	0.838	-1.798	**0.076**	-3.171	0.158
MC×DR-L	-0.105	1.151	-0.092	0.927	-2.394	2.183
	R^2	MSE	F	p		
ZBI→MC	0.275	0.744	4.307	< **0.001**		
MC×DR→PA	0.226	0.722	2.33	**0.015**		

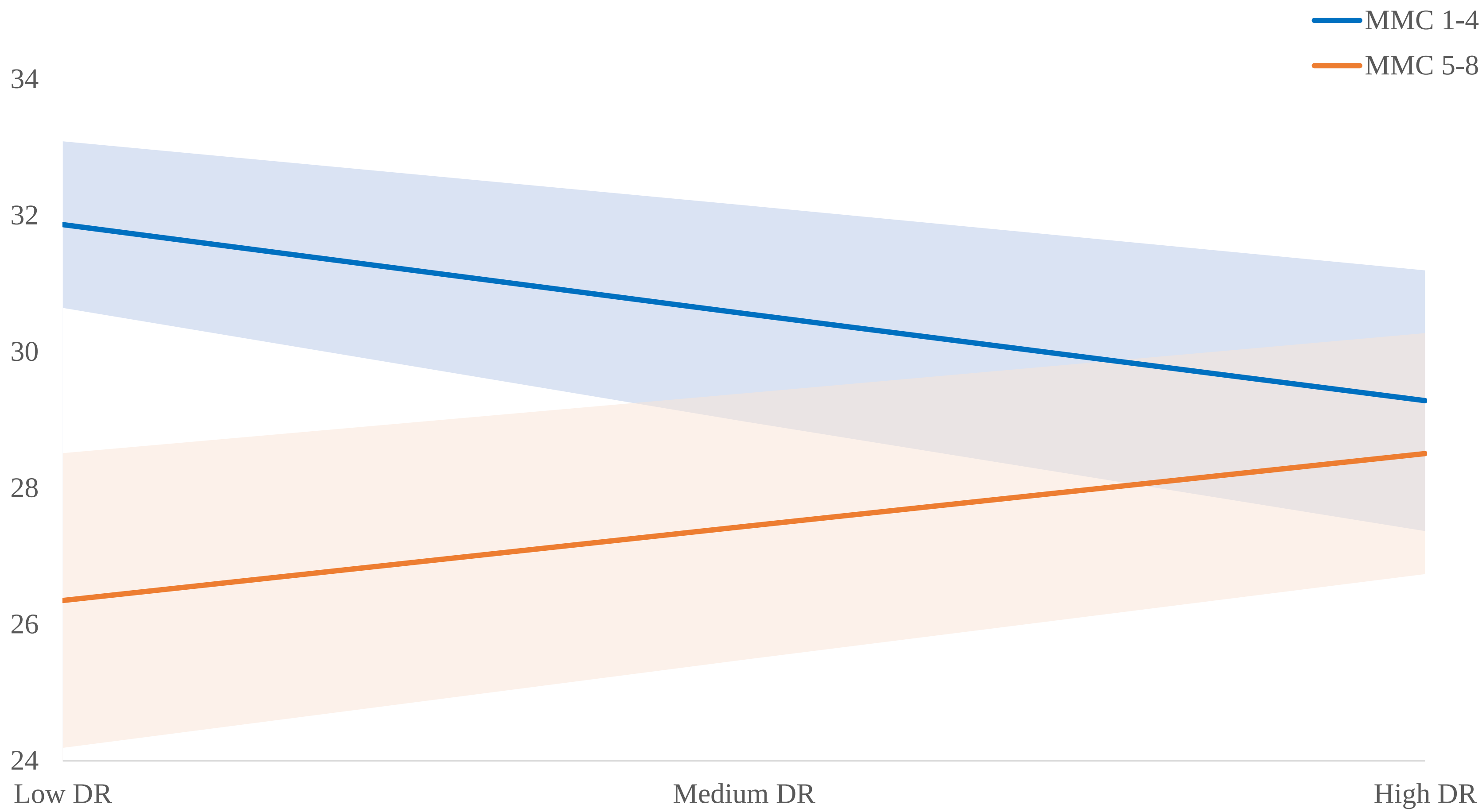

Figure 12. Moderated mediation model in prediction of PA

As shown in Table 40, in the path A of the model, one's meaning-making choice to be flourishing was predicted by non-religious belief marginally ($B = 0.496$, $p = 0.098$), better financial status ($B = -0.350$, $p = 0.001$), and caregiving burden marginally ($B = 0.022$, $p = 0.058$). The path A was statistically robust ($R2 = 0.275$, $p < 0.001$). In path B of the model, one's positive affectivity was predicted by better financial status ($B = 1.333$, $p = 0.002$). After controlling for several covariates (age, gender, education, religion, financial status, living arrangement, and length of care), the meaning-making choice of 1 to 4 significantly predicted improved positive affectivity ($B = -1.533$, $p < 0.001$) while the caregiving stress insignificantly predicted improved positive affectivity ($B = 0.308$, $p = 0.130$). The path B was statistically robust ($R2 = 0.503$, $p < 0.001$). The interaction terms between meaning-making choice and dyadic relationship were significant ($B = 0.062$, $p = 0.047$). More specifically, the meaning-making choices were significantly moderated by a higher level ($p < 0.001$), and a moderate level of dyadic relationship ($p < 0.001$). Even though the lower level of dyadic relationship was not a significant moderator ($p = 0.155$), **Figure 13** showed that people with the meaning-making choice of 1 to 4, in general, reported a higher level of the ESM positive affectivity than the ones with the meaning-making choice of 5 to 8 regardless of dyadic relationship levels. From the perspective of dyadic relationships, a higher level of relationship improved the ESM positive affectivity of the ones with the meaning-making choice of 5 to 8; yet compromised the ESM positive affectivity of the ones with the meaning-making choice of 1 to 4 as the relationship was at a better level. The moderated mediation model was statistically robust for both path A ($R^2 = 0.275$, $p < 0.001$) and path B ($R^2 = 0.503$, $p < 0.001$).

Qualitative interview indicated that a better dyadic relationship motivates one's care attitudes and work ethics. Narratives implied that considering work as a way to earn more money or achieve self-actualization, WDCs with the better dyadic relationship can encourage proactive care, work and personal development (**Appendix V**). In addition, one participant pointed out that better dyadic relationships are associated with higher expectations for PwD, who were believed to be more responsive to care and show positive feedback for what has been done in caregiving. On the other hand, for WDCs who with high meaning-making level already, they already have a decent level of well-being in comparison to WDCs with a lower meaning-making level (**Appendix V**).

Table 40. Moderated mediation model in prediction of ESM PA

	B	S.E.	t	p	LLCI	ULCI
ZBI→MC						
constant	0.695	0.896	0.776	0.440	-1.085	2.476
Age	0.006	0.011	0.537	0.593	-0.016	0.028
Gender	0.302	0.201	1.503	0.136	-0.097	0.700
Education attainment	-0.224	0.154	-1.453	0.150	-0.529	0.082
Religion	0.496	0.296	1.672	**0.098**	-0.093	1.085
Financial status	-0.350	0.100	-3.517	**0.001**	-0.548	-0.152
Living arrangement	-0.276	0.220	-1.255	0.213	-0.713	0.161
Caregiving length	-0.001	0.003	-0.213	0.832	-0.007	0.006
Caregiving burden	0.022	0.012	1.833	**0.058**	-0.026	0.071
MC×DR→PA						
constant	26.480	3.501	7.564	< **0.001**	19.523	33.437
Age	-0.090	0.044	-2.053	**0.043**	-0.177	-0.003
Gender	-0.136	0.777	-0.175	0.862	-1.680	1.408
Education attainment	-0.052	0.596	-0.088	0.930	-1.237	1.132
Religion	0.033	1.180	0.028	0.977	-2.311	2.378
Financial status	1.333	0.416	3.204	**0.002**	0.506	2.159
Living arrangement	0.397	0.863	0.461	0.646	-1.317	2.112
Caregiving length	-0.010	0.013	-0.815	0.418	-0.036	0.015
Caregiving stress	0.308	0.201	1.529	0.130	-0.092	0.708
MC	-1.533	0.401	-3.817	< **0.001**	-2.330	-0.735
DR	0.038	0.032	1.185	0.239	-0.026	0.101
MC×DR	0.062	0.031	2.011	**0.047**	0.001	0.124
MC×DR-H	-2.282	0.549	-4.154	< **0.001**	-3.374	-1.190
MC×DR-M	-1.533	0.401	-3.817	< **0.001**	-2.330	-0.735
MC×DR-L	-0.783	0.546	-1.433	0.155	-1.869	0.303
	R^2	MSE	F	p		
ZBI→MC	0.275	0.744	4.307	< **0.001**		
MC×DR→PA	0.503	10.875	8.109	< **0.001**		

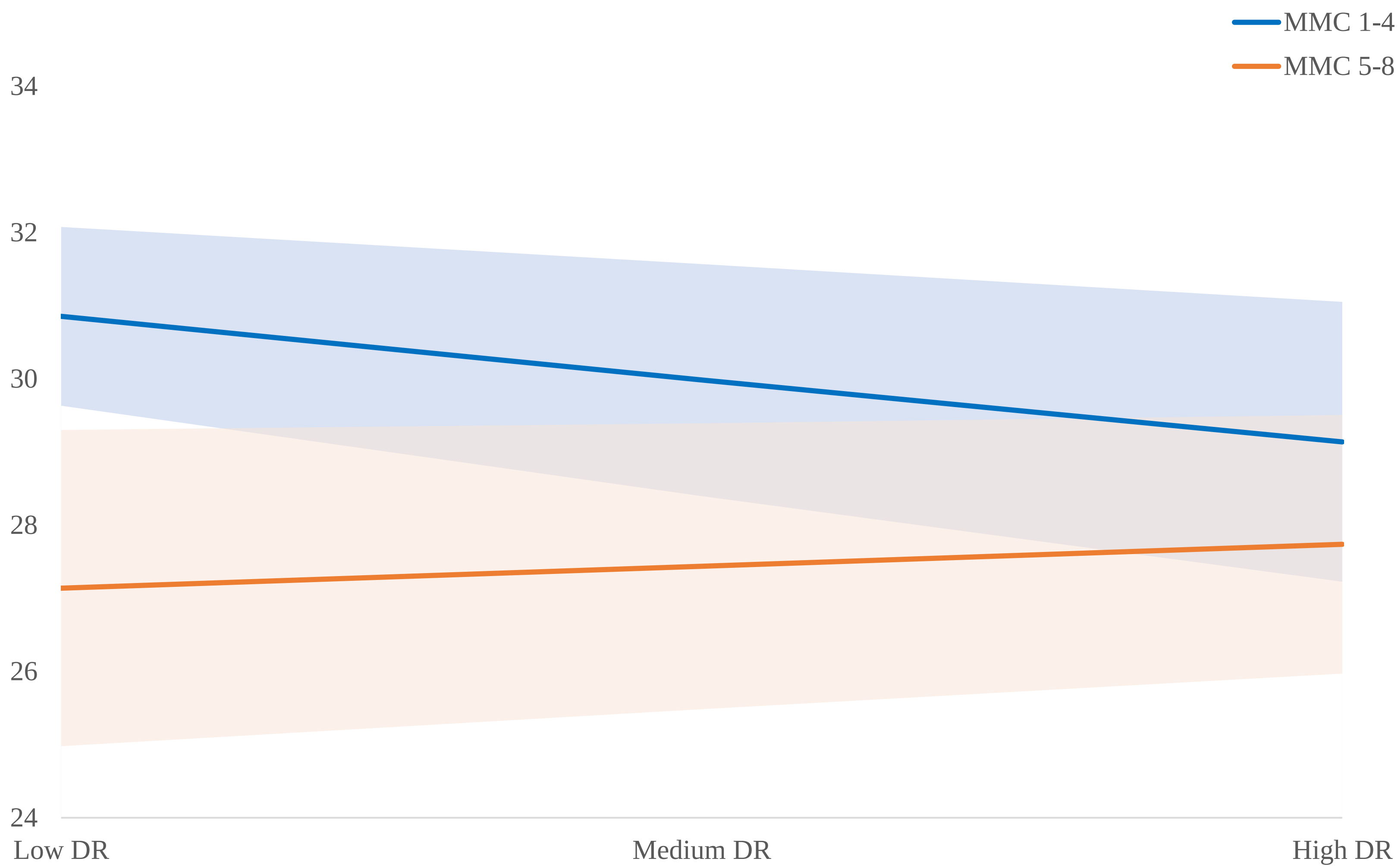

Figure 13. Moderated mediation model in prediction of ESM PA

6.6.11 *H4.2: among participants who have improved DR, the ones with MMC 1 to 4 are more likely to have improved NA than MMC 5 to 8*

As shown in **Table 41**, in the path A of the model, one's meaning-making choice to be languishing was predicted by religious belief marginally (B = 0.496, p = 0.098) and worse financial status (B = -0.350, p = 0.001). Even though the path A was statistically robust (R^2 = 0.275, p < 0.001), caregiving stress did not significantly predict meaning-making choice of 5 to 8 (B = 0.022, p = 0.368). In path B of the model, one's negative affectivity was predicted by non-religious belief (B = -0.787, p = 0.034). After controlling for several covariates (age, gender, education, religion, financial status, living arrangement, and length of care), the meaning-making choice of 5 to 8 significantly predicted worsened negative affectivity (B = 0.364, p = 0.006) while the reduced caregiving stress significantly predicted worsened negative affectivity (B = 0.067, p = 0.037). The path B was statistically robust (R^2 = 0.179, p = 0.093). The interaction terms between meaning-making choice and dyadic relationship were significant (B = -0.128, p = 0.097). More specifically, the meaning-making choices were significantly moderated by a higher level of dyadic relationship (p = 0.095). Even though a moderate and a higher level were not significant moderators (p = 0.376 and p = 0.534), **Figure 14** showed that people with the meaning-making choice of 5 to 8 in general reported a higher level of negative affectivity than the ones with the meaning-making choice of 1 to 4 on a low and moderate level of dyadic relationships. From the perspective of dyadic relationships, a higher level of relationship mitigated negative affectivity of the ones with the meaning-making choice of 5 to 8; yet aggregated negative affectivity of the ones with the meaning-making choice of 1 to 4 as the relationship was at a better level.

Table 41. Moderated mediation model in prediction of NA

	B	S.E.	t	p	LLCI	ULCI
ZBI→MC						
constant	0.695	0.896	0.776	0.440	-1.085	2.476
Age	0.006	0.011	0.537	0.593	-0.016	0.028
Gender	0.302	0.201	1.503	0.136	-0.097	0.700
Education	-0.224	0.154	-1.453	0.150	-0.529	0.082
Religion	0.496	0.296	1.672	0.098	-0.093	1.085
Financial status	-0.350	0.100	-3.517	**0.001**	-0.548	-0.152
Living arrangement	-0.276	0.220	-1.255	0.213	-0.713	0.161
Caregiving length	-0.001	0.003	-0.213	0.832	-0.007	0.006
Caregiving burden	0.022	0.024	0.905	0.368	-0.026	0.071
MC×DR→NA						
constant	3.545	1.098	3.228	**0.002**	1.363	5.727
Age	-0.010	0.014	-0.751	0.455	-0.037	0.017
Gender	-0.129	0.247	-0.522	0.603	-0.620	0.362
Education	-0.097	0.190	-0.513	0.610	-0.475	0.280
Religion	-0.787	0.366	-2.151	**0.034**	-1.514	-0.060
Financial status	-0.068	0.132	-0.513	0.609	-0.331	0.195
Living arrangement	0.255	0.279	0.916	0.362	-0.299	0.809
Caregiving length	0.004	0.004	0.920	0.360	-0.004	0.012
Caregiving stress	0.067	0.032	2.119	**0.037**	0.004	0.130
MC	0.364	0.129	2.821	**0.006**	0.108	0.621
DR	-0.013	0.014	-0.868	0.388	-0.041	0.016
MC×DR	-0.128	0.076	-1.676	**0.097**	-0.279	0.024
MC×DR-H	1.695	1.005	1.688	**0.095**	-0.301	3.692
MC×DR-M	0.581	0.653	0.890	0.376	-0.717	1.880
MC×DR-L	-0.533	0.853	-0.624	0.534	-2.228	1.163
	R^2	MSE	F	p		
ZBI→MC	0.275	0.744	4.307	**< 0.001**		
MC×DR→NA	0.179	1.098	1.000	**0.093**		

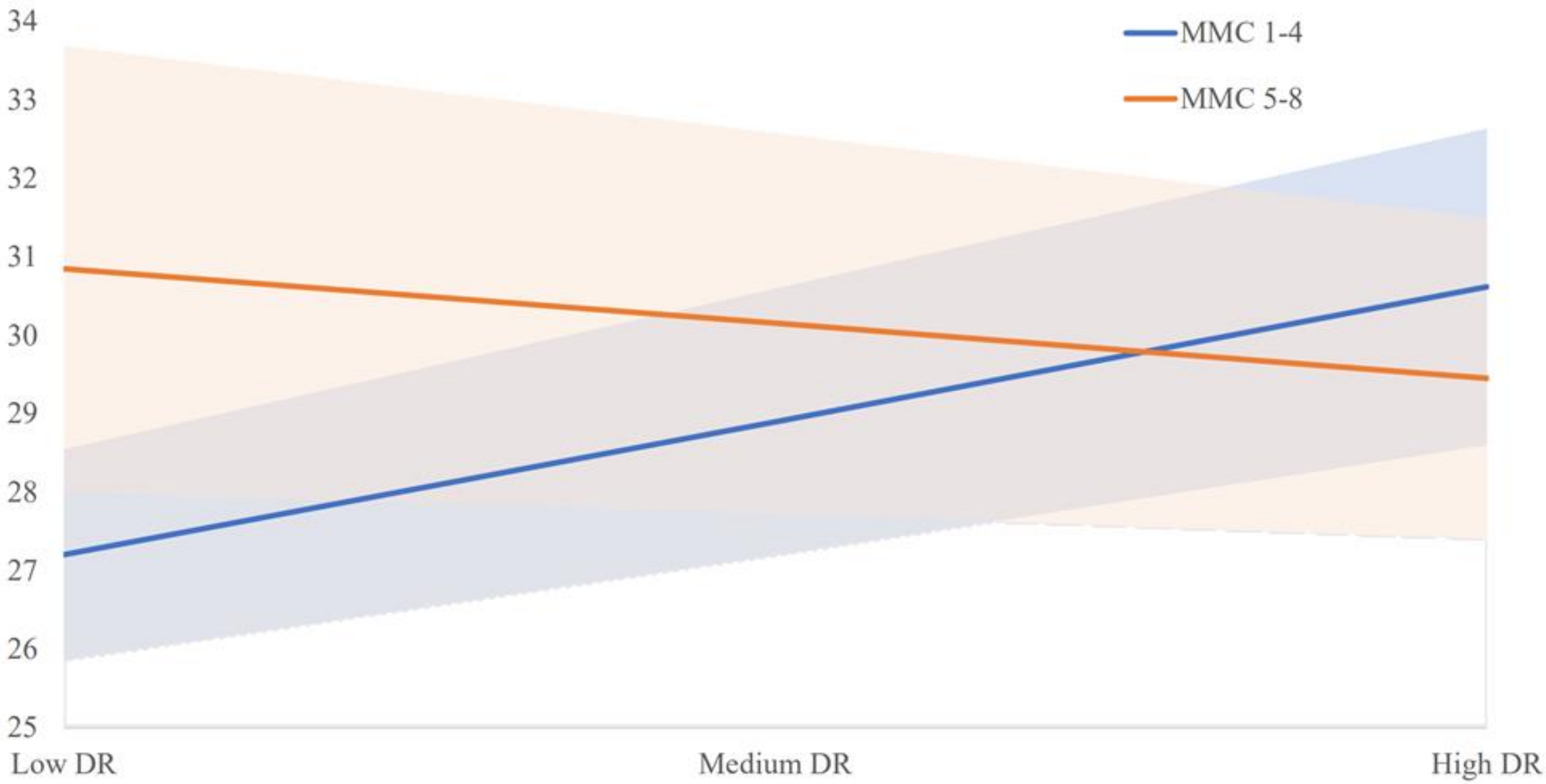

Figure 14. Moderated mediation model in prediction of NA

As shown in **Table 42**, in the path A of the model, one's meaning-making choice to be languishing was predicted by worse financial status (B = -0.350, p = 0.001), religious belief (B = 0.496, p = 0.098) and caregiving burden marginally (B = 0.022, p = 0.058). The path A was statistically robust (R^2 = 0.275, p < 0.001). In the path B of the model, one's negative affectivity was predicted by gender of female (B = -2.857, p = 0.025), worse financial status (B = -2.352, p = 0.001), and caregiving burden (B = 0.370, p = 0.023). After controlling for several covariates (age, gender, education, religion, financial status, living arrangement, and length of care), the meaning-making choice of 5 to 8 significantly predicted worsened negative affectivity (B = 0.581, p = 0.021) while the reduced caregiving stress (B = 0.370, p = 0.023) significantly predicted worsened negative affectivity. The path B was statistically robust (R^2 = 0.372, p < 0.001). The interaction terms between meaning-making choice and dyadic relationship were significant (B = -0.024, p = 0.048). More specifically, the meaning-making choices were significantly moderated by a lower level (p = 0.004), and a moderate level of dyadic relationship (p = 0.006). Even though a higher level of dyadic relationship was insignificant moderators (p = 0.141 and p = 0.405), **Figure 15** showed that people with the meaning-making choice of 5 to 8 in general reported a higher level of negative affectivity than the ones with the meaning-making choice of 1 to 4 on lower and moderate level of dyadic relationships. From the perspective of dyadic relationships, a higher level of relationship mitigated negative affectivity of the ones with the meaning-making choice of 5 to 8; yet aggregated negative affectivity of the ones with the meaning-making choice of 1 to 4 as the relationship was at a better level.

Qualitative interviews indicated that a better dyadic relationship allowed WDCs to comfort oneself. On the contrary, worse dyadic relationships discourage WDCs in their daily care and work performance that is worsened with additional psychological stress. Narratives implied that in the Chinese society valuing filial piety, WDCs should have satisfactory relationships with the family members and were obliged to take care of older adults in the family. Therefore, compromised dyadic relationships hindered one's daily interactions and communications with the PwD. As a result, lowered dyadic relationships negative affected one's well-being overall (**Appendix V**).

Table 42. Moderated mediation model in prediction of ESM NA

	B	S.E.	t	p	LLCI	ULCI
ZBI→MC						
constant	0.695	0.896	0.776	0.440	-1.085	2.476
Age	0.006	0.011	0.537	0.593	-0.016	0.028
Gender	0.302	0.201	1.503	0.136	-0.097	0.700
Education attainment	-0.224	0.154	-1.453	0.150	-0.529	0.082
Religion	0.496	0.296	1.672	**0.098**	-0.093	1.085
Financial status	-0.350	0.100	-3.517	**0.001**	-0.548	-0.152
Living arrangement	-0.276	0.220	-1.255	0.213	-0.713	0.161
Caregiving length	-0.001	0.003	-0.213	0.832	-0.007	0.006
Caregiving burden	0.022	0.012	1.833	**0.058**	-0.026	0.071
MC×DR→NA						
constant	38.573	5.504	7.009	< **0.001**	27.638	49.509
Age	-0.086	0.068	-1.265	0.209	-0.222	0.049
Gender	-2.857	1.250	-2.285	**0.025**	-5.340	-0.373
Education attainment	0.286	0.960	0.298	0.766	-1.622	2.194
Religion	-0.326	1.850	-0.176	0.861	-4.002	3.351
Financial status	-2.352	0.668	-3.524	**0.001**	-3.679	-1.026
Living arrangement	0.360	1.408	0.256	0.799	-2.438	3.157
Caregiving length	0.013	0.021	0.611	0.543	-0.029	0.054
Caregiving stress	0.370	0.161	2.306	**0.023**	0.051	0.690
MC	0.581	0.253	2.296	**0.021**	-0.717	1.880
DR	0.043	0.022	1.955	**0.035**	-0.100	0.186
MC×DR	-0.024	0.015	-1.623	**0.048**	-0.054	0.006
MC×DR-H	0.141	0.169	0.837	0.405	-0.194	0.476
MC×DR-M	-0.064	0.129	2.818	**0.006**	0.107	0.620
MC×DR-L	-0.586	0.198	-2.954	**0.004**	0.192	0.980
	R^2	MSE	F	p		
ZBI→MC	0.275	0.744	4.307	< **0.001**		
MC×DR→NA	0.372	6.922	5.270	< **0.001**		

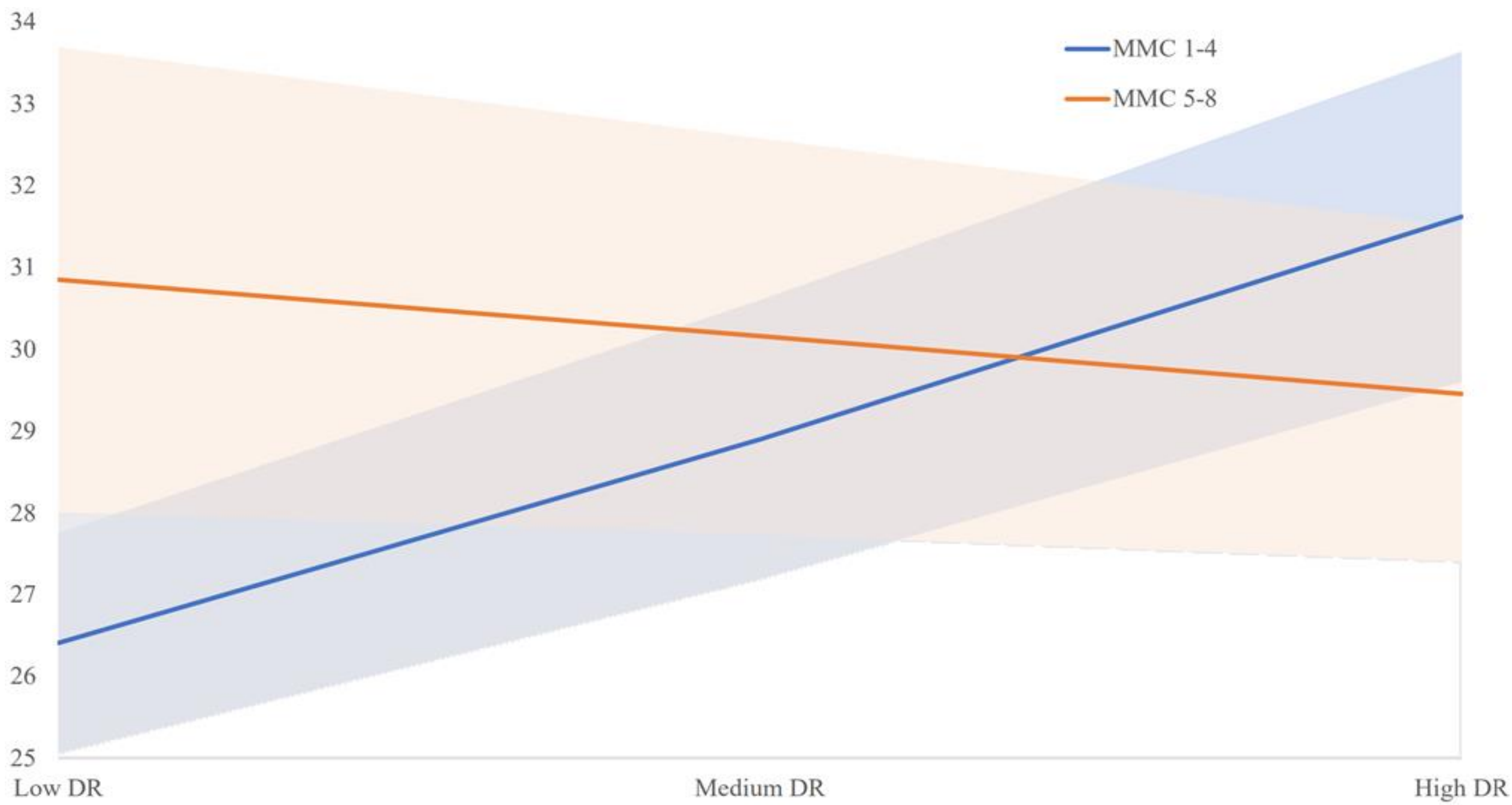

Figure 15. Moderated mediation model in prediction of ESM NA

6.6.12 *H4.3: among participants who have improved DR, the ones with MMC 1, 2, 4 and 6 are more likely to have improved SWB than the rest*

As shown in **Table 43**, in the path A of the model, one's meaning-making choice to be high work meanings was predicted by worse financial status ($B = -0.374$, $p < 0.001$) and employment burden marginally ($B = 0.132$, $p = 0.059$). The path A was statistically robust ($R^2 = 0.269$, $p < 0.001$). In the path B of the model, one's social well-being was predicted by non-religious belief ($B = -7.848$, $p = 0.049$), better financial status ($B = 4.598$, $p = 0.001$). After controlling for several covariates (age, gender, education, religion, financial status, living arrangement, and length of care), the high work meanings significantly predicted better social well-being ($B = -2.532$, $p = 0.052$). The path B was statistically robust ($R^2 = 0.433$, $p < 0.001$). The moderation model showed that the better dyadic relationships in general led to a higher level of social well-being than the other choice ($B = 0.236$, $p = 0.030$; $R^2 = 0.433$, $p < 0.001$) (**Table 43**). The moderation effects showed that caregivers the choice of 1, 2, 4 and 6 had more improved positive affectivity than caregivers with the choice of 5 to 8 when the dyadic relationships were the worst ($B = -5.298$, $p = 0.005$), the moderate ($B = -2.532$, $p = 0.052$), and the best level ($B = 0.233$, $p = 0.058$) (**Figure 16**).

Qualitative interviews indicated that a better dyadic relationship promoted not only one's care performance, but also their work performance and ethics. While having an improved dyadic relationship with the PwD, a WDC can care for the family member with confidence but also could be less distracted in their employment and personal development (**Appendix V**). On the other hand, weak dyadic relationships negatively affected one's perception of daily experiences in care, work and personal life aspects and corresponding well-being level (**Appendix V**).

Table 43. Moderated mediation model in prediction of SWB

	B	S.E.	t	p	LLCI	ULCI
OSI→MC						
constant	1.166	0.945	1.234	0.220	-0.710	3.042
Age	0.006	0.011	0.556	0.580	-0.016	0.028
Gender	0.282	0.201	1.408	0.163	-0.116	0.681
Education	-0.227	0.154	-1.473	0.144	-0.534	0.079
Religion	0.455	0.308	1.479	0.143	-0.156	1.067
Financial status	-0.374	0.096	-3.899	< **0.001**	-0.565	-0.184
Living arrangement	-0.242	0.218	-1.114	0.268	-0.675	0.190
Caregiving length	-0.003	0.011	-0.312	0.756	-0.024	0.018
Employment stress	0.132	0.069	1.913	**0.059**	-0.259	0.264
MC×DR→SWB						
constant	100.599	12.029	8.363	< **0.001**	76.695	124.504
Age	-0.228	0.143	-1.599	0.113	-0.512	0.055
Gender	-1.306	2.502	-0.522	0.603	-6.277	3.666
Education	0.310	1.932	0.161	0.873	-3.529	4.150
Religion	-7.848	3.933	-1.995	**0.049**	-15.664	-0.032
Financial status	4.598	1.316	3.493	**0.001**	1.982	7.214
Living arrangement	-2.105	2.746	-0.766	0.445	-7.562	3.352
Caregiving length	-0.025	0.041	-0.618	0.538	-0.107	0.056
Employment stress	-0.009	0.133	-0.068	0.946	-0.272	0.254
MC	-2.532	1.286	-1.969	**0.052**	-5.089	0.024
DR	-0.080	0.106	-0.755	0.452	-0.291	0.131
MC×DR	0.236	0.107	2.207	**0.030**	0.024	0.448
MC×DR-H	-5.298	1.830	-2.895	**0.005**	-8.935	-1.662
MC×DR-M	-2.532	1.286	-1.969	**0.052**	-5.089	0.024
MC×DR-L	0.233	1.761	3.133	**0.058**	-3.266	3.733
	R^2	MSE	F	p		
OSI→MC	0.269	0.742	4.825	< **0.001**		
MC×DR→SWB	0.433	112.787	6.101	< **0.001**		

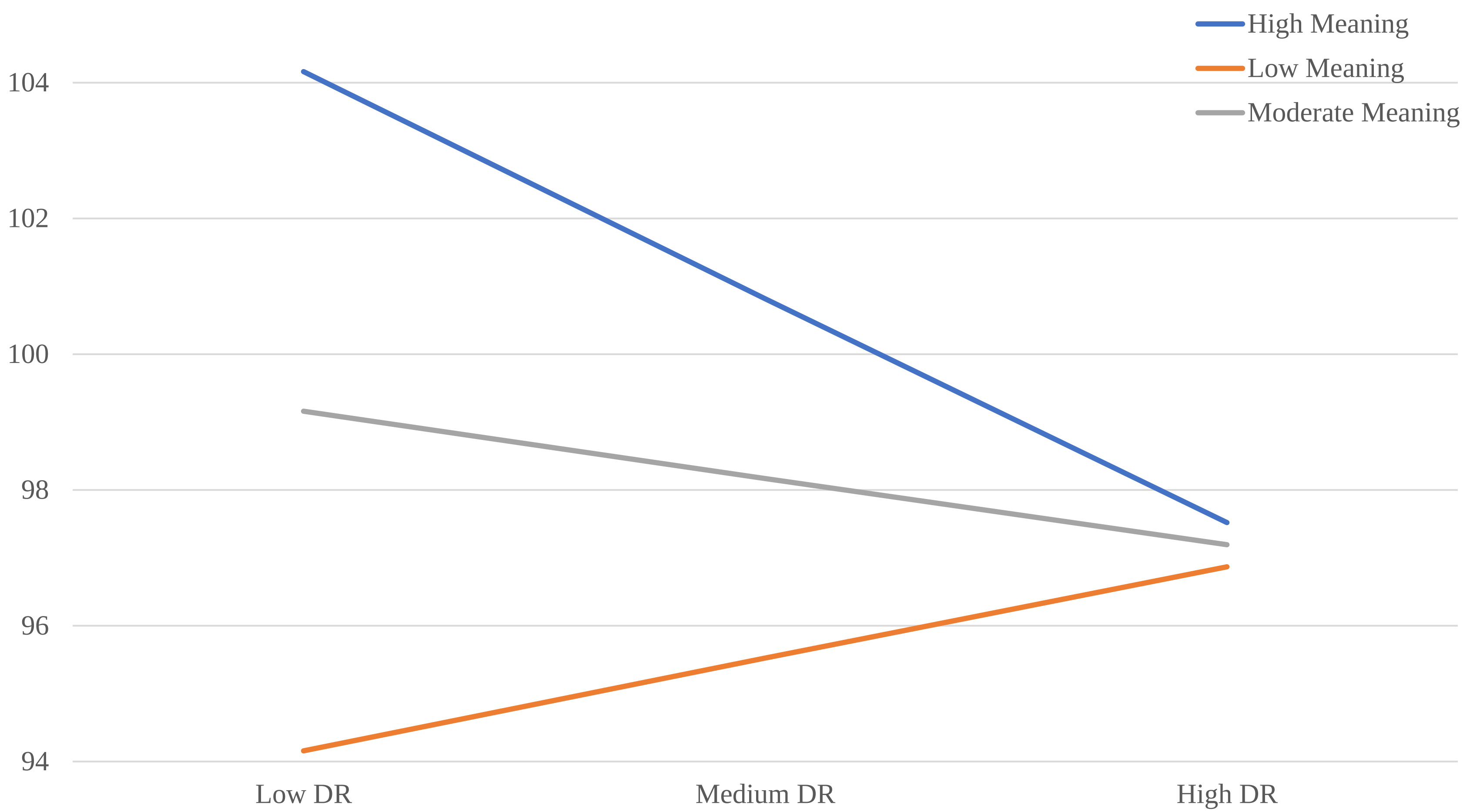

Figure 16. Moderated mediation model in prediction of SWB

To conclude, all hypotheses were supported using either baseline well-being or ESM well-being. Among nine hypotheses testing PANAS, 4 hypotheses (44.4%) were supported using baseline well-being were supported. In comparison, all hypotheses (100%) were supported by models using ESM well-being (**Table 44**).

Table 44. Hypotheses testing

Index	Hypotheses	Baseline WB		ESM WB	
		Statistics	Result	Statistics	Result
H1.1	MC 1-3 ⟷ [H] PA + [L] NA	$p < 0.001$	✓	$p < 0.001$	✓
H1.2	MC 4 ⟷ [H] PA + [M] NA	$p < 0.001$[a]	~~✓~~	$p < 0.001$	✓
H1.3	MC 5-8 ⟷ [H] PA + [L] NA	$p < 0.001$	✓	$p < 0.001$	✓
H2.1	MC 1-4 > MC 5-8 ⟷ [H] PA	$p = 0.009$	✓	$p < 0.001$	✓
H2.2	MC 1-3 < MC 4-8 ⟷ [L] NA	$p < 0.001$	✓	$p = 0.003$	✓
H2.3	MC 1246 > MC 3578 ⟷ [H] SWB	$p < 0.001$	✓		
H3.1	MC 1-4 > MC 5-8 → [↑] PA	$p = 0.114$	✕	$p = 0.001$	✓
H3.2	MC 1-3 > MC 5-8 → [↓] NA	$p = 0.527$	✕	$p = 0.030$	✓
H3.3	MC 1246 > MC 3578 → [↑] SWB	$p = 0.047$	✓		
H4.1	MC 1-4 –([↑]DR)→ [↑] PA	$p = 0.088$	✕	$p = 0.047$	✓
H4.2	MC 1-4 –([↑]DR)→ [↓] NA	$p = 0.097$	✕	$p = 0.048$	✓
H4.3	MC 1246 –([↑]DR)→ [↑] SWB	$p = 0.030$	✓		

Note. [H] - high level; [L] - low level; [M] moderate level; [↑] improved; [↓] declined.
[a] the statistics are questionable due to the limited sample size (n = 6), comparing with ESM (n = 228).

Chapter 7 Discussion

To investigate the meaning-making choices of Chinese WDCs and how the choices affect their hedonic well-being and social well-being in the proposed theoretical framework of meaning-making choice, this chapter focused on the following avenues: 1) The overall findings were comprehensively summarized and compared with existing empirical studies; 2) Theoretical implications; 3) Methodological implications; 4) Practical implications; and 5) Limitations and future recommendations.

7.1 Feasibility and Validity of ESM among Chinese Working Dementia Caregivers

The compliance response rates, usability, feasibility, and ecological validity all indicate that the ESM approach is desirably appropriate for studying the well-being of Chinese WDCs.

The overall response rate at 93.3% shows a good measure of ESM in the target research environment, being consistent with an average ESM study (van Berkel et al., 2019). Even though the overall response rates were slightly lower on weekends, it is reasonable to understand that these working caregivers may spend more time on their PwD and personal business over the weekends to compensate for what they have missed during the workdays (Stanfors et al., 2019). The response rates on work are significantly lower on weekends majorly due to the fact that fewer participants work on weekends leading to insufficient data and unstable statistical results. Literature indicates that only a few occupations require people to work over weekends and further leads to their emotional exhaustion and stress due to work-family conflicts (Cottingham et al., 2020). For cancer caregivers, their efforts in balancing work and care are positively associated with symptoms of depression and anxiety (Hastert et al., 2020).

In addition, people are generally more distracted by other family issues or social needs on weekends besides work. Caregivers usually report adverse experiences balancing their caregiving and employment tasks, such as productivity loss, extra duties, reduced savings, depressive symptoms, and anxious emotions (Hastert et al., 2020). Hourly response preferences to assessments have delineated the general behavioral pattern of Chinese WDCs (Bernstein et al., 2019; Sakka et al., 2019). Response rates of each activity type have expected rush and down hours. For instance, the period between 9am to 4pm is predominately occupied by occupational work. The period of 12pm to 2pm is usually a lunch break that allows participants to respond to the assessment. Response rates on care usually peak from 6pm to 8pm right after work hours, when working caregivers are back at home with their family members living with dementia. The period of 8pm to 10pm is usually used for personal businesses after dinner and care. More evidence indicates that Chinese caregivers have their own schedules balancing work and care tasks throughout the day. Chinese working class usually work diligently everyday, and even on weekends (Y. Yu et al., 2018).

The ESM demonstrates satisfactory usability and feasibility. Mobile-based ESM assessments are generally found easy to use on different operation systems platforms across various digital devices since digital phones and tablets are popular in the Chinese population. China has advanced with digital entertainment and communication platforms following popular digital culture and media production from the Western world (Plantin & De Seta, 2019). A statistic has shown that almost half of the population becomes dependent on smartphones and develops smartphone addiction along with depressive symptoms (X. Yang et al., 2021). Each ESM assessment taking 1 to 2 minutes is feasible to complete during a day for 2 to 3 times. ESM approach is found to be time-saving at each scheduled session and overall progress, in comparison to the paper-based approach. In addition, the ESM method is reported to have better effectiveness and usability than paper-based questionnaires in real-time assessments (Bailon et al., 2019). Also, ESM saves time collecting back paper-based questionnaires. The ESM data was instantly input into qualities or other online databases (Murnane et al., 2016). In terms of usability and feasibility, ESM in comparison to the paper-based assessment is beneficial in several other areas. First, ESM can be more easily conducted on one's mobile device anytime when a participant has spare time of 1 to 2 minutes. On the contrary, paper-based assessments offline meetings cost much more time on commute and communication between a participant and the assessor than the assessment itself (Kowalski et al., 2020). For instance, a working participant can take one minute out of her coffee break to finish the online ESM assessments. However, it is unlikely for the participant to spend an hour out of the company to fill in a one-page survey in paper form. Second, ESM allows instant responses during each activity by recording assessment information on one's digital phone (Braekman et al., 2018). In comparison, the paper-based assessment may not be feasibly performed during work or care right away. Third, ESM is more feasible for non-contacting assessments during the COVID period, whereas paper-based assessments require face-to-face contact. Studies suggest that web-based ESM ensures both desirable sensitivity and specificity (Munsch et al., 2020), recalls self-monitoring process and avoids interpersonal contacts that cause potential virus spread (Schrager et al., 2020). Fourth, ESM assessments are self-monitored whereas the paper-based version is suggested to be administered. Evidence shows that in the self-monitored survey, online versions are suggested to better accommodate with efficacy such needs rather than paper-based questionnaires (Bos et al., 2020). Attrition during collecting or returning questionnaires may exist in real practice settings (Greenlaw & Brown-Welty, 2009). Web-based ESM survey, on the other hand, can accurately record all participants' input into the databases without a secondary data collection procedure (Larson & Csikszentmihalyi, 2014). Last, ESM reduces the memory burden to retrieve retrospective information and reflect on real-time situations over a series of time points. Such measures avoid inaccurate recalls, for instance, from past weeks or two months (Verhagen et al., 2016).

Together with the usability and feasibility, the high ecological validity of ESM indicates the accuracy of the design and evaluation in reflecting the real environment where Chinese dementia working caregivers live (Csikszentmihalyi & Larson, 2014). ESM with a high ecological validity allows real-time examination and validation of what was investigated in the laboratory environment, as well as strengthen the validity of instrument measuring the phenomenon in naturalistic situations in clinical practices and everyday life (Andrade, 2018). ESM, due to its design of nature, can stimulate the natural flow of real-life and avoid biased generalization (Verhagen et al., 2016). Various activities in corresponding settings in WDC eliminate the measurements that lack ecological validity or external validity (Andrade, 2018). There is also decent percentage of care and personal businesses carried out at settings other than outdoors, such as at hospitals, community centers, neighborhoods, or recreational areas. In addition, the majority of events are taken place at home during the COVID period, even including more than 10% of employment is taken from the home. According to empirical evidence, activities taken place both indoors and outdoors with decent ecological validity minimize adverse effects on well-being during stressful events (Stieger et al., 2021). More empirical studies find that ESM measurements show satisfactory ecological validity in its broad application under other circumstances (Myin-Germeys et al., 2018).

7.2 Characteristics portrait of Chinese Working Dementia Caregivers

The portrait of Chinese WDCs is depicted as females in their 40s. Chinese dementia studies have pointed out that the majority of dementia caregivers are females in their 40s and 50s (B. Wu et al., 2019). The retirement age of Chinese women is between 55 and 60, WDCs usually need to face various life stress, such as dementia caregiving and employment (V. W. Q. Lou et al., 2015; Sun, 2014a). Chinese caregivers usually live with their care recipients with the notions of filial piety and family reunion. In a financially competitive society, Chinese family members tend to live together for a more economical daily life and highly accessible resources rather than living separately. In order to better care for PwD, dementia caregivers usually choose to live with their family members and PwD (Sun, 2014a). Advocated by the Chinese government, elderly care demands a society that is "family-based, community-oriented, and social worker-supported" (Cui et al., 2010). Therefore, family members living together enables economic care and living standards (Cepoiu-Martin et al., 2016; Savla et al., 2021). Caregivers in their 40 and 50s are usually married and partnered. There is a small portion of them divorced and widowed leading to a worse condition for caring for PwD because more family members imply more familial support in caregiving (Savla et al., 2021). Caregivers that are non-partnered are less supported in their daily caregiving tasks; especially they need to work full-time during the workdays (Fujihara et al., 2019). WDC of such age groups, in general, are highly educated nowadays. Knowing that WDCs are decently educated, more informative education regarding dementia symptoms and interventions should be provided to this population (Terayama et al., 2018). The recruited Chinese participants are majorly non-religious (Sun, 2014), being inconsistent with the deeply believed notion that “China is very religious” (C. K. Yang, 2020; C. Zhang & Lu, 2020). Religion is considered a type of existential meaning-making approach. Whereas non-religious dementia caregivers may have a stronger will to seek advanced existential meaning-making approach (Terayama et al., 2018). These caregivers are moderately financed. Well-financed dementia families usually decide to send the PwD to daily care or elderly care centers, while poorly dementia families often overlook dementia symptoms, skip diagnosis and avoid prescriptions (Sun, 2014). It is well-known that dementia caregivers suffer from both care burden and employment stress, especially disturbed by PwD’s NPI symptoms (Cheng, 2017; B. Wu et al., 2019). Our data support a similar conclusion that these dementia caregivers generally have a moderately high level of caregiving and work stress. Therefore, a decent level of meaning-making is found among WDCs in settings of care, work and personal life. Sufficient evidence indicates that caregivers have needs to make meaning of their caregiving experiences (Krok et al., 2021; X. Zhang et al., 2018). Working caregivers also tend to make meanings of their employment stress (Fujihara et al., 2019; Pinquart & Sörensen, 2006). Meaningful work is found critical in improving job performance in organizations and enhancing one’s personality at work (Lysova et al., 2019). Another meta-analysis indicates that meaningful work predicts improved work engagement, commitment, and job satisfaction (Allan et al., 2019).

The study distinguishes two major subgroups of flourishing and languishing caregivers. Among two subgroups of caregivers, flourishing caregivers are mainly females (i.e., only half of languishing caregivers are females), more highly educated, better financed, and less been disturbed by PwD's neuropsychiatric symptoms. Many female dementia caregivers tend to make meanings of caregiving experiences in comparison to male caregivers (Teo et al., 2020). Dementia caregivers with a lower educational level have more caregiving burden, whereas the ones with higher education have more tendency to make meanings of caregiving stress (F. Yang et al., 2019). A higher educational level also facilitates one's meaning-making level as life-enhancing knowledge and interventions are more acceptable to these highly educated individuals (Levy et al., 2019). Better hedonic and psychological well-being are empirically reported among caregivers who have a higher meaning-making level (Heintzelman, 2018; Janicke-Bowles et al., 2019; Lavy & Naama-Ghanayim, 2020). Studies show that caregivers suffering from a great caregiving burden while with a lower meaning-making level have more depressive symptoms (F. Yang et al., 2019).

In comparison with caregivers with low working meanings, caregivers with high work meanings have their PwD majorly females, living with family members, have slightly shorter diagnostic history, more in their early stage of dementia, with a bit fewer ADL and IADL issues. In the high work meaning subgroup, both WDCs and their PwD are five years younger, WDCs have better financial status, are less disturbed by PwD's neuropsychiatric symptoms, and with a higher level of meaning in care, work and personal life experiences. According to a systematic review, female caregivers are more susceptible to occupational stress, such as job changes, work arrangements, and motivations (U. Schneider et al., 2013). People with a higher income salary generally make better sense of work stress because they have more choices in seeking useful resources in both informal and formal approaches (Bernstein et al., 2019). Dementia caregivers with a certain meaning-making level are mentally resilient and psychologically ready for stresses from caregiving and working settings (Vos et al., 2022).

7.3 Meaning-making Choices and Well-being

7.3.1 Meaning-making Choices and Classes

The meaning-making choice from 1 to 8 indicates caregivers with full meaning-making have the highest positive affectivity and the lowest affectivity while the ones with non-meaning-making have the lowest positive affectivity and the highest negative affectivity. Among all eight meaning-making choices, the meaning-making choice No.1 has the best hedonic and social well-being as well as the lowest depressive symptoms, while the meaning-making choice No. 8 has the worst hedonic and social well-being as well as one of the highest depressive symptoms. With the highest meaning-making level, caregivers have reduced behavioral problems, work and family strains that could potentially cause psychological well-being (C. Liu et al., 2021). On the other hand, without an appropriate meaning-making level to intervene, dementia caregivers may have overwhelming challenges in both care and work settings that potentially lead to compromised well-being (C. Liu et al., 2021). Other empirical evidence also supports that the class of flourishing caregivers have a higher level of hedonic well-being, and a lower level of depressive symptoms. The class of languishing caregivers has lower hedonic well-being and a higher level of depressive symptoms (Araujo et al., 2017; Greer et al., 2019; C. Liu et al., 2021).

There are a few unexpected cases in the meaning-making choices. First, for those Chinese WDCs who value filial care the most (the meaning-making choice No. 4 – neutral caregivers), they have one of the highest hedonic and social well-being using both baseline and ESM data. Filial piety is commonly reported by Chinese caregivers, and is considered inspiring and motivational to serve a protective purpose of reducing negative affectivity that is generated from stressors, and promoting positive affectivity during caregiving experiences (X. Zhang et al., 2018). Second, even though the neutral caregivers appear to be in a transitional phase between flourishing and languishing caregivers, with high positive affectivity and moderate negative affectivity, they are similar to the meaning-making choice No. 2 and the meaning-making choice No. 3. Next, the languishing caregivers generally have low positive affectivity and high negative affectivity. Empirical evidence indicates that poor meaning-making level leads to worsened well-being. More specifically, caregivers without appropriate meaning-making interventions or meaningful support will develop psychological and physical issues in wellness (Cavanaugh et al., 2020). However, the meaning-making choice No. 5 has the highest positive affectivity and the lowest negative affectivity among languishing caregivers. Evidence indicates that certain caregivers with low meaningful purposes can still find their way, such as coping strategies or rumination, to overcome the stressors (Kamijo & Yukawa, 2018; Milman et al., 2019). Last, according to both cross-sectional and ESM data, the meaning-making choice No. 7 has the lowest negative affectivity. The the meaning-making choice No. 7 refers to the ones with a higher level of personal life meanings and a lower level of care and work meanings. Studies find that caregivers with a certain level of selfishness may appear to have negative feelings, declined cognitive functioning, and negative affectivity in depression (Maki et al., 2020; Waldrop & McGinley, 2020).

7.3.2 Flourishing and Languishing Caregivers

The major classes of caregivers with meaning-making choices include choices of 1 to 4 and choices of 5 to 8, corresponding to flourishing and languishing types, respectively.

Caregivers of these two classes had homogeneous baseline characteristics. The perceptions of care stress and work stress are equivalent across two meaning-making choice classes. Financial status is the only demographic difference between these two populations. Dementia caregiving demands great financial costs, including expensive medications that are not covered by the medical insurance’s compensation, in-home nursing care, social and private elderly care centers and services. Families with financial challenges are greatly suffered and report even lower meaning-making levels and worse quality of life (Sabella & Suchan, 2019).

The findings further demonstrate that these two sub-groups have diversified meaning-making levels of care, work, and personal life, separately leading to corresponding well-being outcomes. Flourishing caregivers in empirical studies have better hedonic well-being because they report a higher level of meaning-making in care and other aspects and life (Keyes, 1998). A flourishing state is a crucial indicator of well-being, according to the Flourishing Through Leisure Model (Greer et al., 2019). On the other hand, languishing individuals report low hedonic and psychological well-being, as well as other mental health issues (Vescovelli et al., 2020).

7.3.3 Meaningful and non-meaningful Working Caregivers

Caregivers with a high meaning-making level on working (choices of 1, 2, 4 and 6 versus choices of 3, 5, 7 and 8) have better social well-being, and improved social well-being with a better dyadic relationship. These two classes had comparable baseline characteristics, including PwD's gender and dementia stage, and caregivers' gender, marital status, education, and religion. The perceptions of care stress and work stress are equivalent across two meaning-making choice classes. (Jolanki, 2015; Sakka et al., 2019).

These two sub-groups have diversified meaning-making levels of care, work, and personal life, distinctively leading to corresponding well-being outcomes. Caregivers transitioning from frustration to flourishing have a higher meaning-making level that leads to positive appraisals and positive well-being (Sabat, 2011). Flourishing caregivers with the meaning-finding tendency in the suffering report better eudaimonic well-being and experience better individual and social well-being (Cogan, 2016; R. M. Ryan et al., 2008). Flourishing individuals engaged in meaningful endeavors at the workplace have fewer intentions to quit the job and improved in-role performance (Redelinghuys et al., 2019). These employees report better job satisfaction, work engagement, positive affectivity, and psychological wellness (Redelinghuys et al., 2019). Theoretically, model of voluntary turnover assumes that dissatisfied employers have a tendency in seeking meanings (Mitchell et al., 2001; Mobley, 1977). In addition, positive affectivity and job satisfaction among flourishing individuals can be explained by the social exchange theory (Blau, 1964) and attraction-selection-attrition theory (B. Schneider, 1987).

In comparison to languishing caregivers, flourishing caregivers generally (with choices of 1 to 3, or 1 to 4) have a better baseline and improved hedonic well-being during the ESM phase. In addition, an improved dyadic relationship can further strengthen hedonic well-being. Relationships in flourishing people enhance one's functioning, experiences of meaningfulness and well-being (Wissing et al., 2021). Eudaimonic relationships with meaningfulness are found beneficial to one's well-being (Finkel & Simpson, 2015; Fowers et al., 2016; White, 2017). According to the multidimensional meaning structure, dyadic relationships serve as a precondition for meaning and social validation that are both associated with one's wellness (Wong, 2012). Empirical evidence also indicates that the relationship is a source of meaning (Fowers et al., 2010), and motivational elements to improve the well-being of languishers (Wong, 2012). Languishers have issues in relationships and conflicts in their motives and goals. Recent studies found improving relationships minimize these conflicts and enhance their well-being (Fowers et al., 2010; Gable & Impett, 2012), and maintain a positive state of health (Ryff & Singer, 2008). Furthermore, flourishers with more meaningfulness possess adequate goals for personal development and growth as well as positive well-being (Wissing et al., 2021).

According to the regression models, in comparison to the low work meaning group, high meaningful working caregivers generally (with choices of 1, 2, 4 and 6) have better social well-being on the baseline and improved states. In addition, an improved dyadic relationship can further strengthen hedonic well-being. Flourishers and languishers perceiving different levels of meanings report dramatically different well-being states. Languishers with a higher expectation of work goals and performance report growing frustration and negative well-being. Flourishers, on the other hand, have clearer meanings assigned to their jobs and report improved positive affectivity (Wissing et al., 2021). Since caregiving significantly affects the length of working hours, workplace choice, and retirement status. Working caregivers experiencing meaningfulness can better balance responsibilities between care and work while maintain one's health status (L. Chen et al., 2019), self-esteem, and well-being (Lavy & Naama-Ghanayim, 2020).

7.4 ESM Advantages in Evaluating Well-being

ESM is advantageous in measuring well-being during a continuously intensive time period, in comparison to the cross-sectional measurement or longitudinal measurement. The cross-sectional measurement records a static state of well-being, assuming one's well-being in all other states remains the same. The longitudinal design usually assumes some changes during a period of time due to exposure stimuli or interventions. ESM allows a clear and accurate depiction of well-being during a short period of time in one's daily living experiences.

One discrepancy in hedonic well-being is found between using cross-sectional data and ESM data. According to the baseline data, the neutral caregivers have the highest positive affectivity than flourishing and languishing caregivers, whereas the flourishing caregivers have higher positive affectivity than the other two classes. In addition, physical well-being is as well important as hedonic and social well-being according to the social interactional paradigm (Sitter, 1974). The social interactional paradigm assumes that the environmental change as a function of change of the system's internal structure. The environmental changes, putting into the context of health, involve physical, psychological and other well-being.

The findings using ESM data support the proposed hypotheses and models in a better way than baseline data and longitudinal data (the changes between baseline and followup well-being). ESM data show that the well-being of the meaning-making choice No. 4 (filial caregivers) has lower positive affectivity and higher negative affectivity of the meaning-making choices No.1 and 3, which partially meet the hypothesis. In addition, languishing caregivers have the worst ESM well-being. Also, caregivers with high work meaning have the highest social well-being. On the contrary, the baseline data indicate well-being of the meaning-making choice No.4 has the highest positive affectivity and the second-lowest negative affectivity among all the meaning-making choices, which do not meet the hypothesis. To conclude, ESM well-being presents a lucid picture of hedonic well-being among Chinese WDCs. ESM data also clearly specify the impacts of meaning-making and/or the dyadic relationship on one's well-being in both mediation models and moderated mediation models, supporting all hypotheses of this study. On the other hand, the cross-sectional data (baseline data, and the difference between follow-up and baseline data for improvement) do not support the hypotheses. Using ESM data, a better dyadic relationship has buffering effects in promoting one's positive affectivity especially among languishing caregivers. On the other hand, the dyadic relationship buffer the negative affectivity of languishing caregivers. Flourishing and neutral caregivers with better dyadic relationships report worse negative affectivity consistently appeared in both cross-sectional and ESM data. Moreover, flourishing and languishing caregiver groups have more distinguishable differences using ESM well-being than using baseline well-being.

7.5 Significance

7.5.1 Meaning-making and Well-being

Meaning-making has been broadly found associated with well-being (Krok et al., 2021). The findings confirm that meaningful care is associated with better well-being (Cavanaugh et al., 2020), and a higher level of positive affectivity (Miao & Gan, 2020). Meaning-making serves as a protective factor against life adversities and reduces negative emotions and affectivity (Cavanaugh et al., 2020; Russo-Netzer et al., 2021). These strong associations can be reasonably explained by an improved quality of life with meanings attached to the events (Russo-Netzer et al., 2021).

Caregivers with a high work meaning level have mild depressive symptoms, moderate social well-being and low positive affectivity and worse negative affectivity. Studies indicate that meaningful work are found with improved life satisfaction and decreased emotional exhaustion, both of which lead to reduced negative affectivity (Clarke, 2019; Hobfoll, 2001; Pines & Zaidman, 2014). Meaningful work as a job resource enhances one's occupational well-being, according to the Conservation of Resources Theory (Kim & Beehr, 2018). Low meanings in work lead to a higher level of burnout experiences and negative well-being (Kim & Beehr, 2018).

A high meaning in personal life among working caregivers is associated with the worst depressive symptoms, the lowest positive affectivity and the highest negative affectivity. Meaninglessness together with the hopelessness of healthcare caregivers contributes to a greater level of depression due to multiple stressors from conflicting roles in occupations, care provision, and life events (Güngör & Uçman, 2020).

Our findings also demonstrate that caregivers with a high meaning-making level in any two or three aspects of their daily activities have a better well-being than the rest. More aspects with a higher meaning-making level lead to a better state of well-being. WDCs being challenged by "double burden" or "triple burden" can benefit from the meaning-making of multiple stressors (Black et al., 2010; F. Chen et al., 2018). It is common sense that people have an innate tendency to maintain a balance between multiple daily activities, such as work-life balance (X. Zhang et al., 2018). Similar psychological needs apply to "care-work balance" or "care-life balance." It is assumed that WDCs have comparatively more enhanced well-being by adopting meaning-making approaches to multiple stressors (Sakka et al., 2019). Multiple sources support the findings that caregivers with a higher meaning-making level in all aspects have minimal depressive symptoms, the highest social well-being, the highest positive affectivity and the lowest negative affectivity. For instance, meaning-making of both care and work, or care and life, is associated with improved wellness, cognition, emotional status, work ethics, reduced work stress, functioning in daily life, and quality of life (Jolanki, 2015; Sakka et al., 2019).

7.5.2 Theoretical Implications

This study is investigated on the basis of the proposed theoretical framework, including the Stress Processing Model (SPM), the Meaning Frame Theory (MFT), Self-Determination

Theory (SDT), the Theory of Meaning and Decision (TMD), the Choice Theory (CT), and the broaden-and-built theory.

The findings suggest that one's health outcome is influenced by not only psychosocial factors, but also primary stressors (i.e., care stressors and occupational stressors), and mediators (i.e., personal choice on meaning-making of events). These elements strongly associated with one's well-being are the critical components in the SPM (Folkman, 1997; Park, 2010).

The mediator within the SPM involves personal choice in meaning-making of multiple dimensions in care, work, and personal life. Our findings indicated meanings made from different sources lead to diverse meaning-making experiences, each of which is dynamically developed. Such a phenomenon aligns with the assumptions of MFT in a way that each meaning-making choice is dynamic and has a developing impact on one's well-being outcome (Noguchi, 2019).

In addition, the moderated mediation models further underline the importance of the dyadic relationship to the improvement of WDC's well-being. These findings support SDT, where the relatedness serves as the innate needs and intrinsic motivation that can potentially benefit people's well-being (Ng et al., 2012). First of all, for WDCs with a lower meaning-making level, better dyadic relationships significantly have a strong positive influence on their well-being. Both parents and children are reported to have better subjective well-being when they are more intimate and close in terms of family relationships in diverse family structures (Dinisman et al., 2017). Families with better familial relationships and interpersonal quality have better physical and emotional health, higher life satisfaction, and salutogenic characteristics (Grevenstein et al., 2019). On the other hand, interestingly and surprisingly enough, a desirable dyadic relationship does not help much with the WDCs who have a higher meaning-making level. Instead, these high-meaning-makers have worse hedonic and social well-being when they have better dyadic relationships. This phenomenon can be explained by the fact that family members with better dyadic relationships may have higher expectations of communications, interactions, and rewarding gestures from the other dyad during the caregiving scenario (Alavi et al., 2011). Another study also points out that if the unmet relationship with high expectations from a family member may influence the emotional state of both dyads (Chiang & Ellis, 2019). Even though WDCs with better dyadic relationships are reporting declining well-being when they have a higher meaning-making level than those with a lower meaning-making level, these WDCs with more meaningfulness still have a slightly higher level of well-being than their counterparts.

The meaning-making choices with diverse meaning-making levels are associated with different levels of well-being. The findings support the theoretical assumptions of TM that intrinsic meaningfulness attached to events enhances one's psychological well-being. Certain choices among all meaning-making patterns result in optimal hedonic and social well-being. The best meaning-making choice for WDC's psychological well-being can be accounted for by the Theory of Meaning and Decision.

Illustrated in **Figure 1** (p.34), the moderating models in the findings fully support the proposed integrated theory in many ways. First, the integration of SPM and MFT connects the meaning-making choices and well-being. Second, SDT and TMD support the phenomenon in which certain meaning-making choices yield an ideal well-being state. Last, the dyadic relationship has a positive impact on enhancing WDCs' hedonic and social well-being, especially when they have a lower meaning-making level. Future studies could also address one's personality and resilience level, which can be potential contributors to the meaningful experiences.

7.5.3 Methodological Implications

7.5.3.1 Robustness of ESM Sampling

Due to the constraints in the funding budget and scheduled timeline, this project aims to recruit 100 participants in total. To maintain a statistical power level at 0.80, the effect sizes in performing a variety of analyses ranged from moderate to high levels.

A decent power level denies mis-specified models, while its moderate effect size guarantees impactful prediction of parameters on the outcome variables, say caregivers' well-being. Variability with satisfactory effect size leads to a greater variability of estimation explained by the predictor variables, and better robustness of the model. Due to the limited sample that was collected, we strived to maximally eliminate missing values, large standard errors, or lower power (Y. A. Wang & Rhemtulla, 2021). The models used in this study included linear regression models, mediation models, and moderated mediation models. These regression models have a wide range of parameters from one independent variable to 18 predictor variables, leading to a higher coefficient of determination (Mueller & Hancock, 2018).

It is noteworthy that our recruited participants were more educated than the general population in other similar studies. The implications generalized to the population with a low educational level should be made with cautions. Furthermore, our sampled Chinese population is non-religious or agnostic (97.7%). The generalization to the population with religious background should be made with deliberation.

7.5.3.2 User experiences of mobile-based ESM for participants and researchers

Response rates in the mobile-based survey commonly range from 20% to 70%, and on average is approximately 11% lower than surveys of other approaches (Bälter et al., 2005; Deutskens et al., 2004). Surveys with lower response rates, especially close to 20%, have the highest accuracy and validity than the ones with higher response rates (Fan & Yan, 2010). Even though we adopted online assessments with our ESM protocol, there are some advantages of the protocol compared to the regular online survey and conventional in-person surveys.

Its protocol design makes the assessment process easier than in-person mode because we used concise and simplified language, avoided vague or biased questions, made a clearer user interface, and used a visually-friendly presentation of the layout for the participant to follow. The user interface is adaptive to different mobile devices, ranging from tablets, smartphones, and portable laptops, as well as both orientations (horizontal and vertical) of the display. In addition, we used the auto-advance function to automatically progress to the next question when the current answer was finished. The survey is user-friendly in mobile phones, making the survey easier to complete at any time. Due to our research questions, we need participants to fill in the survey while providing care, working, or engaging in personal activities. Therefore, the phone-adapted survey is very advantageous for participants to respond to our assessment at any time and any place.

The length of an ESM assessment, not counting baseline and follow-up surveys, was designed to cost 2 to 3 minutes. In real practice, participants spent around 30 to 90 seconds on one ESM assessment. The options for items, such as PANAS items, were designed with default answers on the neutral level for participants easier to pick the mood level that is abnormal and save time to select check the neutral level of other unchanged moods. Such feasibility allows a participant to take assessments during their break or on their commute. Meanwhile, the Qualtrics system records the progress of each assessment from all devices. Therefore, participants who accidentally quit the assessment can come back anytime to pick up what has been left from the last filling-in. New assessments can be offered only when a past assessment is completed. This function guarantees completion of each assessment and avoid partially completed assessments.

The accessibility to assessment power by the Qualtrics platform is much easier than web links provided in other traditional online survey methods (Fan & Yan, 2010). Instead, the accessible link and QR code of the assessment were both reusable in any daily reminder message. The link is clickable and directs to the survey interface that can be reused anytime. The QR code assists transfers between devices in case a participant needs to switch devices.

The costs, excluding incentives, are more economically competitive than the face-to-face approach. The internet coverage, including Wifi and data plan, is prevalent and inexpensive nowadays, besides the low cost of data loading the web-based survey page. Participants are able to access the interface regardless of time and location. Moreover, the memory function of Qualtrics enables the resumption of assessment even when the assessment is accidentally discontinued. This mechanism saves not only more data spent to reload the page, but also time to refill the lost part of an assessment if it is discontinued.

Incentives are secured and convenient for delivery and use. Once a participant finished the entire survey on double-checking, the incentives can be given via online transaction (e.g., WeChat pay or red packets). The transaction is delivered instantly with the official receipt that can be legally tracked. The transaction is as secure as a bank transaction and costs less time to reach the participants' accounts via the electronic wiring system. The electronic incentives are obviously more convenient than mailing the paper-based coupons and bank wiring method. Participants can use the incentives right after we send it out without waiting for one second. Incentives can be availably accepted in any online and offline stores, unlike coupons that are limited to only a selection of stores. An online transaction is more flexible than using a coupon. For instance, some participants devoted more time to the interview besides the survey. We are able to pay the extra incentives for the interview by easily making another amount of incentive transactions over the phone.

The ESM protocol is pandemic adaptive. The entire electronic survey, including assessments and incentives, successfully avoid face-to-face contact during several COVID outbreaks (e.g., February, May, and June in Guangzhou). Electronic signatures, in comparison to the traditional process, made the process even easier. Past protocol often requires the participant to review the consent form and make the signature. During the pandemic, when social distance and no face-to-face contact have been strongly suggested, the informational session and informed consent process can be performed online via video conferences and electronic signatures. The electronic signatures are securely stored and kept confidential.

The data collection is secured and convenient for researchers. First of all, the Qualtrics system is professionally used for social science survey and maximally ensure confidentiality during and after the survey. The data can be directly filled in the built-in survey form that securely encrypts input data and transfers them to the database. The database can only be administered by the researchers with credentials. The online database can be easily backed up by exporting the data to local computers. The system has the version control function, with which one can undo any changes or restore any careless operation. The "force-response" function efficiently prevents accidental or man-made data attrition, which usually costs a great amount of time to retrieve the missing data. The electronic form also saves human resources and time in collecting data. The traditional approach requires either researchers or participants to fill out the paper-based questionnaire first. Then more researchers and research assistants are involved to input all questionnaire data into a database. Then the data input needs to be double-checked by other researchers to verify its accuracy of data input. Data organization is arranged to unify data formats and variable names. In comparison, the electronic survey form allows to input data directly to the electronic database with the "force response" and "data validation" functions that coerce non-missing values, a unified data format, and specified variable names. This mechanism greatly saves a portion of human resources and time. The Qualtrics system offers a data visualization function, with which we can inspect the data abnormality and outliers anytime during the data collection process. Qualtrics provides various data distribution methods, such as the web link, anonymous web link, QR code, email notification and media share. We distributed the electronic survey to potential participants via online chat groups (e.g., WeChat groups, QQ groups, media subscribers) organized by doctors, case managers, and the administration of social service organizations. The data output system supports diversified formats that are usable in SPSS, SAS, STATA, and comma-separated values files that are compatible to R, Python, and Microsoft Excel.

7.5.3.3 Ecological Validity and Reliability in Measurement

The ecological validity demonstrates that our data characteristics represent real-world situations. One noteworthy scenario is a small portion of caregiving occurred at the workplace. Based on our available interview content for cross-validation, there are a few examples supporting this counter-intuitive scenario. One case is a psychiatric doctor whose father is widowed and living with mild symptoms of dementia. He occasionally took his father to the psychiatric wards where he works at. Most time during his work, he frequently needs to take and receive phone calls from his father, who is alone by himself during the daytime. Another case is a female salesperson who cares for her mother alone by herself. She is allowed to bring her mother to the visitors' longue at her workplace. A third example is a male software engineer who often works at home. He and his father care for the mother on a daily basis. Strictly he is often working from home, but he considers his study room as a workplace. Empirical studies also showed that during the COVID pandemic people more often had remote working conditions (Brynjolfsson et al., 2020). Even though working from home has not been quite encouraged, certain occupations (e.g., salesmen, IT workers, financial analysts) may allow home office during extreme pandemic conditions (Fadinger & Schymik, 2020).

Measurements generally demonstrated satisfactory reliability and validity. Most instruments are well-developed ones, such as ZBI, URCS, PANAS, SWB, FMTC, PAC and NPI, and have desirable reliability (≥ 0.78) and validity (≥ 0.777). There are a few prestigious instruments with lower index values, such as PHQ and NPI. Even though we use PHQ to evaluate caregivers' depressive levels, none of these participants are officially diagnosed with depression or have a depressive tendency. The averaged PHQ score of 4.94, ranging from 0 to 10, indicates minimal depressive to mild depressive levels. None of these participants have depressive symptoms at moderate and severe levels. Therefore, PHQ may not be reliably and accurately tested in the population who are relatively mentally healthier. NPI is another well-known inventory. Its NPI severity sub-scale presents well in reliably detecting PwD's dementia symptomology and its disturbance to the caregiver. However, the validity of both sub-scales is only on the verge of 0.70. The reason why our data does not accurately reflect the true nature of symptomatology may be because our sampled population was primarily medicated PwD with medication compliance and professional community assistance. These participants may not present a full spectrum of dementia symptoms in comparison to those who have untreated PwD with behavioral, emotional, cognitive, and other functional symptoms. The NPI severity sub-scale can be less severe and accurate based on our sampled data. Therefore, caregivers may experience a less intense caregiving burden with the help of medication and community services and support. Their NPI disturbance scores and PHQ scores (depressive levels) may be affected as they are not living with the real-world symptoms of PwD. WAMI is a newly developed inventory while its Chinese version has not yet been officially available. Both of the English and Chinese versions lack broad empirical validation (M. F. Steger et al., 2012). Surprisingly, we find high reliability and validity of WAMI in our sampled data. Future studies on this topic should provide an insightful perspective on the revised version of WAMI provided to people who have work and care as their primary daily living activities. ADL and IADL have been widely used in clinical and community settings. Their validity and reliability have not been seriously investigated due to complicated scenarios of application (Holmqvist & Holmefur, 2019). ADL and IADL were used as the inclusion criteria rather than as any independent variables in this project. These two scales are more crucial as screening tools used in clinical practices and community services, and less often used as psychological instruments. It is expected that their reliabilities are less than satisfactory for our sampled participants. OSI is not a frequently used inventory and has relatively low validity. Our sampled caregivers suffer from multiple stressors in daily life and are not the same as the regular working class. A more adapted occupational stress inventory can be validated in this target population.

The categorization of meaning-making choices was developed based on the high/low criteria of meaning-making measurements in each aspect in the present study. Such categorization could be challenging as this is a bit arbitrary and data driven. One possible alternative methodology in future studies, can treat these meaning-making measures as continuous variables as a reference to more accurately group working dementia caregivers with heterogenous meaning-making choices from a statistically-driven perspective. There are also possible statistical approaches that might could address the categorization issues based on continuous variables, such as propensity score analysis, latent profile analysis, or latent class analysis (Gudicha et al., 2016; Mueller & Hancock, 2018), which could be tried out in future.

7.5.4 Limitations

The data collected in the proposed regions may not fully reflect the true population and the issue with less favorable representative power. Big scales, such as SWB (33 items) and FMTC (43 items), may discourage participation and cause attrition in data. ESM using 2 or 3 times per day may not well reflect the actual daily life of participants. The least data points suggested as the threshold could be potentially small (estimated at least 11 participants for 285 data points) and cause insufficient information on demographic and relevant variables may. Missing some control variables may impact our understanding of the issue. In addition, the measurement of well-being has been limited in positive and negative affectivities as well as social well-being. A more comprehensive measuring system can be adopted in future ESM studies for monitoring well-being of working dementia caregivers.

Qualitative studies should be conducted on the perception of meaning-making choices as no empirical evidence was available to support this concept. Interviews on WDCs shall be conducted to understand whether they make choices on meaning-making depending on the situations.

The cost analysis of caregiving support services and community services can be evaluated if the meaning-focused intervention is used to demonstrate the effectiveness of meaning-making choices for WDCs. Moreover, cultural differences shall be addressed in the framework and cost-benefit analysis in future studies.

The biased sample causes not only inaccurate PHQ and NPI measurement, but also fails to truly reflect the real living circumstances of PwD and their caregivers. The current channel in collecting data has been through the hospitals and the communities. In future studies, we hope to have standardized early screening for dementia so that services and research can be popularized in the community to capture more accurate data.

Chapter 8 Conclusion

Choices of meaning-making in various daily living activities are crucial in enhancing WDCs' hedonic and social well-being. In addition, to improve a WDC's well-being, a better dyadic relationship sustain a satisfactory well-being level when a WDC has a higher meaning-making level, and a stronger enhancement effect when with a lower meaning-making level. Taking an interesting yet practical view, this study uses an integrative design and ESM measurement to evaluate WDCs' well-being and contribute to the caregiving experiences in the era of healthy aging: 1) meaning-making of all three aspects can have the best well-being level; 2) the proposed meaning-making choice theory indicates that WDCs' well-being is determined accordingly by the hierarchical meaning-making levels (i.e., the choice on meaning-making of more aspects have a better outcome than that of fewer aspects); 3) ESM provides a more accurate picture on WDCs' well-being during a period of time, rather than one-time measurement; 4) dyadic relationship is a critical factor to improve one's well-being somehow at all meaning-making levels.

First, among all possible meaning-making choices, meaning-making of care, work, and personal experiences is associated with a high level of baseline hedonic and social well-being. In addition, WDCs with meaning-making of all three aspects have the best hedonic and social well-being during the ESM period. For flourishing caregivers with meaning-making of two or three aspects, they have much higher hedonic well-being than languishing caregivers. Caregivers with meaning-making choices of 1, 2, 4 and 6 have higher social well-being. On the other hand, we find WDCs with meaning-making of one or none aspects, they have much lower hedonic well-being. For meaning-making choices of 3, 5, 7 and 8, they have lower social well-being. In practice, meaning-oriented counseling and caregiver support services can be provided to WDCs with low hedonic and social well-being.

Second, the study has proposed the very first model on meaning-making choice, which paves a preliminary understanding on meaning-making choices among care, work, and personal life. Based on all eight choices categorized, future studies can examine specific categories that have been identified in this study, such as filial caregivers, self-centered caregivers, workaholic caregivers, etc. Furthermore, in-depth analysis can be conducted in the future to conclude more generalized categories, such as what we have found in this study – languishing caregivers, and high work-meaning caregivers. Moreover, other aspects of daily living activities can be potentially explored in the future, such as physical activities, social interactions, and dyadic communications (Cain et al., 2018; Krause, 2007; X. Zhang et al., 2018).

Third, this study proves that ESM is a useful tool for examining hedonic well-being in a more stable and accurate way than cross-sectional data. In the mediation models and moderated mediation models, the improved hedonic well-being is also found to be more reasonably associated with the associated factors and better delineated by the proposed theoretical models. In this study, we use the mean score across the 14-day ESM period for each item of hedonic well-being. Better statistical approaches can be used in future studies. For instance, the median absolute deviation, also known as MAD, refers to an effective calculation of variability within the longitudinal or panel data. MAD has been broadly used in disciplines of astronomy, biology, engineering, geography, environmental sciences, other social sciences and natural sciences (Gorard, 2013). MAD is advantageous in revealing relative conciseness, measurement efficacy, daily experiences, and eliminating distortion and outliers that are commonly observed from the calculation of standard deviation (Konno & Koshizuka, 2005). MAD is not used in this study due to our main research focus on the validation of the proposed theoretical models. We somehow do not address the within-person variances too much for this sake. Of course, future studies on within-personal variances and fluctuations in hedonic well-being can try MAD approach or other ESM-adaptive approaches.

Last, the dyadic relationship being part of the proposed theoretical framework plays an important role in moderating the impact of meaning-making choices on both hedonic and social well-being. Better dyadic relationships have significant positive influences on the well-being of WDCs with a lower meaning-making level. Counterintuitively, high-meaning-makers have worsened hedonic and social well-being when they have better dyadic relationships. Even though WDCs with better dyadic relationships are reporting declining well-being when they have a higher meaning-making level than those who with a lower meaning-making level, these WDCs with more meaningfulness still have a slightly higher level of well-being than their counterparts.

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Appendix

Appendix I. Ethical approval

THE UNIVERSITY OF HONG KONG

香 港 大 學

October 8, 2020

Mr. Shuangzhou CHEN
PhD Student
Department of Social Work and Social Administration

Dear Mr. CHEN,

Application for Ethics Approval
HREC's Reference Number: EA200080

I refer to your application for ethics approval of your project entitled "Well-beings of meaning-making choice profiles among working caregivers of dementia parents".

2. I am pleased to inform you that the application has been approved by the Human Research Ethics Committee (HREC) regarding the ethical aspect of the above-mentioned research project, and the expiration date of the ethical approval is October 7, 2024.

3. Please be reminded of the following points concerning the approved project:

(a) The HREC reference number of your project (EA200080) has to be shown in all materials sent to potential and actual participants to enable participants to link the materials to an approved project.

(b) You should report to the HREC any amendments and new information on the project using the prescribed form ('Application for Amendment of an Approved Project' downloadable from the Research Services website (http://www.rss.hku.hk/integrity/ethics-compliance/hrec)). Application for extension should be submitted well before the initially approved expiration date.

(c) Any deviation from the study protocol or compliance incident that has occurred during a study and may adversely affect the rights, safety or well-being of any participant or breaches of confidentiality should be reported to the HREC within 15 calendar days from the first awareness of the deviation/incident by the Principal Investigator.

Yours sincerely,

Secretary
Human Research Ethics Committee

THE REGISTRY
POKFULAM ROAD, HONG KONG. TEL:(852)2859 2111 FAX:(852)2858 2549

Appendix II. Consent form

Informed Consent Form

Project Title: Meaning-making and well-being of Chinese working dementia caregivers

You are invited to participate in a research study conducted by Mr. Shuangzhou Chen in the Department of Social Work and Social Administration at the University of Hong Kong.

PURPOSE OF THE STUDY

The purpose of this study is to explore adult child dementia caregivers' well-being and daily living experiences in balancing care, work and personal leisure life; and understand how they make sense of these experiences; and facilitate the development of supportive services and policies that positively affect dementia caregivers.

PROCEDURES

This study recruits participants who are adult child caregivers of parents living with dementia. This study consists of three phases: 1) a baseline phase, which records the basic information of both caregivers and their care recipients as well as the meaning-making and well-being of caregivers will take approximately 40 to 60 minutes to finish during one face-to-face interview or online survey; 2) the second phase of a longitudinal study for 14 days using Experience Sampling Methods (ESM). Participants will receive reminder text messages or calls on their mobile phones for the online survey 2 to 3 times per day depending on their activities (care, work, or personal leisure life). The ESM will be conducted using online survey only and each survey will take 3 minutes to finish; and 3) a follow-up phases after ESM period will take 15 to 20 minutes to finish via face-to-face interviews or online surveys.

POTENTIAL RISKS / DISCOMFORTS AND THEIR MINIMIZATION

There are no potential or foreseeable risks involved for participation. This study is completely safe and will not cause and discomfort during any stage of the survey. Participants can opt out from this study anytime at their discretion without any negative consequences. On the other hand, there is no direct benefits to you even though your participation will greatly benefit our understanding of dementia caregivers and their daily living experiences and the development of policy implication for better supportive services for these caregivers.

COMPENSATION FOR PARTICIPATION

Once your participation completes, a monetary token will be given as a compensation for your time spent and support to our study.

POTENTIAL BENEFITS

This study potentially benefits researchers, practitioners, and caregivers with a clearer understanding about the experiences of care, work and personal leisure life and how they make sense of these experiences; and provides a fundamental work for informing further development of supportive services and policies for dementia caregivers.

CONFIDENTIALITY

This study promises strict confidentiality and that the information obtained in the study will be used for research purposes only. Confidentiality will be practiced while private info will be handled carefully. Sensitive information will be maximally avoided during the study while the recorded personal or sensitive information will be encrypted in the archived folder only for the use of reference. Research ID (not including any personal identifier) will be created as the identifier of each case for further follow-up studies. Follow-up protocol (depending on the specific design) will be complied. Any data will be stored in password protected laptop, external hard drive and back-up server in laboratory. Data will be backed up monthly in laboratory server. At least 3 copies will be made.

DATA RETENTION

All data will be retained for at least 5 year after the end of the project. Data will be reused to validate the research findings when requested. Dara will also be used to conduct follow-up study if there is any. Data can be deleted after 5 years since publication. Dataset will be kept in password-protected laptop, external hard drives and laboratory server before deletion.

PARTICIPATION AND WITHDRAWAL

Your participation is voluntary. This means that you can choose to stop at any time without negative consequences.

QUESTIONS AND CONCERNS

If you have any questions about the research, please feel free to contact Mr. Shuangzhou Chen (phone: 852-56152374/+86-13815254846; email: chenshuangzhou@hku.hk). If you have questions about your rights as a research participant, contact the Human Research Ethics Committee, HKU (2241-5267).

SIGNATURE

I ____________________(Name of Participant) understand the procedures described above and agree to participate in this study.

____

Signature of Participant:
Date of Preparation:
HREC Approval Expiration date:
HREC Reference Number:

Appendix III. Recruitment flyer

Appendix IV. Questionnaire

《失智症长者的照顾者经历》调查流程介绍

在线问卷方式（供个人参考）

受访者根据提供的二维码（手机扫描）或者收到的问卷链接（电脑上）进行填写。本问卷调查包含三个部分，分别在三个不同的时间填写：

1） 前期调查（第一次调查或者访谈时一次性填写，总共会花费 30-40 分钟完成）：筛查问题，知情同意书和基线调查；

2） 经验取样法调查（每天根据活动安排做 2-3 次，每次花费 1-2 分钟完成）；

3） 后期跟踪调查（当受访者完成 14 天的经验取样法调查后将在第 15 天进行后期调查，大概会花费 5-10 分钟）。

填写每部分问卷时请按照在线问卷中的指导语进行。

机构发放纸质版方式（供机构工作人员参考）

每一个文件袋中都有 5 份文件或问卷，使用前先检查文件是否齐全：

A. 前期筛查问题（1 分钟左右完成）；

B. 知情同意书（1 分钟左右完成）；

C. 基线调查（30-40 分钟完成）；

D. 经验取样法调查（每天 2-3 次在线问卷；每次需要 1-2 分钟）；

E. 后期跟踪调查（5-10 分钟左右完成）。

A，B 和 C 问卷需要在第一次的调查或者访谈时一起填写，总时间会花费 30-40 分钟完成，取决于受访者的状态。可使用纸质问卷，也可选择在线（用手机或电脑）完成。当受访者完成 A，B 和 C 三份问卷后，由机构工作人员检查是否填写完整，有无缺漏填的题目。并核对每份问卷第一页右上角的受访者编号是否一致。**D 问卷**会根据之前受访者填写的时间安排表在接下来的 14 天内每天通过手机发送问卷的链接进行 2-3 次的问卷调查，每次花费 1-2 分钟完成。D 问卷的纸质版作为参考可以预先给受访者熟悉问题（每次都是问相同的问题），填写纸质的问卷无效；必须于在线问卷（用手机或电脑）上完成。**E 问卷**用于 14 天的经验取样法调查后将在第 15 天进行后期调查，需要 5-10 分钟。可使用纸质问卷，也可选择在线（用手机或电脑）完成。机构可以根据之前的受访者编号，找出对应文件袋中的相同编号的问卷给受访者填写。填写完成后收回到相应编号的文件袋中存档。

受访者选择纸质版的问卷需要由研究员再次验收；受访者完成的电子版则由研究员随时检查。若问卷填写中发现有问题，机构工作人员需要直接联系受访者要求继续完成或修改问卷；若没有问题，研究员登记并联系受访者发放现金券或转账作为报酬。

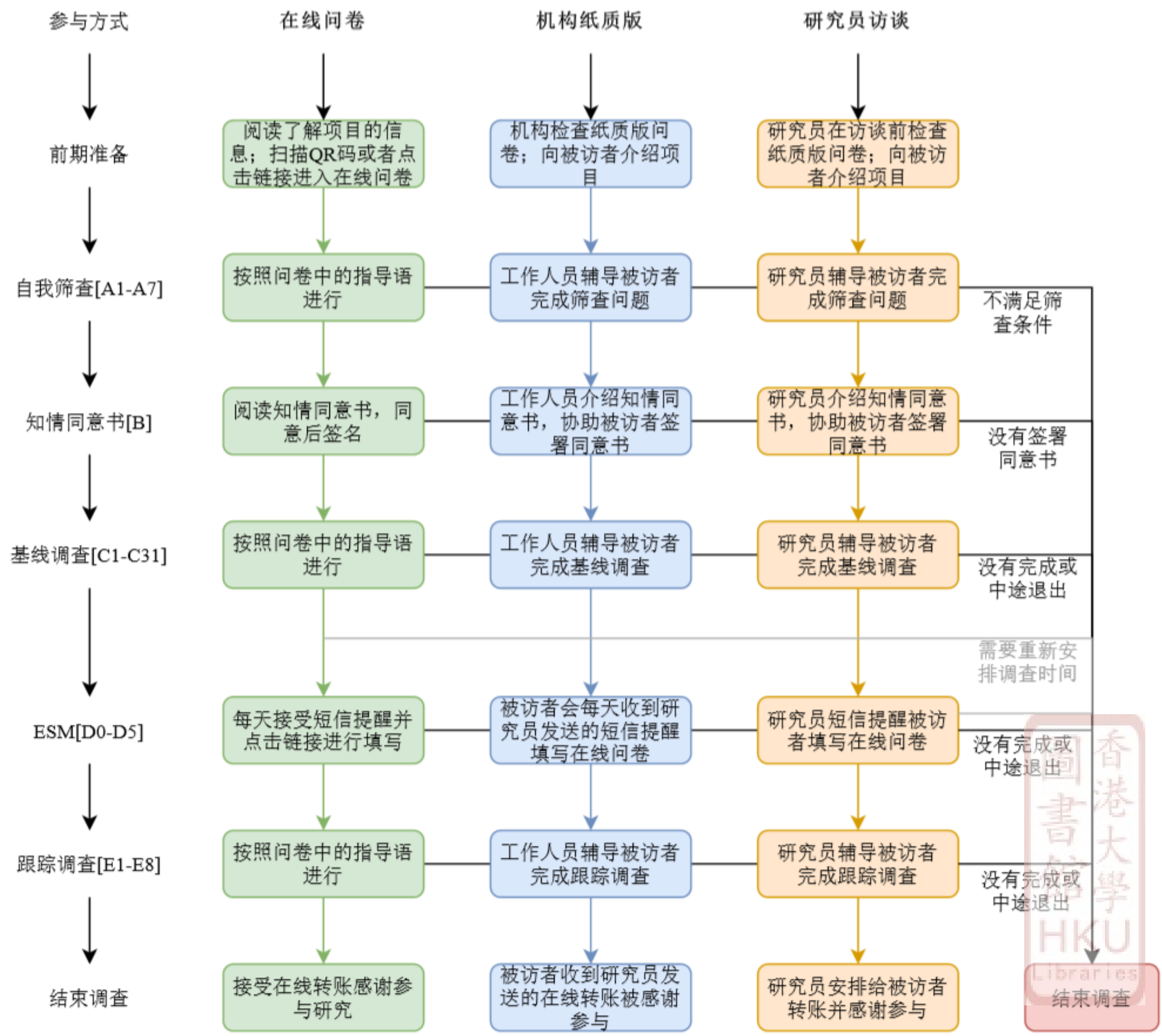
研究员访谈方式（供研究员参考）

研究员随身携带整套纸质版问卷以供说明参考之用：A.前期筛查问题（需要 1 分钟），B.知情同意书（需要 1 分钟），C.基线调查（需要 30-40 分钟），D.经验取样法调查（每天 2-3 次，每次需要 1-2 分钟），E.后期跟踪调查（需要 5-10 分钟）。研究员应强烈建议受访者用手机直接在线填写，

并在其身旁随时准备提供讲解。如果受访者在线填写出现困难，研究员可以以面对面的形式给受访者提供纸质版问卷并读给他/她听，同时用自己的手机或者电子设备帮助受访者填写在线调查。

《失智症长者的照顾者经历》调查流程图

个人直接做在线问卷，通过机构做纸质版问卷，或者通过和研究员一对一访谈进行调查请参考以下流程图：



失智症长者的成年子女照顾者的生活经历和幸福感 问卷调查

尊敬的先生/女士：

本人是来自香港大学社会科学学院社会工作及社会行政学系的博士生，正在进行一项关于失智症（老年痴呆症）长者的成年子女照顾者（以下简称“照顾者”）生活经历和幸福感的调查的研究。此研究为了解成年子女照顾者对于生活、工作和照顾意义的想法，以推动对于失智症照顾者所提供的支持服务及政策的发展。

如果您有在广州照顾患有失智症（老年痴呆症）的父母，并且有使用智能手机的习惯，欢迎前来参加我们的研究。当您通过前期自我筛查检查并满足本研究的招募条件后，可以开始正式的问卷调查。

本调查问卷包含三个阶段：第一阶段的问卷调查（可自选纸笔填写或手机在线填写）将向您询问基本情况，大概需要 30-40 分钟（手机填写比较快）。第二阶段的调查将持续 14 天，根据您已提供的工作、照顾以及个人休闲娱乐体育活动的时间安排，每天向您发送 2-3 次短信或电话提醒和问卷的在线链接，每次问卷的填写时间大概需要 1-2 分钟。第三阶段的跟踪调查（可自选纸笔填写或手机在线填写）将在第二阶段结束后的第 15 天进行，大概需要 5-10 分钟（手机填写比较快）。完成所有阶段的问卷后，本人或研究员将会和您继续保持联系，并通过现金券或转账方式赠予您 300 元人民币表示感谢。

希望您能对此研究给予支持，并愿意参与本次的问卷调查。请联系机构，本人、或者直接扫描下方提供的二维码直接开始参与调查。如果您对本次研究有任何问题或意见，请随时联系本人(电话 13815254846；邮件 chenshuangzhou@hotmail.com)。

陈霜洲先生
香港大学社会科学学院
2020 年 9 月 1 日

A.前期自我筛查

您好，本次研究旨在了解失智症照顾者的照顾经验和幸福感。请先完成以下的筛查条件问题。如果满足以下描述的条件，请在对应的问题后面勾选“是”；反之勾选“否”。如果有任何一项为“否”即表明您不符合本次研究的招募条件，十分抱歉您将无法参与后续的研究。如果满足全部条件请同时填写知情同意书并可开始下一步的调查。

编号	筛查条件	是	否
A1	受照顾老人被医学诊断为失智症（老年痴呆症）		
A2	在过去一个月内，受照顾老人有任何一项症状评估量表(NPI)中提到的情况：1)妄想，2)幻觉，3)烦燥/攻击行为，4)抑郁/情绪低落，5)焦虑，6)情绪高涨/兴奋，7)情绪淡漠/冷漠，8)抑制解除，9)易怒/情绪波动，10)异常的动作行为，11)睡眠异常，12)缺乏食欲（**可参考表格下方的具体解释）		
A3	您和受照顾的老人都是广州居民（有广州户口，或者在广州居住满半年并持有广州居住证）		
A4	您是成年（18 岁及以上）子女照顾者，并没有因为照顾向老人收取费用		
A5	您**每周**有 8 小时或以上的时间用在照顾老人		
A6	您**每周**有 8 小时或以上的时间在工作		
A7	您**每天**有 30 分钟或者以上的时间用在个人的休闲、娱乐或体育活动上		

1) **妄想**：老人认为其他人正偷他/她的东西，或计划去伤害他/她。
2) **幻觉**：老人听到不存在的声音或跟不存在的人说话。
3) **烦燥/攻击行为**：老人有情绪或者行为上的冲动。
4) **抑郁/情绪低落**：老人是否会悲伤、情绪低落或哭泣等表现。
5) **焦虑**：老人有因为跟您分开而变得不安或其他焦虑的病征，如呼吸困难、叹气、不能放松或过度紧张。
6) **情绪高涨/兴奋**：老人没有理由地显得过分兴奋。
7) **情绪淡漠/冷漠**：老人对平时常做的事、或对家人和朋友安排的活动计划表现出兴趣减少。
8) **抑制解除**：老人不加思考而作出冲动行为。例如他/她有跟陌生人随性谈话；或者不顾他人感受地去评论他人。
9) **易怒/情绪波动**：老人变得没有耐性、脾气暴躁、容不得拖延等待。
10) **异常的动作行为**：老人有重复的动作或行为；如在家中漫无目的地踱步；或反复地玩弄衣钮，绳子；或作出其他重复的行为。
11) **睡眠异常**：老人在晚上大吵大闹，或过早醒来(比他/她以往的起床的时间更早)，或在白天睡得太多。
12) **缺乏食欲**：老人的体重有显着的减轻或增加。喜欢的食物种类有变化。

B.知情同意书

项目课题：失智症长者的成年子女照顾者的生活经历和幸福感（博士研究项目）

项目目的：为了解失智症（老年痴呆症）长者的成年子女照顾者对于生活、工作和照顾意义的想法，并推动对于这些照顾者相关的支持服务及政策的发展。[更多关于项目的内容和背景可以参考项目资料或询问调查员]

调查过程概述：本研究招募失智症长者的子女照顾者。该研究包含前期的基线研究、中期的经验取样法(ESM)研究以及后期的跟踪研究。在征得您同意的情况下，我们会定时通过手机短信或者电话提醒您，并附带在线调查问卷的链接给您填写。本调查问卷包含三个阶段，第一阶段的问卷调查将向您询问基本情况，大概需要 30-40 分钟。第二阶段的调查将持续 14 天，根据您已提供的工作、照顾以及个人休闲娱乐体育活动的时间安排，每天向您发送 2-3 次短信或电话提醒和问卷的在线链接，每次问卷的填写时间大概需要 1-2 分钟。第三阶段的跟踪调查将在第二阶段结束后的第 15 天进行，大概需要 5-10 分钟。完成所有阶段的问卷后，研究员将会和您继续保持联系，并通过现金券或转账方式赠予您 300 元人民币表示感谢。[更多关于项目的流程可以参考项目资料或询问调查员]
受访者获益及风险：本研究的问卷调查不会对您构成日常生活以外的额外风险。该研究没有和您产生直接的利益关系，但会帮助我们更加深入的了解失智长者的照顾者的真实生活感受和经历以及他们的幸福感状况。您提供的信息将为长者之照顾者的支持服务及政策的发展及完善提供宝贵的资料，也为学术研究提供了分析的基础。为表示您对我们研究支持的感谢，在完成整个调查之后，我们将支付给每位受访者 300 元人民币。请务必留下您的联络资料，我们将在研究结束后联系您以现金券或转账的方式给您答谢。

参与原则：参加本调查期间，您不需要支付任何费用。您可以自由地决定是否愿意参加该研究，或在研究过程中中途退出。您不会因为拒绝参加或者退出研究而受到伤害或者惩罚。这项研究绝对安全，不会令您产生不安。这次问卷调查纯属自愿参与性质，您可随时终止回答问卷，有关决定将不会引致任何不良后果。如果有研究问题令您感到不安，可以和我们沟通或拒绝回答。

保密原则：本研究严格对您的个人资料、信息以及隐私进行保密。研究结束后，我们不会没有经过您的同意开展上述提到研究以外的任何研究，同时伦理委员会应被允许在必要时检查研究数据。因此，如果我们需要扩展研究内容，将会再次跟您联系并得到您的认可，而且我们届时还会单独获得伦理委员会的批准。没有您的同意或伦理委员会的批准，这些研究不能也不会在任何条件下进行。研究过程中，每一位受访者都会被分配一个特定号码，所收集的资料绝对保密，且只供作学术研究用途。 所有研究报告内容都不会披露任何受访者的身份，所有数据会在研究报告发布五年后销毁。有需要的话，每个受访者都有权利获得其个人的数据以及公开报告的研究结果。您有权对您个人资料进行保密，如在本项研究中或与本项研究有关的个人资料的收集、保管、保留、管理、控制、使用（分析或比较）、在内外转让、不披露、消除和/或任何方式处理。如有任何问题，您可以咨询本研究所属单位香港大学的隐私资料私隐专员或致电该办公室（电话号码：852-28272827），来帮助您认识和了解依法保护隐私的意义，以及共同监督并保护您的个人资料。

问题和疑虑：如果您对本次研究有任何问题及疑虑，请联系陈霜洲先生（电话：13815254846）及其导师楼玮群教授（电话：852-28315334/852-39174835; 电邮：wlou@hku.hk）。如果您想知道更多有关受访者的权益，请联络研究所属的香港大学研究伦理操守委员会（电话号码：852-22415267）。如您明白以上内容，请在下方签署您的姓名以表示同意参与本调查。非常感谢您的帮助。

[如同意请在背面的签名页签名]

B.知情同意书签名

研究者声明

“我已告知该受访者研究项目的研究目的、步骤、风险及获益情况,给予他/她足够的时间阅读《知情同意书》、与其他人讨论咨询，并解答了其有关研究的问题。我已告知该受试者当遇到与研究相关的问题，或遇到与自身权利或权益相关问题时可随时与陈霜洲先生（电话：13815254846）或香港大学研究伦理委员会（电话号码：852-22415267）联系。我已告知该受访者可以随时退出本研究。我已告知该受试者将得到这份《知情同意书》的副本，上面包含我和他/她的签名。”

受访者姓名（工整书写）　　　　日期

受访者签名

受访者声明

“我已被告知该研究项目的研究目的、步骤、风险及获益情况,并有足够的时间阅读《知情同意书》、与其他人讨论咨询，并得到了相关问题的回复并对答复很满意。我已被告知当遇到与研究相关的问题，或遇到与自身权利或权益相关问题时可随时与陈霜洲先生（电话：13815254846）或香港大学研究伦理委员会（电话号码：852-22415267）联系。我已被告知可以随时退出本研究。我已被告知将得到这份《知情同意书》的副本，上面包含我和访问员的签名。”

受访者姓名（工整书写）　　　　日期

受访者签名

受访者联系方式

手机号码：____________________
家庭电话号码：____________________
紧急联系人电话号码：____________________

C.基线问卷

受照顾老人的信息

（如果父母亲都有失智症，请根据患病更严重的一个老人的情况来填写以下问卷）

C1. 请问老人的性别是：☐男 ☐女
C2. 请问老人的出生年份是：________ 年 (注：以身份证为准)
C3. 请问老人是您的？
☐父亲 ☐母亲 ☐配偶的父亲 ☐配偶的母亲
C4. 请问老人目前居住的地方是？
☐在家居住

☐在医院住院
☐在养老院/养老机构/老年公寓/护养院/护理院/护老院/福利院/敬老院居住
C5. 如果老人在家居住（否则可跳过该题），请问目前和老人同住的有多少人？_______人
（注：同住的人包括家人、保姆和朋友。如果老人独居请填写 0）
C6. 老人何时由医生确诊为失智症（老年痴呆症）？ ________ 年（如：2009 年）
C7. 请问老人**目前是否能独立完成**日常生活及社区生活方面的活动？ 请在适当的方格内打勾☑

日常生活 ADLs		**是**	**否**
C7_1	吃饭	1	0
C7_2	冲凉(洗澡)	1	0
C7_3	穿衣服	1	0
C7_4	从床或凳起身	1	0
C7_5	大小便控制	1	0
C7_6	去厕所 (能否走到厕所、穿脱裤子等)	1	0
社区生活 IADLs			
C7_7	购物	1	0
C7_8	做饭	1	0
C7_9	吃药	1	0
C7_10	财务管理	1	0
C7_11	外出通勤安排	1	0
C7_12	洗衣服	1	0
C7_13	接打电话	1	0
C7_14	做家务	1	0

C8. 老人的失智症现处于哪个时期？（请选择最接近的选项）
0＝早期 : 近期记忆力衰退，思考能力下降。
1＝中期 : 情绪不稳定，容易发怒，作息日夜颠倒，四处走动，日常生活需要家人协助。
2＝晚期 : 认不出家人、大小便失禁、说话和理解力迟钝、完全丧失自我照顾能力。

C9. 请问在**过去一个月内**，您照顾的长者出现以下情况的**老人失智症症状的严重程度**：

0:没有发生或者由于失智症晚期没有明显症状
1:轻微；症状明显可见，但相对于患病初期没有出现明显的行为改变
2:中等；症状非常明显，相对于患病初期有出现明显的行为改变
3:严重；症状十分明显，相对于患病初期有非常明显的行为改变

以及**对您的困扰程度**（失智症状对您情绪和心理的影响）
0:完全没有困扰
1:极少困扰，可妥善处理负面情绪
2:轻微困扰，一般可妥善处理负面情绪
3:中等困扰，不能时常轻易妥善处理负面情绪
4:严重困扰，处理负面情绪非常困难
5:极度或非常严重困扰，不能处理负面情绪

请根据情况，分别在**最后两栏中填入对应的数字**。

NPI	症状	说明	老人症状严重程度（0–3 分）	对您的困扰程度（0–5 分）
C9_1	妄想	老人是否觉得其他人会偷他/她的东西，或打算伤害他/她？		
C9_2	幻觉	老人是否会听到不存在的声音？是否有和不存在的人说话？		
C9_3	烦躁/攻击行为	老人是否有情绪或者行为上的冲动？		
C9_4	抑郁/情绪低落	老人是否会悲伤、情绪低落或哭泣等表现？		
C9_5	焦虑	老人是否有因为跟您分开而变得不安？他或她是否有其他焦虑的病征，如呼吸困难、叹气、不能放松或过		
C9_6	情绪高涨/兴奋	老人是否会没有理由地显得过分兴奋吗？		
C9_7	情绪淡漠/冷漠	老人对平时常做的事、或对家人和朋友安排的活动计划表现出兴趣减少吗		
C9_8	抑制解除	老人有否不加思考而作出冲动行为吗？例如他/她有跟陌生人随性谈话，或者不顾他人感受地去评论他人？		
C9_9	易怒/情绪波动	老人是否变得没有耐性、脾气暴躁、容不得拖延等待？		
C9_10	异常的动作行为	老人有否做出重复的动作或行为；如在家中漫无目的地踱步；或反复地玩弄衣钮、绳子；或作出其他重复		
C9_11	睡眠	老人在夜间大吵大闹？是否在早上过早醒来(比他/她以往的习惯更早)？在白天睡得太多吗？		
C9_12	食欲或饮食失调	老人的体重有否有明显的减轻或增加？他/她喜欢的食物有任何改变吗？		

照顾者和受照顾者的关系

C10. 请您回想在照顾前后和老人的关系，并分别按照以下描述的情况**打分（填写 1-7 的数字）**:
非常不同意(1)，不同意(2)，有些不同意(3)，中立态度(4)，有些同意(5)，同意(6)，非常同意(7)

	单维关系亲密量表（照顾前+照顾后）URCS-1/2	照顾之前 和老人的关系 （1-7 分）	照顾之后 和老人的关系 （1-7 分）
C10_1	我与他/她关系很亲近		
C10_2	当我们分开时，我会很想念他/她		
C10_3	他/她与我会彼此披露了重要的个人事物		
C10_4	他/她和我有紧密的关系		
C10_5	他/她和我想花时间在一起		
C10_6	我肯定我与他/她的关系		
C10_7	他/她是我生活中的一个优先事项		
C10_8	我和他/她一起做了很多事情		
C10_9	当我有空时, 我选择单独和他/她有相处时间		
C10_10	我会想很多有关他/她的事		
C10_11	我和他/她的关系对我的生命很重要		
C10_12	当我作出重大决定时，我会考虑到他/她		

照顾者信息

以下问题是有关**您自己**

C11. 您的性别 ☐男 ☐女

C12. 出生年份 ____________年(注：以身份证为准)

C13. 婚姻状况 ☐未婚 ☐已婚 ☐伴侣 ☐丧偶 ☐分居 ☐离婚

C14. 教育程度 ☐未有接受过正规教育 ☐小学 ☐中学 ☐大专 ☐大学 ☐硕士或以上

C15. 宗教信仰 ☐无宗教 ☐天主教 ☐基督教 ☐伊斯兰教 ☐佛教 ☐道教 ☐其他:_____

C16. 您从何年何月开始照顾该老人？ ________年________月 （若忘记月份，请写上 6 月）

「照顾」的意思是指为老人提供在日常生活(包括吃饭、洗澡、穿衣、从床或凳起身、大小便控制、去厕所等)，或者社区生活(包括购物、做饭、吃药、财务管理、外出通勤安排、洗衣服、接打电话、做家务等) 方面的协助，或是照看/监督/情绪支持: 包括照看或雇人照顾老人、与老人倾诉心事，鼓励老人。

C17. 您目前和老人同住吗？☐ 是 ☐ 否

C18. 您是最主要的照顾者吗？(除保姆外提供最多照顾的人) ☐ 是 ☐ 否

C19. 请问还有哪些人来分担照顾老人的责任：

☐ 没有

☐ 有，可多选 ☐ 老人的配偶 ☐ 朋友 ☐ 保姆（24 小时）

☐ 老人儿子 ☐ 老人女婿 ☐ 钟点工

☐ 老人女儿 ☐ 老人儿媳 ☐ 其他(请注明:____________)

C20. **目前**，您认为您有能力应付照顾老人的经济需要吗？

☐完全不能应付 ☐多数不能应付 ☐仅仅能应付 ☐多数能应付 ☐完全能应付

C21. 您用了多少时间去适应照顾者的角色及生活？（如果选择「已适应」，请填写适应所花的时长；如果选择「还没适应」，则说明原因）

☐ 已适应，用了______个月 ☐ 还没适应，原因：________

C22. **目前**，为了照顾该老人，您**有多经常**有以下的感受，请在最能描述您情况的选项方格内打勾☑:

	照顾压力 ZBI	**从不**	**很少**	**一般**	**经常**	**非常频密**
C22_1	您有没有觉得为了照顾老人占用您很多时间而没有时间留给自己？	1	2	3	4	5
C22_2	您有没有觉得照顾老人同您的家庭或者工作上的责任有冲突？	1	2	3	4	5
C22_3	您有没有觉得您在老人身边的时候您会觉得紧张或者有压力？	1	2	3	4	5
C22_4	您有没有觉得您不确定如何应对老人的问题？	1	2	3	4	5

C23. **目前**，在工作中您是否有以下的感受，请在最能描述您情况的选项方格内打勾☑:

	工作压力 Job Stress	**非常不同意**	**不同意**	**中立态度**	**同意**	**非常同意**
C23_1	我的工作十分复杂而且工作量很大	1	2	3	4	5
C23_2	担心工作中的个人安全	1	2	3	4	5
C23_3	经常加班	1	2	3	4	5
C23_4	责任重大，害怕被问责	1	2	3	4	5
C23_5	我的工作内容没有被清晰告知	1	2	3	4	5
C23_6	有时领导给我不同的工作要求	1	2	3	4	5
C23_7	有时我在同一时间被安排不同的工作岗位	1	2	3	4	5
C23_8	和同事有矛盾或者不快	1	2	3	4	5
C23_9	感觉被孤立	1	2	3	4	5
C23_10	缺乏上级支持	1	2	3	4	5
C23_11	绩效工资制不好	1	2	3	4	5
C23_12	表现好没有得到公司的认可	1	2	3	4	5
C23_13	我担心自己将来的职业发展	1	2	3	4	5
C23_14	工作不稳定，担心下来没有项目做	1	2	3	4	5
C23_15	我的权益没有得到保护	1	2	3	4	5
C23_16	我感觉家庭的经济负担大	1	2	3	4	5
C23_17	工作经常出差不能尽家庭的责任	1	2	3	4	5
C23_18	家人不能很好的理解和支持我的工作	1	2	3	4	5

C24.请问目前照顾老人有没有使您有以下感受，并在最能描述您情况的选项方格内打勾☑:

	照顾意义 FMTC (provisional meaning)	**非常不同意**	**不同意**	**中立态度**	**同意**	**非常同意**
C24_1	我享受与老人一起，如果他/她离开了我，我会想念他/她	1	2	3	4	5
C24_2	我会铭记我拥有的祝福（珍惜我所拥有的东西）	1	2	3	4	5
C24_3	照顾老人给予我生命的目的和意义	1	2	3	4	5
C24_4	我珍惜与老人过去的记忆和经历	1	2	3	4	5
C24_5	我是一个坚强的人	1	2	3	4	5
C24_6	照顾他人使我感觉良好和感到能帮助他人	1	2	3	4	5
C24_7	来自老人的拥抱和“我爱您”使我感到一切都是值得的	1	2	3	4	5
C24_8	我是一名斗士	1	2	3	4	5
C24_9	我很高兴我能在这里照顾老人	1	2	3	4	5
C24_10	与其他亲近的人交谈，能恢复我对自己能力的信心	1	2	3	4	5
C24_11	即使在我生活中遇上困难，我仍期待着未来	1	2	3	4	5
C24_12	照顾他人帮助我学习了关于自己的新事物	1	2	3	4	5
C24_13	无论每年生活质量如何，都是一种祝福（好运）	1	2	3	4	5
C24_14	我未必会选择现在我所处的情况，但照顾他人使我感到满足	1	2	3	4	5
C24_15	每一天都是祝福（每天都有好运或者被老天保佑）	1	2	3	4	5
C24_16	这是我的地方，我必须做到最好	1	2	3	4	5
C24_17	我比我想象的要强大得多	1	2	3	4	5
C24_18	我每天都知道我们将会一起度过美好的一天	1	2	3	4	5
C24_19	照顾他人使我成为一个更强大、更好的人	1	2	3	4	5

C25. 在过去一个月内，请问照顾长者有没有使您有以下列感受，请在最能描述您情况的选项打勾☑:

	照顾意义 PAC	**非常不同意**	**不同意**	**中立态度**	**同意**	**非常同意**
C25_1	使我感到自己更加有用	1	2	3	4	5
C25_2	使我对自己感觉良好	1	2	3	4	5
C25_3	使我觉得自己被人需要	1	2	3	4	5
C25_4	使我觉得自己被人感激	1	2	3	4	5
C25_5	使我觉得自己很重要	1	2	3	4	5
C25_6	使我觉得自己很坚强自信	1	2	3	4	5
C25_7	给自己的生活带来更多的意义	1	2	3	4	5
C25_8	使我能够学习更多技能	1	2	3	4	5
C25_9	使我更加感激生活	1	2	3	4	5
C25_10	使我对生活的态度更加积极	1	2	3	4	5
C25_11	使我与他人的关系更加坚固	1	2	3	4	5

C26. 工作对不同的人可能意味着很多不同的事情。以下项目询问您目前如何看待工作在自己的生活中的作用。请在最能描述您情况的选项方格内打勾☑

	工作意义 WAMI	**非常不同意**	**不同意**	**中立态度**	**同意**	**非常同意**
C26_1	我找到了有意义的职业	1	2	3	4	5
C26_2	我认为我的工作能帮助我个人成长	1	2	3	4	5
C26_3	我的工作对世界真的没有任何影响	1	2	3	4	5
C26_4	我了解自己的工作能增进我生命的意义	1	2	3	4	5
C26_5	我心中明白是什么让我的工作有意义	1	2	3	4	5
C26_6	我知道我的工作对世界产生了积极的影响	1	2	3	4	5
C26_7	我的工作可以帮助我更好地了解自己	1	2	3	4	5
C26_8	我发现工作有令人满意的目的	1	2	3	4	5
C26_9	我的工作帮助我了解周围的世界	1	2	3	4	5
C26_10	我所做的工作有更大的目标	1	2	3	4	5

C27. 请仔细思考一下, 是什么让您感到您的生命和存在是重要并有意义的。请在最能描述您情况的选项方格内打勾☑

	生活意义 MLQ	**非常不同意**	**不同意**	**有些不同意**	**中立态度**	**有些同意**	**同意**	**非常同意**
C27_1	我明白我生命的意义	1	2	3	4	5	6	7
C27_2	我正在寻找能让我生活感到有意义的东西	1	2	3	4	5	6	7
C27_3	我总是在寻求, 以便找到我生活的目的	1	2	3	4	5	6	7
C27_4	我生活有一个明确的目的	1	2	3	4	5	6	7
C27_5	我心中有数是什么让我的生活有意义	1	2	3	4	5	6	7
C27_6	我已经发现了一个令人满意的生活目的	1	2	3	4	5	6	7
C27_7	我一直在寻找某些可以让我的生命感到重要的东西	1	2	3	4	5	6	7
C27_8	我在为我的生活寻找一个目的或者使命	1	2	3	4	5	6	7
C27_9	我的生活没有明确的目的	1	2	3	4	5	6	7
C27_10	我正在寻找生命的意义	1	2	3	4	5	6	7

C28. 请根据照顾、工作和个人生活的不同情况下，最能描述您情况的选项方格内打勾☑

	整体和情境意义的差异 Meaning Discrepancy	**完全没有影响**	**基本没有影响**	**有一点影响**	**有一定影响**	**比较有影响**	**有很多影响**	**非常有影响**
C28_1	照顾老人怎样影响到您的世界观*?	1	2	3	4	5	6	7
C28_2	工作怎样影响到您的世界观?	1	2	3	4	5	6	7
C28_3	个人休闲娱乐体育活动怎样影响到您的世界观?	1	2	3	4	5	6	7
C28_4	照顾老人怎样影响到您认为重要的每日目标和计划?	1	2	3	4	5	6	7
C28_5	工作怎样影响到您认为重要的每日目标和计划?	1	2	3	4	5	6	7
C28_6	个人休闲娱乐体育活动怎样影响到您认为重要的每日目标和计划?	1	2	3	4	5	6	7

*<u>**世界观**</u>：理解这个世界运行以及事物发生的方式

C29. 请您根据自己当下的感受选择最适合的程度来回答以下关于积极和消极的情感。请在最能描述您情况的选项方格内打勾☑

情感量表	PANAS	没有感觉	有一点感觉	有适度的感觉	有相当感觉	有强烈感觉
C29_1	有趣的	1	2	3	4	5
C29_2	忧伤的	1	2	3	4	5
C29_3	兴奋的	1	2	3	4	5
C29_4	苦恼的	1	2	3	4	5
C29_5	坚强的	1	2	3	4	5
C29_6	有罪恶感的	1	2	3	4	5
C29_7	惊吓的	1	2	3	4	5
C29_8	有敌意的	1	2	3	4	5
C29_9	热忱的	1	2	3	4	5
C29_10	自豪的	1	2	3	4	5
C29_11	易怒的	1	2	3	4	5
C29_12	警觉的	1	2	3	4	5
C29_13	羞愧的	1	2	3	4	5
C29_14	激励的	1	2	3	4	5
C29_15	焦虑的	1	2	3	4	5
C29_16	坚决的	1	2	3	4	5
C29_17	小心翼翼的	1	2	3	4	5
C29_18	紧张不安的	1	2	3	4	5
C29_19	活跃的	1	2	3	4	5
C29_20	害怕的	1	2	3	4	5

C30. **过去两周内**，您**有多少天**因为要照顾长者而被以下的问题困扰，请在最能描述您情况的选项方格内打勾☑

	PHQ-9	完全没有（0 天）	几天（1–6 天）	一半及以上的时间（7–12 天）	几乎每天（13–14 天）
C30_1	做事缺乏兴趣或乐趣	0	1	2	3
C30_2	感到低落、沮丧、或绝望	0	1	2	3
C30_3	难以入睡、容易睡醒、或过度睡眠	0	1	2	3
C30_4	感到疲倦或精力不足	0	1	2	3
C30_5	食欲不振或过度饮食	0	1	2	3
C30_6	觉得自己很差劲、觉得自己是个失败者，使自己或家人失望	0	1	2	3
C30_7	难以集中精神，例如读书或看电视	0	1	2	3
C30_8	他人能察觉到自己的动作或说话缓慢；或更经常徘徊、心绪不宁或坐立不安	0	1	2	3
C30_9	有自残或要死的想法	0	1	2	3

C31. 请您根据自己当下的感受选择最适合的选项来回答以下有关社会幸福感的问题。请在最能描述您情况的选项方格内打勾☑

	社会幸福感 SWB	**明显不符合**	**不符合**	**有些不符合**	**介于中间**	**有些符合**	**符合**	**明显符合**
C31_1	社会正在不断发展进步着	1	2	3	4	5	6	7
C31_2	社会在不断改善我的生活状况	1	2	3	4	5	6	7
C31_3	世界正在变得越来越美好	1	2	3	4	5	6	7
C31_4	我对社会的发展充满信心	1	2	3	4	5	6	7
C31_5	我并不认为这个世界很复杂	1	2	3	4	5	6	7
C31_6	我能理解世界上发生的事情	1	2	3	4	5	6	7
C31_7	我对社会上出现的很多事感兴趣	1	2	3	4	5	6	7
C31_8	我了解这个社会运行的规则	1	2	3	4	5	6	7
C31_9	我和其他人保持着很亲密的关系	1	2	3	4	5	6	7
C31_10	这个社会令我感到舒适	1	2	3	4	5	6	7
C31_11	这个社会让我有安全感	1	2	3	4	5	6	7
C31_12	我总是能得到别人的关怀	1	2	3	4	5	6	7
C31_13	我相信人性是善良的	1	2	3	4	5	6	7
C31_14	我赞同“人人为我，我为人人”	1	2	3	4	5	6	7
C31_15	人们彼此之间应该相互信任	1	2	3	4	5	6	7
C31_16	人无完人，我们不能苛责别人	1	2	3	4	5	6	7
C31_17	我为社会创造着价值	1	2	3	4	5	6	7
C31_18	我的日常活动是有意义的	1	2	3	4	5	6	7
C31_19	我对这个社会是有积极意义的	1	2	3	4	5	6	7
C31_20	我为这个社会做出了自己的贡献	1	2	3	4	5	6	7

C32. 在**最近一个月内**，您**平均每天**用在以下活动的**时间（填写数字）**：

		周一至周五	周末
a.	照顾老人：	每天_______小时	每天_______小时
b.	工作：	每天_______小时	每天_______小时
c.	个人休闲娱乐体育活动：	每天_______小时	每天_______小时

C33. 请思考一下您第二天在照顾、工作和个人休闲娱乐体育活动方面的时间安排，并在各个事项下方勾选☑相应的时间安排。如果某事项没有安排（例如：工作），请在该事项下方的第一行“没有安排”栏中勾选☑。

事项 时间段	照顾	工作	个人休闲、娱乐和体育活动
没有安排			
清晨			
00:00 - 6:00			
上午			
6:00 - 7:00			
7:00 - 8:00			
8:00 - 9:00			
9:00 - 10:00			
10:00 - 11:00			
11:00 - 12:00			
下午			
12:00 - 13:00			
13:00 - 14:00			
14:00 - 15:00			
15:00 - 16:00			
16:00 - 17:00			
17:00 - 18:00			
晚上			
18:00 - 19:00			
19:00 - 20:00			
20:00 - 21:00			
21:00 - 22:00			
22:00 - 23:00			
23:00 - 24:00			

如果有特殊的时间安排，请在下方空白处用文字解释说明：

____

受访者联系方式

手机号码：______________________________

家庭电话号码：__________________________

紧急联系人电话号码：____________________

D.经验取样法调查问卷

（此纸质问卷填写无效，仅供提前了解问题。请扫描右方二维码完成电子问卷）

信息记录

D0_1 参与研究时填写的电话号码：__________
D0_2 今天的日期：__________
D0_3 现在的时间：__________

Chapter 9 经验取样法/生态瞬时问题

D1. 收到提醒短消息时您在哪里？

☐ 家里 [D1_1]
☐ 工作单位 [D1_2]
☐ 户外 [D1_3]
☐ 其他，并填写具体的地方：____________ [D1_4]

D2. 收到提醒短消息时您在做什么？

☐ 照顾老人（符合研究要求的老人） [D2_1]
☐ 照顾其他人 [D2_2]
☐ 工作 [D2_3]
☐ 体育活动（非照顾老人期间） [D2_4]
☐ 休闲娱乐活动（非体育运动，非照顾老人期间） [D2_5]
☐ 其他，并填写具体的事项：____________ [D2_6]

D3. 当前从事的事情是否给您带来压力？

☐没有压力 ☐有一点压力 ☐有一些压力 ☐有很多压力 ☐压力非常大

D4. 若您在目前“照顾”/“工作”/“个人娱乐休闲活动”的时间段内仍然感受到压力时是否<u>会使用</u>或<u>打算以后再尝试</u>以下的方式来应对感受到的压力？如果回答“是”请在相应的选项中打勾☑；回答“否”则不用勾选。

日常应对量表 Daily Coping Inventory		**目前会使用**	**以后尝试**
D4_1	把注意力从问题转向其他东西或进行其他活动		
D4_2	尝试从不同的角度去看待问题使其看起来更能忍受		
D4_3	思考问题的解决方案，收集相关的信息，或做一些事来解决它		
D4_4	对于问题用情绪表达来减少紧张、焦虑或挫折的情绪		
D4_5	承认问题已经发生，但对此毫无办法		
D4_6	从爱人、朋友或专家那里寻求情感支持		
D4_7	下意识地做一些放松		
D4_8	寻求精神安慰和支持		

如果有其他的应对方式，请在下方空白处补充：

积极消极情感量表

D5. 根据您当下自己的感受选择最适合的程度来回答以下关于积极和消极的情感。请在最能描述您情况的选项方格内打勾☑

	PANAS	没有感觉	有一点感觉	有适度的感觉	有相当感觉	有强烈感觉
D5_1	有趣的	1	2	3	4	5
D5_2	忧伤的	1	2	3	4	5
D5_3	兴奋的	1	2	3	4	5
D5_4	苦恼的	1	2	3	4	5
D5_5	坚强的	1	2	3	4	5
D5_6	有罪恶感的	1	2	3	4	5
D5_7	惊吓的	1	2	3	4	5
D5_8	有敌意的	1	2	3	4	5
D5_9	热忱的	1	2	3	4	5
D5_10	自豪的	1	2	3	4	5
D5_11	易怒的	1	2	3	4	5
D5_12	警觉的	1	2	3	4	5
D5_13	羞愧的	1	2	3	4	5
D5_14	激励的	1	2	3	4	5
D5_15	焦虑的	1	2	3	4	5
D5_16	坚决的	1	2	3	4	5
D5_17	小心翼翼的	1	2	3	4	5
D5_18	紧张不安的	1	2	3	4	5
D5_19	活跃的	1	2	3	4	5
D5_20	害怕的	1	2	3	4	5

如果有任何特别的事件导致您有以上这些感觉，请写在下方：

D6. 请思考一下您下来一天在照顾、工作和个人休闲娱乐体育活动方面的时间安排，并在各个事项列中勾选☑相应的时间安排。如果某事项没有安排（例如：工作），请在该事项下方的第一行“没有安排”栏中勾选☑。**<u>一天只用填写一次，如果今天已经填写过就不必再次填写</u>**。

时间段 \ 事项	照顾	工作	个人休闲、娱乐、体育活动
[没有安排]			
清晨			
00:00 - 6:00			
上午			
6:00 - 7:00			
7:00 - 8:00			
8:00 - 9:00			
9:00 - 10:00			
10:00 - 11:00			
11:00 - 12:00			
下午			
12:00 - 13:00			
13:00 - 14:00			
14:00 - 15:00			
15:00 - 16:00			
16:00 - 17:00			
17:00 - 18:00			
晚上			
18:00 - 19:00			
19:00 - 20:00			
20:00 - 21:00			
21:00 - 22:00			
22:00 - 23:00			
23:00 - 24:00			

如果有特殊的时间安排，请在下方空白处用文字解释说明：

E.跟踪调查问卷

照顾者和受照顾者的关系

E1. 请您回想**最近两周**您和老人的关系，请在最能描述您情况的选项方格内打勾☑:

	单维关系亲密量表 URCS-3	非常不同意	不同意	有些不同意	中立态度	有些同意	同意	非常同意
E1_1	我与他/她关系很亲近	1	2	3	4	5	6	7
E1_2	当我们分开时，我会很想念他/她	1	2	3	4	5	6	7
E1_3	他/她与我会彼此披露了重要的个人事物	1	2	3	4	5	6	7
E1_4	他/她和我有紧密的关系	1	2	3	4	5	6	7
E1_5	他/她和我想花时间在一起	1	2	3	4	5	6	7
E1_6	我肯定我与他/她的关系	1	2	3	4	5	6	7
E1_7	他/她是我生活中的一个优先事项	1	2	3	4	5	6	7
E1_8	我和他/她一起做了很多事情	1	2	3	4	5	6	7
E1_9	当我有空时, 我选择单独和他/她有相处时间	1	2	3	4	5	6	7
E1_10	我会想很多有关他/她的事	1	2	3	4	5	6	7
E1_11	我和他/她的关系对我的生命很重要	1	2	3	4	5	6	7
E1_12	当我作出重大决定时，我会考虑到他/她	1	2	3	4	5	6	7

寻找意义问题

E2. **在<u>调查期间</u>，**请问照顾老人有没有使您有以下列感受，请在最能描述您情况的选项方格内打勾☑:

	照顾意义 FMTC (provisional meaning)	**非常不同意**	**不同意**	**中立态度**	**同意**	**非常同意**
E2_1	我享受与老人一起，如果他/她离开了我，我会想念他/她	1	2	3	4	5
E2_2	我会铭记我拥有的祝福（珍惜我所拥有的东西）	1	2	3	4	5
E2_3	照顾老人给予我生命的目的和意义	1	2	3	4	5
E2_4	我珍惜与老人过去的记忆和经历	1	2	3	4	5
E2_5	我是一个坚强的人	1	2	3	4	5
E2_6	照顾他人使我感觉良好和感到能帮助他人	1	2	3	4	5
E2_7	来自老人的拥抱和“我爱您”使我感到一切都是值得的	1	2	3	4	5
E2_8	我是一名斗士	1	2	3	4	5
E2_9	我很高兴我能在这里照顾老人	1	2	3	4	5
E2_10	与其他亲近的人交谈，能恢复我对自己能力的信心	1	2	3	4	5
E2_11	即使在我生活中遇上困难，我仍期待着未来	1	2	3	4	5
E2_12	照顾他人帮助我学习了关于自己的新事物	1	2	3	4	5
E2_13	无论每年生活质量如何，都是一种祝福（好运）	1	2	3	4	5
E2_14	我未必会选择现在我所处的情况，但照顾他人使我感到满足	1	2	3	4	5
E2_15	每一天都是祝福（每天都有好运或者被老天保佑）	1	2	3	4	5
E2_16	这是我的地方，我必须做到最好	1	2	3	4	5
E2_17	我比我想象的要强大得多	1	2	3	4	5
E2_18	我每天都知道我们将会一起度过美好的一天	1	2	3	4	5
E2_19	照顾他人使我成为一个更强大、更好的人	1	2	3	4	5

E3. **在<u>调查期间</u>**，请问照顾长者有没有使您有以下感受，请在最能描述您情况的选项方格内打勾☑

	照顾意义 PAC	**非常不同意**	**不同意**	**中立态度**	**同意**	**非常同意**
E3_1	使我感到自己更加有用	1	2	3	4	5
E3_2	使我对自己感觉良好	1	2	3	4	5
E3_3	使我觉得自己被人需要	1	2	3	4	5
E3_4	使我觉得自己被人感激	1	2	3	4	5
E3_5	使我觉得自己很重要	1	2	3	4	5
E3_6	使我觉得自己很坚强自信	1	2	3	4	5
E3_7	给自己的生活带来更多的意义	1	2	3	4	5
E3_8	使我能够学习更多技能	1	2	3	4	5
E3_9	使我更加感激生活	1	2	3	4	5
E3_10	使我对生活的态度更加积极	1	2	3	4	5
E3_11	使我与他人的关系更加坚固	1	2	3	4	5

E4. 工作对不同的人可能意味着很多不同的事情。以下项目询问您如何看待工作在自己的生活中的作用。请您如实指出**在<u>调查期间</u>**，是否同意以下的关于您和您的工作的每条陈述。请在最能描述您情况的选项方格内打勾☑

	工作意义 WAMI	**非常不同意**	**不同意**	**中立态度**	**同意**	**非常同意**
E4_1	我找到了有意义的职业	1	2	3	4	5
E4_2	我认为我的工作能帮助我个人成长	1	2	3	4	5
E4_3	我的工作对世界真的没有任何影响	1	2	3	4	5
E4_4	我了解自己的工作能增进我生命的意义	1	2	3	4	5
E4_5	我心中明白是什么让我的工作有意义	1	2	3	4	5
E4_6	我知道我的工作对世界产生了积极的影响	1	2	3	4	5
E4_7	我的工作可以帮助我更好地了解自己	1	2	3	4	5
E4_8	我发现工作有令人满意的目的	1	2	3	4	5
E4_9	我的工作帮助我了解周围的世界	1	2	3	4	5
E4_10	我所做的工作有更大的目标	1	2	3	4	5

E5. **在<u>调查期间</u>**，请仔细思考一下，是什么让您感到您的生命和存在是重要并有意义的。请在最能描述您情况的选项方格内打勾☑

	生活意义 MLQ	非常不同意	不同意	有些不同意	中立态度	有些同意	同意	非常同意
E5_1	我明白我生命的意义	1	2	3	4	5	6	7
E5_2	我正在寻找能让我生活感到有意义的东西	1	2	3	4	5	6	7
E5_3	我总是在寻求, 以便找到我生活的目的	1	2	3	4	5	6	7
E5_4	我生活有一个明确的目的	1	2	3	4	5	6	7
E5_5	我心中有数是什么让我的生活有意义	1	2	3	4	5	6	7
E5_6	我已经发现了一个令人满意的生活目的	1	2	3	4	5	6	7
E5_7	我一直在寻找某些可以让我的生命感到重要的东西	1	2	3	4	5	6	7
E5_8	我在为我的生活寻找一个目的或者使命	1	2	3	4	5	6	7
E5_9	我的生活没有明确的目的	1	2	3	4	5	6	7
E5_10	我正在寻找生命的意义	1	2	3	4	5	6	7

E6. 请根据照顾、工作和个人生活的不同情况下来回答以下的问题。请在最能描述您
在调查期间情况的选项方格内打勾☑

	整体和情境意义的差异 Meaning Discrepancy	完全没有影响	基本没有影响	有一点影响	有一定影响	比较有影响	有很多影响	非常有影响
E6_1	照顾老人怎样影响到您的世界观*?	1	2	3	4	5	6	7
E6_2	工作怎样影响到您的世界观?	1	2	3	4	5	6	7
E6_3	个人休闲娱乐体育活动怎样影响到您的世界观?	1	2	3	4	5	6	7
E6_4	照顾老人怎样影响到您认为重要的每日目标和计划?	1	2	3	4	5	6	7
E6_5	工作怎样影响到您认为重要的每日目标和计划?	1	2	3	4	5	6	7
E6_6	个人休闲娱乐体育活动怎样影响到您认为重要的每日目标和计划?	1	2	3	4	5	6	7

***世界观**: 理解这个世界运行以及事物发生的方式

幸福感问题

E7. 请根据您当下自己的感受选择最适合的程度来描述以下关于积极和消极的情感。
请在最能描述您情况的选项方格内打勾☑

情感量表	PANAS	没有感觉	有一点感觉	有适度的感觉	有相当感觉	有强烈感觉

E7_1	有趣的	1	2	3	4	5
E7_2	忧伤的	1	2	3	4	5
E7_3	兴奋的	1	2	3	4	5
E7_4	苦恼的	1	2	3	4	5
E7_5	坚强的	1	2	3	4	5
E7_6	有罪恶感的	1	2	3	4	5
E7_7	惊吓的	1	2	3	4	5
E7_8	有敌意的	1	2	3	4	5
E7_9	热忱的	1	2	3	4	5
E7_10	自豪的	1	2	3	4	5
E7_11	易怒的	1	2	3	4	5
E7_12	警觉的	1	2	3	4	5
E7_13	羞愧的	1	2	3	4	5
E7_14	激励的	1	2	3	4	5
E7_15	焦虑的	1	2	3	4	5
E7_16	坚决的	1	2	3	4	5
E7_17	小心翼翼的	1	2	3	4	5
E7_18	紧张不安的	1	2	3	4	5
E7_19	活跃的	1	2	3	4	5
E7_20	害怕的	1	2	3	4	5

E8. 请根据您当下的感受选择最适合的选项来回答以下关于社会幸福感的问题。请在最能描述您情况的选项方格内打勾☑

	社会幸福感 SWB	**明显不符合**	**不符合**	**有些不符合**	**介于中间**	**有些符合**	**符合**	**明显符合**
E8_1	社会正在不断发展进步着	1	2	3	4	5	6	7
E8_2	社会在不断改善我的生活状况	1	2	3	4	5	6	7
E8_3	世界正在变得越来越美好	1	2	3	4	5	6	7
E8_4	我对社会的发展充满信心	1	2	3	4	5	6	7
E8_5	我并不认为这个世界很复杂	1	2	3	4	5	6	7
E8_6	我能理解世界上发生的事情	1	2	3	4	5	6	7
E8_7	我对社会上所发生的很多事感兴趣	1	2	3	4	5	6	7
E8_8	我了解这个社会运行的规则	1	2	3	4	5	6	7
E8_9	我和其他人保持着很亲密的关系	1	2	3	4	5	6	7
E8_10	这个社会令我感到舒适	1	2	3	4	5	6	7
E8_11	这个社会让我感到安全感	1	2	3	4	5	6	7
E8_12	我总是能得到别人的关怀	1	2	3	4	5	6	7
E8_13	我相信人性是善良的	1	2	3	4	5	6	7
E8_14	我赞同“人人为我，我为人人”	1	2	3	4	5	6	7
E8_15	人们彼此之间应该相互信任	1	2	3	4	5	6	7
E8_16	人无完人，我们不能苛责别人	1	2	3	4	5	6	7
E8_17	我为社会创造着价值	1	2	3	4	5	6	7
E8_18	我的日常活动是有意义的	1	2	3	4	5	6	7
E8_19	我对这个社会是有积极意义的	1	2	3	4	5	6	7
E8_20	我为这个社会做出了自己的贡献	1	2	3	4	5	6	7

E9. 访谈问题

1. 为什么您想寻找照顾、工作和个人生活的意义？
2. 寻找意义有没有影响到您的幸福感体验？若有的话，它是如何影响您的幸福感？

受访者联系方式

手机号码：______________________________

Appendix V. Interviews (samplers)

Q1. Do caregivers make different meaningful choices among care, work and personal life.

P1: "Of course, people make different choices all the time. They can't decide what to make meaning of, For caregivers their primary tasks are to care. However we still need to work sometimes we don't have time to care or to work. Full time job can be troublesome because sometimes if my father needs me it's hard to stay with him during the workdays. The busy schedule affects my motivations in work and sometimes upset me. Work stress in return can also influence my attitude and feelings when I at home caring my father. I don't have time for too much of my personal business let alone learning or studying or getting another degree. I am fully occupied by care and work already. So I need to comfort myself so that I can continue with my work especially. I always say to myself he's my father I need to take care of him no matter what happens and even I lost my job. I'm not going to make too much money from my job. My job only helps me get some money to buy medicines and some community care for my father. So these are meaningful to me. I don't have hope for my own personal business because that's not my concern even though I hope to have my own time either doing exercise or watching movies with my friends. I don't even have time to get enough sleep. But now I don't have any other ways. I have to persuade myself to either give up on care or work or spend more time for myself. I would imagine other people can make difference meaningful choices and are different from mine because my father has serious illnesses and he needs constant care from me. For other families with mild dementia they don't need so much time to care because the patient they can still live independently somehow so that their caregivers can spend more time in their work or their personal interests."

P2: "I am still optimistic about the current situation. My mother was lucky to have and early diagnosis so that we can take her to the doctor very soon and she's on medications since day one of her diagnosis. She has been doing fine. So far she can eat and goes to toilet by herself sometimes she needs some assistance for going outside and sometimes a little bit shopping.

Because she likes to go outside she doesn't want to stay at home because she would feel boring. I am still having full time job because it's not easy for me to get to my current status and I still hope that I can get promoted next year. So that's my hope for my work and also I hope my carer can help my mother to get better or not get even worse. So her body and mind can stay healthy stay out. I guess that's the most comfort to me. I like to travel a lot when I was young. I still like to spend some time with my husband and son on weekends to travel a little bit, or going to the park or some recreational things to make some fun. I personally believe some relax is necessary in our daily lives because if you get too stressful. You won't be able to do anything. If you get happier you can do things more actively and happily."

Q2. Do more meanings in life help promote your well-being?

P1: "Yeah I think it's helpful. I have tried many things to upbeat myself. Like talking to my friends or go to the counselors. However I can't do that all the time. Since I am getting busier I don't have too much time to hang out with my friends. Once I joined a workshop where all the participants are dementia caregivers. I learned that many of them need some kind of religious belief. The religion can help them to make sense of the sufferings and comfort themselves. I have no other choices I mean. Now I sleep 4 to 5 hours every day and have rest of the time busy doing care and work. I only have one father and I cannot let him go. I need to take good care of him. We always say filial to our parents. I am just doing the right thing that everybody would do. I think every family they have different issues with elderly people. In our family it's about dementia. Maybe other families it's about cancer or heart diseases or other brain diseases. I think when illnesses come to older people their families suffer. When you find other people are having similar situations and problems it won't be that sad because you won't fail alone. There are so many people are facing the same things and same issues. We need to find some way to help them and also help ourselves. You will feel much better if you are not alone and people are all thinking to deal with it. You are supported and we should together fight against this disease."

P2: “I am not sure about meaningful life in this way. I have been optimistic all the time. I think that every issues we'll have solutions. There it's always some way to conquer the difficulties. For me my mother is having dementia and I know that she will become more severe in the next few years. I hope that I will learn enough knowledge and care skills to cope with her severity and symptoms. I personally don't have any religious belief. I trust myself undoing everything. So with my mother. I need to work hard to help my mother to get through it's difficult times as well as myself to sustain my job and also my personal life with my own core family. I believe everyone will have some kind of chronic diseases in their old age. I am not afraid that I will get measured in the future. I only hope that with sufficient family support and Social Security I can spend a good quality of life during my late life. And for my mother she is lucky enough to get Social Security and medical care to cover her expenses on medications and hospital visits. And she was early diagnosed so that she can get early treatment and interventions. The reason why care my mother is meaningful is that we need to take positive attitude towards any difficulties especially for our family members. Because family reunion is a critical thing in Chinese society. We should take care of our family members. And we will be taking good care of as well. Promoting our family members health it's meaningful as well as tackling problems is helpful.”

Q3. Does the dyadic relationship help promote your well-being when you have found some meanings in care, work and life?

P1: “you mean the relationship between me and my father yes I think it does help. Can sacrifice anything for my father because he's my father. I would do the same thing to my family members. So I think the relationship is very important. The better the relationship we have, better care can be provided. Is an important figure during my growing up. I certainly have a good relationship with him. I admire him and care for him. 'cause we are so close I want him to have a good life. So I work hard to help him and to help the family because he is an important figure in our family. I think the dyadic relationship is very crucial two daily caregiving experiences because I care for my father. However for work and personal life the relationship appears not that important. There are some cases where the relationship might influence my work performance or personal life. For example if there is any conflict between me and my father that experience will impact my work performance on the second day. Or bad relationship well reduce my interests in my personal interests period. For me with a good relationship with my father I can perform better in my daily caregiving and working performance. If I can Weld balance my care and work at the same time I would definitely feel so much better and a higher level of well being."

P2: "I think the dyadic relationship is definitely useful in helping promoting my well being. I have a good relationship with my mother. And I care for her so much. She is very important to me and to our family. We would do anything to help her if she had troubles. If anything that can promote her well-being well certainly promote mine. Good relationship with my mother or other family members are very important. If the family is harmonious we can work and live happily all together. If not, it is difficult for us to concentrate on our daily activities. Good relationships with family support enables family members to work even harder in their daily life. Bad relationship would deviate people's attention from their care and work. There are a few cases where a desirably good relationship can compromise my daily activities. For example, you should know as well, like a kid, if you have a high expectation on him or care about how he thinks about you, you will feel more stressful when he does something inappropriately or wrong. It is similar for an older adult. If you expect too much from her attitude and reactions to your care, you will feel much more lost due to the huge differences between your expected reactions and actual reactions from your mother."